

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

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Information and Entertainment Systems - Edge & MKX

SPECIFICATIONS

TORQUE SPECIFICATIONS

Description	Nm	lb-in
AM/FM antenna base bolt	7	62
AM/FM antenna cable bolt	12	106
Satellite radio antenna bolt	7	62
Satellite digital audio receiver system (SDARS) module nuts	7	62
Subwoofer amplifier screws	5	44
Subwoofer speaker bolts	7	62

DESCRIPTION AND OPERATION

INFORMATION AND ENTERTAINMENT SYSTEM

Audio Line-Up

The following audio systems are available:

- Base
 - AM/FM single CD with MP3
 - 4 speakers
- Premium - Edge
 - AM/FM 6-CD changer with MP3
 - 4 speakers
- Audiophile
 - AM/FM 6-CD changer with MP3
 - Four 2-way speakers and a subwoofer (9 speakers total)
- Premium - MKX
 - AM/FM 6-CD changer with MP3
 - 6 speakers in 6 locations (includes separate tweeter speakers)
- THX® sound
 - AM/FM 6-CD changer with MP3
 - 12 speakers in 10 locations
 - Audio digital signal processing (DSP) module (located behind the subwoofer)
 - Also available with navigation audio system
- Navigation

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

- Navigation audio control module (ACM)
- Voice-activation (includes microphone in the inside rear view mirror)
- Available with Premium (MKX), Audiophile (Edge), or THX® sound system (MKX)

All audio systems include the following features:

- Speed sensitive volume
- Accessory delay
- MP3 capability
- Audio input jack

The following options are also available:

- Satellite audio system
 - Available with all audio systems
- DVD entertainment system
 - Available with all audio systems
 - Not available on vehicles with panoramic roof
- SYNC
 - Optional on Edge, standard on MKX
- Steering wheel controls
 - Standard on all audio systems except base
 - Unique steering wheel controls for vehicles with the SYNC system

Antenna

The antenna system consists of the antenna and the antenna cable. The antenna cable connects the ACM to the antenna, located on the front of the roof.

Audio Input Jack - Vehicles Without The SYNC System

The audio input jack is used to connect a portable MP3 device to the vehicle. When connected, the audio files from the MP3 player can be broadcast through the vehicle speakers.

Satellite Audio System

The satellite audio system provides satellite-broadcast audio. The satellite antenna is mounted on the rear spoiler. The satellite audio system consists of the following components:

- Satellite digital audio receiver system (SDARS) module (located behind the liftgate scuff plate trim panel)
- Satellite radio antenna
- Satellite radio antenna cable

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

DVD Entertainment System

The DVD entertainment system is available on all vehicles except the base Edge and vehicles equipped with a panoramic roof. The DVD entertainment system includes the following:

- Rear entertainment module (RETM)
- A/V ports for connecting a gaming system
- 2 wireless headphones
- A remote control

THX® Audio System

The THX® audio system provides theatre quality sound inside the vehicle. The THX® audio system is only available for the MKX, and includes the following components:

- DSP module
- Front center imaging speaker (also referred to as the instrument panel speaker)
- 2 mid-woofers in the front doors
- 2 tweeter speakers in the front doors
- 2 co-axial speakers in the rear doors
- 2 rear mid-tweeters in the D-pillars
- A subwoofer

Navigation System

The navigation system audibly and visually guides the user to a selected destination. The system uses a global positioning system (GPS) antenna, a vehicle speed input, and other sensor information to accurately establish the vehicle position. The ACM provides the visual and audible instructions of the maneuvers required to arrive at the pre-entered destination.

The only external component of the navigation system is the GPS antenna, located in the instrument panel.

SYNC System

The SYNC system is comprised of the following:

- Accessory protocol interface module (APIM), located underneath the floor console
- SYNC-specific steering wheel controls
- Microphone located in the rear view mirror
- Universal serial bus (USB) port, located in the center console
- Audio input jack, located in the center console

The SYNC system is a hands-free communications and entertainment system that allows the following interactions:

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

- Send and receive phone calls via a Bluetooth®-enabled phone
- Send and receive text messages via a Bluetooth®-enabled phone
- Connect media devices (such as an iPod® or USB device) in order to play audio files
- Play media files via a Bluetooth®-enabled audio device

Not all features will be available with every phone.

For additional information on audio system operation, including the SYNC system, refer to the Owner's Literature.

DIAGNOSTIC TESTS

INFORMATION AND ENTERTAINMENT SYSTEM

Special Tools

Illustration	Tool Name	Tool Number
 ST1137-A	73III Automotive Meter	105-R0057 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	
 ST2574-A	Flex Probe Kit	105-R025C or equivalent
 ST3051-A	Universal Serial Bus (USB) Male-A To Male-A Cable	CCMUSB2-AM-AM-10

Principles of Operation

NOTE: The smart junction box (SJB) is also known as the generic electronic module (GEM).

Audio Control Module (ACM)

The audio system can be operated while the key is in RUN or ACC. The accessory delay feature allows the audio system to be operated for a preset period of time after the key is turned to OFF and a front door has not been opened. The audio control module (ACM) sends AC voltage audio signals to the speakers or to the amplifier, depending on vehicle configuration. The ACM can produce its own DTCs, which can be communicated to the scan tool through the medium speed controller area network (MS-CAN), or can be retrieved through the ACM self-test by simultaneously pressing presets 3 and 6. The ACM communicates via the MS-CAN.

THX® Audio System

The THX® audio system provides theatre quality sound in the vehicle. Audio signals are sent to the audio digital signal processing (DSP) module, which processes the audio signals and sends them to the 13 speakers: 4 door speakers, 2 tweeter speakers, 3 front/center speakers, 2 rear/center speakers, and 2 subwoofer speakers. The DSP module communicates via the MS-CAN.

Navigation

The navigation module is integrated in the ACM and controls the operation and the interface between the user, the vehicle subsystems, and the external components. The navigation ACM communicates via the MS-CAN and can be diagnosed with a scan tool.

The vehicle navigation system guides the user to a pre-entered destination. A map DVD stored in the navigation ACM is used for calculating the vehicle course. The ACM display audibly and visually instructs the user of the maneuvers required to arrive at the destination entered.

A voice recognition system allows the user to interface with the system without using the touch screen. A microphone located in the interior rear view mirror provides a direct input to the navigation ACM.

In order to calculate initial vehicle position, the global positioning system (GPS) antenna is used to track several available satellites simultaneously.

The wheel pulse signal and transmission gear selected signal are received by the navigation ACM through the MS-CAN. These signals are used to increase the accuracy of the displayed course.

Satellite Audio

The satellite audio system consists of a satellite digital audio receiver system (SDARS) module, a satellite radio antenna, and an antenna cable. The satellite radio antenna receives digital audio signals and sends them to the SDARS module, where the signals are converted to analog and sent to the ACM, the rear entertainment module (RETM) (if equipped with a DVD entertainment system), or the accessory protocol interface module (APIM) (if equipped with the SYNC system). The SDARS module communicates via the MS-CAN.

SYNC System

The SYNC system allows interaction with several types of customer devices, including mobile phones and

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

media devices. The system is comprised of the APIM, a microphone (located in the interior rear view mirror), the universal serial bus (USB) cable and port, and the audio input jack. The APIM contains an on-board Bluetooth® chipset, which enables certain wireless devices to interact with the system.

The APIM consists of 2 internal modules: the consumer interface processor (CIP) and the vehicle interface processor (VIP). The modules are not replaceable individually, but can be flashed independently, if required.

The CIP interfaces with all of the inputs to the APIM. The CIP contains an analog-to-digital- to-analog converter, as well as the Bluetooth® chipset. Any consumer-available application upgrades that are available are loaded directly to the CIP through the USB port.

The VIP provides an interface between the CIP and the vehicle. The main functions of the VIP are controlling the APIM power management and translating both inbound and outbound signals over the controller area network (CAN).

The APIM can receive inputs from the following audio sources:

- USB port
- Audio input jack
- SDARS module
- Bluetooth®

The USB port can be used for connecting a media device (such as an iPod®) with the device's available cable, or for directly plugging in a portable mass storage device (such as a "thumb drive"). When playing media files stored on a mass storage device, the SYNC system will only play files that do not have digital rights management (DRM) protection. The USB port can also be used for uploading vehicle application upgrades.

The USB port is powered by the APIM, so no external power source is needed to power a device plugged into the USB port.

The audio input jack can be used for connecting a media device (such as an iPod®) utilizing a 1/8-inch audio jack. When a device is connected through the audio input jack, only the speaker volume can be controlled by the ACM. All other functions (such as seek, fast forward, pause, etc.) must be carried out on the device itself.

If the vehicle is equipped with satellite audio, the audio signals from the SDARS module are sent to the APIM, rather than directly to the ACM. The APIM then relays the satellite audio signals to the ACM, or to the RETM if the vehicle is equipped with a DVD entertainment system. The RETM then relays the signals to the ACM.

The Bluetooth® interface can accommodate both Bluetooth®-enabled mobile phones and Bluetooth®-enabled media devices. Any Bluetooth® device used with the SYNC system must first be paired with the system before it is operational.

Bluetooth® is a secure, short-range radio frequency that allows devices to communicate wirelessly through radio waves. The operating range of a Bluetooth® signal is a maximum of 32 feet.

Only one Bluetooth® phone and one Bluetooth® device can be connected to the system at any one time. If an

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

additional device of either type is paired with the system and made active, the APIM will disconnect any active connection and establish a connection with the new device.

It is important to understand that not all mobile phones will have the same level of features when interacting with the SYNC system. For a list of compatible phones, .

In addition to audio information, metadata may also be sent to the APIM from a device plugged into the USB port. Metadata consists of such information as artist, album title, song title, and genre. The metadata is used by the APIM to create indexes that can be used to sort for particular music, based on customer preference. Not all USB devices will send metadata to the APIM; also, no metadata is transferred when a device is connected through the audio input jack. When a new media device is connected to the SYNC system, the APIM will automatically index the information. This may take up to several minutes (depending on the amount of data on the device), and is considered normal operation. When a device that was previously connected to the SYNC system is reconnected, the APIM will update the index (rather than creating a new one), which reduces the amount of time needed to create the index.

The APIM receives both stereo and mono sound inputs, and can also transmit both stereo and mono sound. The mono function is used to receive the microphone input, and to send sound to the ACM for voice prompts, the text-to-speech (TTS) feature, ring tones, and any audio received through a connected mobile phone. The TTS feature speaks information so that it does not have to be read from the display.

The APIM communicates via the MS-CAN.

Voice Recognition For Vehicles With Navigation And The SYNC System

When a vehicle is equipped with both the navigation system and the SYNC system, the interaction between the audio system components is unique for voice recognition.

Both the navigation ACM and the APIM utilize voice-recognition, but in order for the modules to determine which one should be acknowledging the voice command, the voice command set is different. This allows the modules to identify the intended source when the VOICE button is pressed. The VOICE (an icon of someone speaking), MEDIA, and PHONE (an icon of a phone and an up-and-down arrow) buttons are wired to the navigation ACM, rather than to the APIM (as they are in non-navigation audio systems).

When the audio system enters voice recognition mode, the ACM sends a CAN message to the APIM to enter voice recognition mode. The microphone is wired directly to the APIM, which relays the microphone input to the ACM through dedicated wiring. Control of the voice functions is given to whichever module is indicated by the first voice command.

Steering Wheel Controls

The steering wheel controls consist of a series of resistors. Each steering wheel control switch function corresponds with a specific resistance value within the switch. When a switch is pressed, the ACM (or the APIM) monitors the change in reference voltage to determine the requested function.

Audio Input Jack - Vehicles Without The SYNC System

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

The audio input jack allows for a portable MP3 player to be connected to the vehicle audio system. When a portable MP3 player is connected, audio from the MP3 player can be played through the vehicle speakers.

Antenna - AM/FM

The roof-mounted antenna receives both AM and FM radio waves. The audio signals are then sent to the ACM through the antenna cable. An antenna module is integral to the roof-mounted antenna base.

Subwoofer

The Audiophile subwoofer is powered by its own amplifier. The subwoofer in the THX® system is powered by the THX® amplifier. The enable/clip detection circuit carries out 2 functions: to turn the amplifier on, and to indicate to the ACM when the subwoofer output distortion signal reaches a damaging level. The ACM then clips the audio output signal to prevent damage to the subwoofer.

DVD Entertainment System

When a video disc is inserted into the RETM and the PLAY button is pressed, an image is displayed on the screen and a stereo signal is transmitted to the ACM. The DVD entertainment system allows for single/dual-play mode. The RETM communicates via the MS-CAN.

Single-Play Mode

In single-play mode, any audio input routed through the RETM is broadcast through all of the vehicle speakers. These audio inputs include the RETM, satellite audio, SYNC audio, and peripheral gaming devices plugged into the RETM.

Dual-Play Mode (All Audio Systems Except THX®)

In dual-play mode, the ACM continues to control the output to the front speakers, while the RETM mutes the audio to the rear speakers. The RETM continues to broadcast the rear audio source to the headphones. When the headphones are set to channel A, all audio sources can be accessed. When the headphones are set to channel B, only the DVD audio and auxiliary audio sources (such as a gaming system) can be sent to the headphones.

Dual-Play Mode (THX®)

In dual-play mode, the DSP module continues to play the front audio source over the front speakers while muting the rear speakers. The RETM continues to broadcast the rear audio source to the headphones. When the headphones are set to channel A, all audio sources can be accessed. When the headphones are set to channel B, only the DVD audio and auxiliary audio sources (such as a gaming system) can be sent to the headphones.

Noise Suppression Equipment

The radio frequency suppression equipment reduces interference transmitted through the speakers by the engine ignition and electrical systems. When installing any new radio suppression equipment components, make sure that a good contact is made at all connections.

Audio Signals

Audio signal flow varies greatly depending on vehicle content.

Stereo signals contain left and right channel information, and are used for most audio signals.

A mono signal is used for the microphone input to the APIM. The mono signal is also used for the voice prompts, the TTS feature, ring tones, and any audio received through a connected mobile phone. These audio signals are output from the APIM to the ACM. The mono and stereo outputs from the APIM utilize separate circuits.

A digital signal is used to transmit data from a media device connected through the USB port to the APIM. The APIM then converts the signal to analog and relays the signal to the ACM.

A wireless signal is used to broadcast audio signals from a Bluetooth® device to the APIM. As with a digital signal sent through the USB port, the APIM converts the wireless audio signal to analog and relays it to the ACM.

Network Communication

The following audio system components communicate via the medium speed controller area network (MS-CAN):

- Audio control module (ACM)
- Navigation ACM
- Satellite digital audio receiver system (SDARS) module
- Rear entertainment module (RETM)
- Audio digital signal processing (DSP) module
- Accessory protocol interface module (APIM)

The following messages are utilized by the audio system:

Message	Transmitting Module	Receiving Module (s)	Audio System Function
Ignition Switch Position	Smart junction box (SJB)	ACM RETM DSP module APIM	Indicates ignition switch position for power-up/shut-down.
Illumination Dimmer Level	SJB	ACM RETM	Controls the backlight intensity, based on the position of the dimmer switch.
Transmission Selector (PRNDL) Range	Instrument cluster (IC)	ACM	For the navigation system, this signal is used for more accurate navigation

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

			tracking.
Accessory Delay Status	SJB	ACM RETM APIM	When active, this signal allows the audio component to be operated after the vehicle is shut off and a door has not been opened.
Volume Cutback	Parking aid module (PAM)	ACM	When active, this signal causes the ACM to reduce the audio output when the parking aid tone is active.
Vehicle Speed	IC	ACM DSP module	Used by the ACM (or DSP module) for the speed-compensated volume function.
Navigation Radio Rolling Wheel Count	IC	ACM	For the navigation system, this signal is used for more accurate navigation tracking.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

VISUAL INSPECTION CHART

Mechanical	Electrical
• Audio control module (ACM) • Antenna • Rear entertainment module (RETM) • Steering wheel controls • Audio input jack • Universal serial bus (USB) port • Speaker(s) • Noise suppression equipment	• Smart junction box (SJB) fuse(s): ○ 3 (15A) (RETM, accessory protocol interface module [APIM]) ○ 18 (20A) (audio digital signal processing [DSP] module) ○ 28 (5A) (ACM hot in START) ○ 38 (20A) (subwoofer or DSP module) ○ 39 (20A) (ACM) ○ 41 (15A) (delayed

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

accessory)

3. If the fault is visually evident, correct the cause (if possible) before proceeding to the next step.

NOTE: **Make sure to use the latest scan tool software release.**

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: **The vehicle communication module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM.**

5. If the scan tool does not communicate with the VCM:

- Check the VCM connection to the vehicle.
- Check the scan tool connection to the VCM.
- Refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.

6. If the scan tool does not communicate with the vehicle:

- Verify the ignition key is in the ON position.
- Verify the scan tool operation with a known good vehicle.
- Refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.

NOTE: **Do not press any buttons on the ACM while the ACM is carrying out the self-test.**

7. Carry out the network test.

- If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
- If the network test passes, retrieve and record the continuous memory DTCs.

8. Carry out the self-test diagnostics for the ACM.

9. If the DTCs retrieved are related to the concern, go to DTC Charts.

10. If no DTCs related to the concern are retrieved, go to the Speaker Walk-Around Test, the Audio Control Module (ACM) Self-Diagnostic Mode, or the Satellite Audio Bezel Diagnostic Test.

Speaker Walk-Around Test

NOTE: **To enter the speaker walk-around test or ACM self-diagnostic mode, the ACM must be on and in radio tuner (AM/FM) mode.**

1. To enter the speaker walk-around test, simultaneously press the ACM preset buttons 3 and 6.
2. The speaker walk-around test stops at each speaker and applies sound to each speaker for about 1 to 2 seconds. Each speaker is tested and displayed on the ACM in the following sequence:

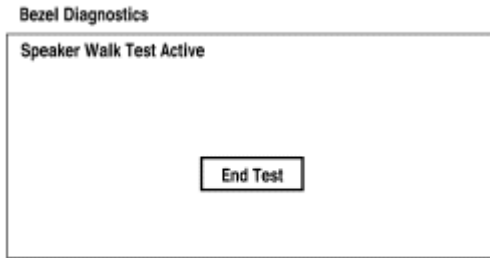
- Except THX®: RF, LF, LR, RR, SUBWOOFER.
 - THX®: CSA FRONT, TWEETER LF, TWEETER RF, CENTER REAR, SPEAKER LF, SPEAKER RF, SPEAKER LR, SUBWOOFER, SPEAKER RR.
3. To exit the speaker walk-around test, turn the key to OFF, turn the ACM off, press preset button 1 for diagnostics (ACMs without navigation), or press "End Test" (navigation ACM).

Audio Control Module (ACM) Self-Diagnostic Mode - All ACMs Without Navigation

NOTE: To enter the ACM self-diagnostic mode, the ACM must be turned on and in radio tuner (AM/FM) mode.

1. To enter the following tests, press the desired preset button while in the speaker walk-around test or while in the ACM self-diagnostic mode.
2. The self-diagnostic mode has the following functions available:
 - Preset button 1 = On-Demand Self-Test. This button runs the on-demand self-test.
 - Pressing the MENU DOWN button allows scrolling of any DTCs found during the test while in this menu.
 - Preset button 2 = Display Continuous DTCs. This button enables viewing of any continuous DTCs that have been logged.
 - Pressing the MENU UP button allows scrolling of any DTCs while in this menu.
 - While continuous DTCs are being displayed, pressing the EJECT button will clear all present DTCs.
 - Preset button 3 = Signal Strength Test. This button displays the signal strength.
 - Preset button 4 = Software Version Display. This button displays the ACM software version.
 - Pressing the MENU UP button allows scrolling of all audio subsystem software versions while in this menu.
 - Preset button 5 = Display Test. This test illuminates all the display segments for 5 seconds, then turns all segments off.
 - Preset button 6 = Configuration Status. This button enables ACM configuration status.
 - Pressing the MENU button displays the ACM part number while in this menu.
3. To exit the self-diagnostic mode, turn the ACM off or turn the key OFF.
4. If the concern remains and the fault is not detected, go to **Symptom Chart - Audio System** to continue diagnostics.

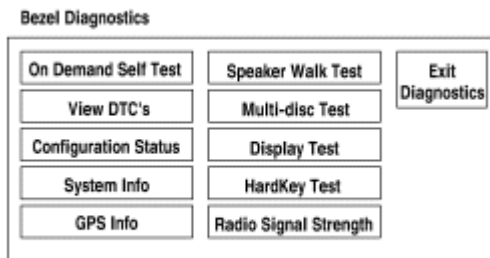
Audio Control Module (ACM) Self-Diagnostic Mode - Navigation ACM



N0026636

Fig. 1: Bezel Diagnostics Screen Display - Speaker Walk Test Active
Courtesy of FORD MOTOR CO.

1. To enter the self-diagnostic mode, press "End Test" during the speaker walk-around test.



N0026635

Fig. 2: Bezel Diagnostics Screen Display - On-Screen Buttons
Courtesy of FORD MOTOR CO.

2. The following diagnostic modes are available using the on-screen buttons:
 - On Demand Self Test - provides internal self-test diagnostics and displays all the DTCs resulting from the self-test.
 - View DTC's - provides a list of all the DTCs currently stored in memory.
 - Configuration Status" - displays the current unit configuration.
 - System Info - provides the navigation module part number and software information.
 - GPS Info - provides satellite information and vehicle current information.
 - Speaker Walk Test - carries out a speaker walk-around test.
 - Multi-disc Test - carries out a test of the CD player mechanism.
 - Display Test - allows the screen colors to be checked, and allows individual touch sectors of the display screen to be tested.
 - Hardkey Test - checks the operation of any ACM button.
 - Radio Signal Strength - carries out a test of the AM/FM antenna signal.
 - Exit Diagnostics - exists bezel diagnostics and returns to the previous function.
3. To exit the self-diagnostic mode, turn the ACM off, turn the key to OFF, or press "Exit Diagnostics".
4. If the concern remains and the fault is not detected, go to **Symptom Chart - Navigation System**.

Satellite Audio Bezel Diagnostic Test

NOTE: To enter the satellite audio bezel diagnostic test, the ACM must be on and in SAT mode.

1. To enter the satellite audio bezel diagnostic test, simultaneously press and hold the AUX button and preset button 2 (except navigation), or the SOUND button and preset button 1 (navigation).
2. Upon entering the self-test, the ACM produces 2 continuously alternating tones of different pitch, one for the right channel, then one for the left.
3. The test continues by displaying any DTCs currently present. If no DTCs are present, NO DTCS will be displayed. If there are DTCs present, the ACM will auto-scroll through the list of active DTCs.
4. Historical DTCs can be viewed by pressing the AUX button and preset button 2 (except navigation), or the SOUND button and preset button 1 (navigation) simultaneously while in the active DTC mode.
 - If any DTCs are present, pressing the AUX button and preset button 2 (except navigation), or the SOUND button and preset button 1 (navigation) will prompt CLEAR DTCS? on the ACM.
 - To clear historical DTCs, press preset buttons 1, 2, and 3 consecutively within 4 seconds.
 - To exit historical DTCs (with or without clearing DTCs), press the AUX button and preset button 2 (except navigation), or the SOUND button and preset button 1 (navigation) simultaneously.
5. If no historical DTCs are present, pressing the AUX button and preset button 2 (except navigation), or the SOUND button and preset button 1 (navigation) simultaneously while in active DTC mode will display the DLP software version.
6. To exit the satellite audio bezel diagnostic test, press the AUX button and preset button 2 (except navigation), or the SOUND button and preset button 1 (navigation) simultaneously while the DLP software version is displayed, or turn the ACM off.
7. If the concern remains and the fault is not detected, go to **Symptom Chart - Audio System**.

DTC Charts

NOTE: For all U-codes listed in this chart, if no related symptoms are observed, disregard the DTC.

NOTE: For all U-codes listed in this chart, if DTC B1318 is present, diagnose it before diagnosing the U-code DTC.

AUDIO CONTROL MODULE (ACM) DTC CHART

DTC	Description	Action
B1117	Audio Steering Wheel Button Stuck	VERIFY that no steering wheel control buttons are stuck and that no steering wheel control button is pressed during the self-test. If no concern is found, go to Pinpoint Test F .
		NOTE: This DTC applies to the navigation DVD mechanism, not the DVD entertainment system.

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

B1119 (Navigation only)	Audio Disc DVD Player Thermal Shutdown	The audio control module (ACM) was over temperature. Navigation operation will resume after the ACM cools. This is normal operation. CLEAR the DTCs.
B1136 (Navigation only)	Audio Steering Wheel Switch #2 Circuit Failure	Go to Pinpoint Test F .
B1140 (Navigation only)	Map Disk Invalid	REMOVE the map DVD and CLEAN it. INSPECT for scratches, fingerprints, or damage, and REPAIR as necessary. VERIFY a valid map DVD is being used. If no concern is found, INSERT a known good map DVD. REPEAT the self-test. • If the system operates correctly, the concern was caused by a damaged or invalid map DVD. • If DTC B1140 is retrieved again, INSTALL a new ACM. REFER to <u>Audio Control Module (ACM)</u>. TEST the system for normal operation.
B1154	Subwoofer #1 Short to Ground	Go to Pinpoint Test E .
B1155	Subwoofer #1 Short to Battery	Go to Pinpoint Test E .
B1158	Subwoofer #2 Not Connected	Go to Pinpoint Test E .
B1318	Battery Voltage Low	CLEAR the DTCs. TEST the charging system. REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article. • If the charging system is OK, the DTC was set by a previous low voltage condition. No concern is found at this time. • If the charging system is not OK, REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article to diagnose the low battery voltage condition.
B1342	ECU is Faulted	CLEAR the DTCs. REPEAT the self-test. If DTC B1342 is retrieved again, INSTALL a new ACM. REFER to <u>Audio Control Module (ACM)</u> . TEST the system for normal operation.
B2103	Antenna Not Connected	Go to Pinpoint Test A .
B2204 (Navigation only)	GPS Antenna Connection Open or Short	Go to Pinpoint Test M .
B2274	Phone Transceiver Active Circuit Failure	DISREGARD the DTC. CLEAR the DTCs.
	Audio Reverse Aid Mute	

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

B2384	Input Ckt. Failure	DISREGARD the DTC. CLEAR the DTCs.
B2404	Audio Steering Wheel Switch Circuit Failure	Go to Pinpoint Test F .
B2405	Audio Disc CD Player Thermal Shutdown Fault	The ACM was over temperature. Audio operation will resume after the ACM cools. This is normal operation. CLEAR the DTCs.
B2406	Audio Disc CD Player Internal Fault	NOTE: DTC B1342 will also be set. CLEAR the DTCs. REPEAT the self-test. If DTC B2406 is retrieved again, INSTALL a new ACM. REFER to <u>Audio Control Module (ACM)</u>. TEST the system for normal operation.
B2477	Module Configuration Failure	REFER to <u>MODULE CONFIGURATION</u> article.
B2633 (Navigation only)	Driver-Front Microphone Circuit Failure	Go to Pinpoint Test R .
B2656	DVD (Digital Versatile Disk) Error	NOTE: DTC B1342 will also be set. NOTE: This DTC refers to the navigation DVD mechanism, not the DVD entertainment system. CLEAR the DTCs. REPEAT the self-test. If DTC B2656 is retrieved again, INSTALL a new ACM. REFER to <u>Audio Control Module (ACM)</u>. TEST the system for normal operation.
B2913 (Audiophile only)	Audio Subwoofer Not Connected	Go to Pinpoint Test E .
B2924	Audio Button Stuck	NOTE: For the navigation ACM, the hardkey test can be used to isolate a stuck button. VERIFY no ACM buttons are stuck and that no ACM button is pressed during the self-test. CLEAR the DTCs. REPEAT the self-test. If DTC B2924 is retrieved again, INSTALL a new ACM. REFER to <u>Audio Control Module (ACM)</u>. TEST the system for normal operation.
B2965	Audio System Speaker Circuit Fault	Go to Pinpoint Test B .
C1992	Vehicle Speed Circuit Failure	DISREGARD the DTC. CLEAR the DTCs.

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

P0182	Reverse Input Circuit	DISREGARD the DTC. CLEAR the DTCs.
U0140	Lost Communication With Body Control Module (GEM)	If a backlighting concern is present, REFER to <u>INSTRUMENT CLUSTER AND PANEL ILLUMINATION</u> article. If the accessory delay does not function correctly, REFER to <u>WIPERS AND WASHERS</u> article.
U0155	Lost Communication With Instrument Panel Cluster (IC) Control Module	If the speed sensitive volume does not operate correctly, go to Pinpoint Test Y . If the navigation is inaccurate, go to Pinpoint Test P .
U0159	Lost Communication With Parking Assist Control Module (PAM)	Go to Pinpoint Test Q .
U0193	Lost Communication With Digital Audio Control Module (SDARS)	For vehicles without a DVD entertainment system, go to Pinpoint Test H . For vehicles with a DVD entertainment system, go to Pinpoint Test I .
U0196	Lost Communication With Entertainment Control Module - Rear (AUX)	CARRY OUT the network test using the scan tool. • If the scan tool communicates with the rear entertainment module (RETM), go to <u>Symptom Chart - Audio System</u> to diagnose the observed symptom. • If the scan tool does not communicate with the RETM, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
U0197	Lost Communication With Telephone Control Module	Go to Pinpoint Test S .
U0238	Lost Communication With Digital Audio Control module "D" (DSP)	CARRY OUT the network test using the scan tool. • If the scan tool communicates with the audio digital signal processing (DSP) module, go to <u>Symptom Chart - Audio System</u> to diagnose the observed symptom. • If the scan tool does not communicate with the DSP module, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
U0249	Lost Communication With Entertainment Control Module - Rear "B" (RCU)	DISREGARD the DTC. CLEAR the DTCs.
U2050	No Application Present	REFER to <u>MODULE CONFIGURATION</u> article.
U2051	One or More Calibration Files Missing/Corrupt	REFER to <u>MODULE CONFIGURATION</u> article.
U2473	Unexpected Vehicle Speed (VSS)	Go to Pinpoint Test P .

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

NOTE: For all U-codes listed in this chart, if no related symptoms are observed, disregard the DTC.

NOTE: For all U-codes listed in this chart, if DTC B1318 is present, diagnose it before diagnosing the U-code DTC.

AUDIO DIGITAL SIGNAL PROCESSING (DSP) MODULE DTC CHART

DTC	Description	Action
B1158	Subwoofer #2 Not Connected	Go to Pinpoint Test C .
B1317	Battery Voltage High	CLEAR the DTCs. TEST the charging system. REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article. • If the charging system is OK, the DTC was set by a previous over-charge condition. No concern is found at this time. • If the charging system is not OK, REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article to diagnose the over-charging condition.
B1318	Battery Voltage Low	CLEAR the DTCs. TEST the charging system. REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article. • If the charging system is OK, the DTC was set by a previous low voltage condition. No concern is found at this time. • If the charging system is not OK, REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article to diagnose the low battery voltage condition.
B1342	ECU is Faulted	CLEAR the DTCs. REPEAT the self-test. If DTC B1342 is retrieved again, INSTALL a new audio digital signal processing (DSP) module. REFER to <u>Audio Digital Signal Processing (DSP) Module</u>. TEST the system for normal operation.
B2477	Module Configuration Failure	REFER to <u>MODULE CONFIGURATION</u> article.
B2913	Audio Subwoofer Not Connected	Go to Pinpoint Test C .
B2925	Subwoofer Speaker Short Circuit	If the symptom is no audio from all speakers, go to Pinpoint Test D . If the symptom is no audio from the subwoofer, go to Pinpoint Test C .
		NOTE:

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

B292A	Subwoofer #2 Speaker Short Circuit	Despite the DTC description, this DTC applies to the THX® amplifier clip/enable circuit, not the subwoofer. Go to Pinpoint Test D .
B2965	Audio System Speaker Circuit Fault	If the symptom is no audio from all speakers, go to Pinpoint Test D . If the symptom is no audio from one or more (but not all) speakers, go to Pinpoint Test C .
C1992	Vehicle Speed Circuit Failure	DISREGARD the DTC. CLEAR the DTCs.
U0155	Lost Communication With Instrument Panel Cluster (IC) Control Module	CARRY OUT the network test using the scan tool. • If the scan tool communicates with the IC, go to <u>Symptom Chart - Audio System</u> to diagnose the observed symptom. • If the scan tool does not communicate with the IC, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
U0184	Lost Communication With Radio (ACM)	CARRY OUT the network test using the scan tool. • If the scan tool communicates with the ACM, go to <u>Symptom Chart - Audio System</u> to diagnose the observed symptom. • If the scan tool does not communicate with the ACM, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
U0196	Lost Communication With Entertainment Control Module - Rear (AUX)	CARRY OUT the network test using the scan tool. • If the scan tool communicates with the rear entertainment module (RETM), go to <u>Symptom Chart - Audio System</u> to diagnose the observed symptom. • If the scan tool does not communicate with the RETM, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
U2050	No Application Present	REFER to <u>MODULE CONFIGURATION</u> article.
U2051	One or More Calibration Files Missing/Corrupt	REFER to <u>MODULE CONFIGURATION</u> article.

NOTE: For all U-codes listed in this chart, if no related symptoms are observed, disregard the DTC.

NOTE: For all U-codes listed in this chart, if DTC B1318 is present, diagnose it before diagnosing the U-code DTC.

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

SATELLITE DIGITAL AUDIO RECEIVER SYSTEM (SDARS) MODULE DTC CHART

DTC	Description	Action
B1031	SDARS Satellite Antenna Open	Go to Pinpoint Test J .
B1032	SDARS Satellite Antenna Short	Go to Pinpoint Test J .
B1318	Battery Voltage Low	CLEAR the DTCs. TEST the charging system. REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article. • If the charging system is OK, the DTC was set by a previous low voltage condition. No concern is found at this time. • If the charging system is not OK, REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article to diagnosis the low battery voltage condition.
B1342	ECU is Faulted	CLEAR the DTCs. REPEAT the self-test. If DTC B1342 is retrieved again, INSTALL a new satellite digital audio receiver system (SDARS) module. REFER to <u>Satellite Digital Audio Receiver System (SDARS) Module</u>. TEST the system for normal operation.
B2477	Module Configuration Failure	REFER to <u>MODULE CONFIGURATION</u> article.
U0184	Lost Communication With Radio (ACM)	For vehicles without a DVD entertainment system, go to Pinpoint Test H . For vehicles with a DVD entertainment system, go to Pinpoint Test I .
U0196	Lost Communication With Entertainment Control Module - Rear (AUX)	Go to Pinpoint Test I .
U0197	Lost Communication With Telephone Control Module	For vehicles without a DVD entertainment system, go to Pinpoint Test H . For vehicles with a DVD entertainment system, go to Pinpoint Test I .
U0249	Lost Communication With Entertainment Control Module - Rear "B" (RCU)	DISREGARD the DTC. CLEAR the DTCs.
U0255	Lost Communication With Front Controls Interface Module	DISREGARD the DTC. CLEAR the DTCs.
U0256	Lost Communication With Front Display Interface Module	DISREGARD the DTC. CLEAR the DTCs.
U2050	No Application Present	REFER to <u>MODULE CONFIGURATION</u> article.

NOTE: For all U-codes listed in this chart, if no related symptoms are observed, disregard the DTC.

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

NOTE: For all U-codes listed in this chart, if DTC B1318 is present, diagnose it before diagnosing the U-code DTC.

REAR ENTERTAINMENT MODULE (RETM) DTC CHART

DTC	Description	Action
B1119	Audio Disc DVD Player Thermal Shutdown	The rear entertainment module (RETM) was over-temperature. Operation will resume once the RETM has cooled. This is normal operation. CLEAR the DTCs.
B111F	LCD Thermal Shutdown	The LCD screen was over-temperature. Operation will resume after the LCD screen cools. This is normal operation. CLEAR the DTCs.
B1317	Battery Voltage High	CLEAR the DTCs. TEST the charging system. REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article. • If the charging system is OK, the DTC was set by a previous over-charge condition. No concern is found at this time. • If the charging system is not OK, REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article to diagnosis the overcharging condition.
B1318	Battery Voltage Low	CLEAR the DTCs. TEST the charging system. REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article. • If the charging system is OK, the DTC was set by a previous low voltage condition. No concern is found at this time. • If the charging system is not OK, REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article to diagnosis the low battery voltage condition.
B1342	ECU is Faulted	CLEAR the DTCs. REPEAT the self-test. If DTC B1342 is retrieved again, INSTALL a new RETM. REFER to <u>Rear Entertainment Module (RETM)</u> . TEST the system for normal operation.
B2477	Module Configuration Failure	REFER to <u>MODULE CONFIGURATION</u> article.
B2656	DVD (Digital Versatile Disk) Error	CLEAR the DTCs. REPEAT the self-test. If DTC B2656 is retrieved again, INSTALL a new RETM. REFER to <u>Rear Entertainment Module (RETM)</u> . TEST the system for normal operation.
B2924	Audio Button Stuck	VERIFY no RETM buttons are stuck and that no RETM button is pressed during the self-test. CLEAR the DTCs. REPEAT the self-test. If DTC B2924 is retrieved again,

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

		INSTALL a new RETM. REFER to Rear Entertainment Module (RETM) . TEST the system for normal operation.
U0140	Lost Communication With Body Control Module (GEM)	If a backlighting concern is present, REFER to <u>INSTRUMENT CLUSTER AND PANEL ILLUMINATION</u> article. If the accessory delay does not function correctly, REFER to <u>WIPERS AND WASHERS</u> article.
U0184	Lost Communication With Radio (ACM)	CARRY OUT the network test using the scan tool. • If the scan tool communicates with the ACM, go to <u>Symptom Chart - Audio System</u> to diagnose the observed symptom. • If the scan tool does not communicate with the ACM, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
U0193	Lost Communication With Digital Audio Control Module (SDARS)	Go to Pinpoint Test I .
U2050	No Application Present	REFER to <u>MODULE CONFIGURATION</u> article.

NOTE: For all U-codes listed in this chart, if no related symptoms are observed, disregard the DTC.

NOTE: For all U-codes listed in this chart, if DTC B1318 is present, diagnose it before diagnosing the U-code DTC.

ACCESSORY PROTOCOL INTERFACE MODULE (APIM) DTC CHART

DTC	Description	Action
B1038	Microphone Input Circuit Failure	Go to Pinpoint Test W .
B1117	Audio Steering Wheel Button Stuck	Go to Pinpoint Test F .
B1136	Audio Steering Wheel Switch #2 Circuit Failure	Go to Pinpoint Test F .
B1317	Battery Voltage High	CLEAR the DTCs. TEST the charging system. REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article. • If the charging system is OK, the DTC was set by a previous high voltage condition. No concern is found at this time. • If the charging system is not OK, REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article to diagnosis the over-charging condition.

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

B1318	Battery Voltage Low	CLEAR the DTCs. TEST the charging system. REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article. - If the charging system is OK, the DTC was set by a previous low voltage condition. No concern is found at this time. - If the charging system is not OK, REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article to diagnosis the low battery voltage condition.
B1342	ECU is Faulted	CLEAR the DTCs. REPEAT the self-test. If DTC B1342 is retrieved again, INSTALL a new accessory protocol interface module (APIM). REFER to <u>Accessory Protocol Interface Module (APIM)</u>. TEST the system for normal operation.
B2477	Module Configuration Failure	REFER to <u>MODULE CONFIGURATION</u> article.
U0184	Lost Communication With Radio (ACM)	Go to Pinpoint Test S .
U0196	Lost Communication With Entertainment Control Module - Rear (AUX)	Go to Pinpoint Test S .
U0249	Lost Communication With Entertainment Control Module - Rear "B" (RCU)	DISREGARD the DTC. CLEAR the DTCs.
U0255	Lost Communication With Front Controls Interface Module	DISREGARD the DTC. CLEAR the DTCs.
U0256	Lost Communication With Front Display Interface Module	DISREGARD the DTC. CLEAR the DTCs.
U2050	No Application Present	REFER to <u>MODULE CONFIGURATION</u> article.
U261C	USB #1 Device Error	Go to Pinpoint Test U .

Symptom Chart

Symptom Chart - Audio System

Condition	Possible Sources	Action
No communication with the audio control module (ACM)	- Fuse - Wiring, terminals or connectors - ACM	REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
No communication with the audio digital signal processing (DSP) module	- Fuse - Wiring, terminals or connectors	REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

	DSP module	
No communication with the satellite digital audio receiver system (SDARS) module	- Fuse - Wiring, terminals or connectors - SDARS module	REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
No communication with the rear entertainment module (RETM)	- Fuse - Wiring, terminals or connectors - RETM	REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
No communication with the accessory protocol interface module (APIM)	- Fuse - Wiring, terminals or connectors - APIM	REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
The satellite audio is inoperative/does not operate correctly - vehicles without DVD entertainment	- Wiring, terminals or connectors - APIM (if equipped) - SDARS module - ACM	Go to Pinpoint Test H .
The satellite audio is inoperative/does not operate correctly - vehicles with DVD entertainment	- Wiring, terminals or connectors - RETM - APIM (if equipped) - SDARS module - ACM	Go to Pinpoint Test I .
The ACM display is blank, but the ACM operates	ACM	INSTALL a new ACM. REFER to <u>Audio Control Module (ACM)</u>. TEST the system for normal operation.
The ACM backlighting does not operate correctly	- Backlighting system concern - Wiring, terminals or connectors - ACM	REFER to <u>INSTRUMENT CLUSTER AND PANEL ILLUMINATION</u> article.
The RETM backlighting does not operate correctly	- Backlighting system concern - Wiring, terminals or connectors - RETM	REFER to <u>INSTRUMENT CLUSTER AND PANEL ILLUMINATION</u> article.

2008 Ford Edge SE

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The CD player is inoperative/does not operate correctly	- CD - ACM	- INSPECT the CD for scratches, fingerprints, a loose paper label, incorrect format, or damage. INSERT a known good CD and TEST the system. - If the system operates correctly, the concern was caused by a damaged CD. - If the system does not operate correctly, INSTALL a new ACM. REFER to <u>Audio Control Module (ACM)</u>. TEST the system for normal operation.
Poor reception - AM/FM	- Wiring, terminals or connectors - Antenna - Antenna cable - Charging system - Ignition system - Noise suppression equipment - ACM	Go to Pinpoint Test A.
Continuous seek/scan in AM/FM	- RDS/PTY function setting - Antenna - Antenna cable - Noise suppression equipment - Antenna module - ACM	Go to Pinpoint Test A.
Poor reception - satellite audio	- Obstructions to the line of sight - Satellite radio antenna - Satellite radio antenna cable - SDARS module	Go to Pinpoint Test J.
No sound while in satellite	- Satellite radio antenna - Satellite radio	Go to Pinpoint Test J.

2008 Ford Edge SE

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radio mode	antenna cable	
	• SDARS module	
• Poor quality/distorted/no sound from one or more speakers (not all speakers) - all audio systems except THX®	• Wiring, terminals or connectors • Speaker • RETM (if equipped) • ACM	• Go to Pinpoint Test B .
• Poor quality/distorted/no sound from one or more speakers (not all speakers) - THX® audio system	• Fuse • Wiring, terminals or connectors • Speaker • Subwoofer • DSP module • ACM	• Go to Pinpoint Test C .
• Poor quality/distorted/no sound from all speakers - all audio systems except THX®	• Wiring, terminals or connectors • ACM	NOTE: If DTC B2965 is set in the ACM, the cause of the concern is a short to ground or short to voltage in the circuitry to an individual speaker. Go to Pinpoint Test B . • With the key in any position except START, MEASURE the voltage between the ACM C290a-15, circuit CBP13 (GY/BN), harness side and ground. • If any voltage is present, REPAIR the circuit. TEST the system for normal operation. • If no voltage is present, INSTALL a new ACM. REFER to <u>Audio Control Module (ACM)</u>. TEST the system for normal operation.
• Poor quality/distorted/no sound while in SYNC mode	• Wiring, terminals or connectors • APIM • ACM	• Go to Pinpoint Test X .
	• Fuse	

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

Poor quality/distorted/no sound from all speakers - THX® audio system	- Wiring, terminals or connectors - DSP module - ACM	Go to Pinpoint Test D .
The subwoofer is inoperative/does not operate correctly - Audiophile	- Fuse - Wiring, terminals or connectors - Subwoofer - ACM	Go to Pinpoint Test E .
The subwoofer is inoperative/does not operate correctly - THX® audio system	- Fuse - Wiring, terminals or connectors - Speaker - DSP module - ACM	Go to Pinpoint Test C .
The video system is inoperative/does not operate correctly	- Damaged DVD - RETM setting - RETM	Go to Pinpoint Test K .
The video system is inoperative/does not operate correctly - no audio	- Wiring, terminals or connectors - RETM - ACM	Go to Pinpoint Test K .
- No auxiliary headphone audio - all audio systems except THX® - No auxiliary headphone audio - THX®	- Headphones - RETM - Wiring, terminals or connectors - Headphones - RETM	- INSTALL a set of known good headphones and OPERATE the audio system in radio tuner (AM/FM) dual-play mode. - If the system operates correctly, the concern was caused by inoperative headphones. - If the system does not operate correctly, INSTALL a new RETM. REFER to <u>Rear Entertainment Module (RETM)</u>. TEST the system for normal operation. - Go to Pinpoint Test L .

2008 Ford Edge SE

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The video system remote control does not function correctly	- Remote control batteries - Video system concern	INSTALL new batteries in the remote control. TEST the system for normal operation. If the concern is still present, CONTACT an authorized audio repair facility for further diagnostic test procedures.
The steering wheel controls are inoperative/do not operate correctly	- Wiring, terminals or connectors - Clockspring - Steering wheel controls - ACM - APIM (if equipped)	Go to Pinpoint Test F .
The speed sensitive volume does not operate correctly	- Instrument cluster (IC) - ACM	Go to Pinpoint Test Y
The audio input jack is inoperative/does not operate correctly	- Wiring, terminals or connectors - Audio input jack - ACM - APIM (if equipped)	Go to Pinpoint Test J .
The audio does not reduce when the parking aid tone sounds	- Parking aid system concern - ACM	Go to Pinpoint Test Q .
Voice recognition is inoperative/does not operate correctly - vehicles without the SYNC system	- Fuse - Wiring, terminals or connectors - Microphone (part of interior rear view mirror) - Steering wheel controls - ACM	Go to Pinpoint Test R .
Voice recognition is inoperative/does not operate correctly - vehicles with the SYNC system	- Fuse - Wiring, terminals or connectors - Microphone (part of interior rear view mirror) - Steering wheel	Go to Pinpoint Test W .

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

- controls
- ACM (if equipped with navigation)
- APIM

Symptom Chart - Navigation

Condition	Possible Sources	Action
No global positioning system (GPS) antenna signal	- GPS antenna - ACM	Go to Pinpoint Test M.
Navigation screen only displays Dearborn, Michigan	GPS antenna	Go to Pinpoint Test M.
The position cursor is inaccurate	- GPS antenna - Vehicle speed sensor (VSS) signal concern - ACM	Go to Pinpoint Test P.
The audible switch feedback is inoperative	- Audible switch feedback setting - ACM	Go to Pinpoint Test N.
Unable to read map disc	- Map DVD - ACM	- REMOVE the map DVD and CLEAN it. INSPECT for scratches, fingerprints, or damage, and REPAIR as necessary. VERIFY a valid map DVD is being used. If no concern is found, INSERT a known good map DVD. TEST the system. - If the system operates correctly, the concern was caused by a damaged or inoperable map DVD. - If the system does not operate correctly, INSTALL a new ACM. REFER to <u>Audio Control Module (ACM)</u>. TEST the system for normal operation. - INSPECT the instrument panel center trim panel for

2008 Ford Edge SE

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The screen does not open/close	- Trim panel interference - ACM	interference with the navigation screen, and REPAIR as necessary. If no interference is found, INSTALL a new ACM. REFER to <u>Audio Control Module (ACM)</u>. TEST the system for normal operation.
The voice guidance is inoperative/does not operate correctly	- Incorrect system setting - Wiring, terminals or connectors (THX® only) - DSP module (THX® only) - ACM	Go to Pinpoint Test O.
- The display screen is inoperative - Voice recognition is inoperative/does not operate correctly - vehicles without the SYNC system	- Wiring, terminals or connectors - ACM - Fuse - Wiring, terminals or connectors - Microphone (part of interior rear view mirror) - Steering wheel controls - ACM	- VERIFY the operation of the AM/FM audio. - If the AM/FM audio system does not operate correctly, go to <u>Symptom Chart - Audio System</u>. - If the AM/FM audio system operates correctly, INSTALL a new ACM. REFER to <u>Audio Control Module (ACM)</u>. TEST the system for normal operation. - Go to Pinpoint Test R.
Voice recognition is	- Fuse - Wiring, terminals or connectors - Microphone (part of interior rear view	

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

inoperative/does not operate correctly - vehicles with the SYNC system	mirror) • Steering wheel controls • ACM (if equipped with navigation) • APIM	• Go to Pinpoint Test W .
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Symptom Chart - SYNC System

Condition	Possible Sources	Action
• The SYNC system is completely inoperative	• Audio system concern • Communication network concern • Customer error • Customer device • APIM	• Go to Pinpoint Test S .
• Unable to pair Bluetooth® device	• Incompatible Bluetooth® device • Customer error • Customer Bluetooth® device • APIM	• Go to Pinpoint Test T .
• An individual Bluetooth® device feature is inoperative • The universal serial bus (USB) port is inoperative/does not operate correctly	• Customer device compatibility • Customer USB device • USB cable and port • APIM	NOTE: If a Bluetooth® device is able to pair with the SYNC system, there are no concerns with the APIM. • INSTRUCT the customer to review the interoperability list on the . Not all phone features are available through SYNC. This is normal operation. • Go to Pinpoint Test U .
• The steering wheel controls are inoperative/do not operate correctly	• Wiring, terminals or connectors • Steering wheel controls • ACM	• Go to Pinpoint Test F .

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

	• APIM	
• The audio input jack is inoperative/does not operate correctly	• Wiring, terminals or connectors • Audio input jack • Audio control module (ACM) • APIM	• Go to Pinpoint Test G .
• The SYNC system audible prompts are inoperative/do not operate correctly • During a phone call, no incoming audio is heard in the vehicle	• Wiring, terminals or connectors • Customer setting • ACM • APIM • Wiring, terminals or connectors • Customer setting • ACM • APIM	NOTE: The audible prompts include the text-to-speech (TTS) feature, voice prompts, and ring tones. • Go to Pinpoint Test V . • Go to Pinpoint Test V .
• Voice recognition is inoperative/does not operate correctly	• Fuse • Wiring, terminals or connectors • Microphone (part of interior rear view mirror) • Steering wheel controls • ACM (if equipped with navigation) • APIM	• Go to Pinpoint Test W .
• During a phone call, no outgoing audio is heard on the outside device	• Fuse • Wiring, terminals or connectors • Microphone (part of interior rear view mirror) • APIM	• Go to Pinpoint Test W .

Pinpoint Tests

Pinpoint Test A: Poor Reception - AM/FM

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and

connector information.

Normal Operation

The antenna receives AM and FM radio waves. The antenna module is integral with the roof-mounted antenna and receives voltage from the audio control module (ACM) through circuit CME44 (YE/GN).

The noise suppression equipment reduces engine ignition and electrical systems interference transmitted through the speakers.

- DTC B2103 (Antenna Not Connected) - sets during the on-demand self-test when the signal strength is less than the configured minimum value. The ACM must be tuned to AM during the self-test for this DTC to set. Also, the DTC will not set when the ACM self-diagnostic mode is run.

This pinpoint test is intended to diagnose the following:

- RDS/PTY function setting
- Wiring, terminals or connectors
- Antenna
- Antenna cable
- Charging system
- Ignition system
- Noise suppression equipment
- ACM

PINPOINT TEST A: POOR RECEPTION - AM/FM

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

A1 VERIFY THE RDS/PTY SETTING

- Verify the RDS/PTY function is set to ALL.
 - For non-navigation audio systems, press MENU repeatedly until RBDS (ON/OFF) appears on the display, then press the TUNE UP or TUNE DOWN button until ALL is displayed.
 - For navigation audio system, while in FM mode, press SHOW OPTIONS, make sure RDS is active, then press SET PTY and select ALL.
- Press the SEEK DOWN or SEEK UP button to scan for a good channel.
- **Is a good channel found?**

YES : The system is operating correctly at this time. The cause of the concern was that no channel in the selected category was found.

NO : Go to A2.

A2 CHECK THE AUDIO SYSTEM RECEPTION

2008 Ford Edge SE

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- Check the ACM reception with the engine running, and with the engine off.
- **Does the poor reception only occur with the engine running?**

YES : Go to A3.

NO : Go to A6.

A3 CHECK THE NOISE SUPPRESSION EQUIPMENT

- Check all required noise suppression equipment and the radio frequency interference suppression bond for security, cleanliness, and metal-to-metal contact.
- **Are the connections clean, secure, and in metal-to-metal contact?**

YES : Go to A4.

NO : CLEAN and SECURE the antenna connections, or INSTALL new noise suppression equipment as needed. TEST the system for normal operation.

A4 CHECK THE GENERATOR

- Key in OFF position.
- Check the generator by disconnecting the wiring harness from the voltage regulator.
- Start the engine.
- Operate the audio system in radio tuner (AM/FM) mode.
- **Is the reception OK?**

YES : INSTALL a new generator. REFER to **GENERATOR AND REGULATOR** article. TEST the system for normal operation.

NO : Go to A5.

A5 CHECK THE IGNITION CIRCUITS

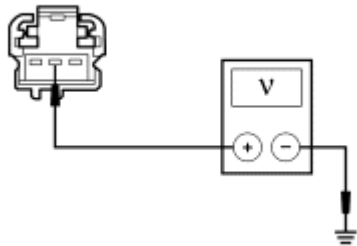
- Key in OFF position.
- Connect the wiring harness to the voltage regulator.
- Check the ignition circuits for correct routing, grounding, and integrity of the connections.
- **Are the ignition components OK?**

YES : USE a jumper cable to ground various parts of the vehicle to the frame (for example: engine, fenders, quarter panels, stone deflectors, air cleaner, body sheet metal). When the noise is eliminated, PROVIDE a permanent ground where necessary. TEST the system for normal operation.

NO : REPAIR the ignition system as necessary. TEST the system for normal operation.

A6 CHECK CIRCUIT CME44 (YE/GN) FOR VOLTAGE

- Key in OFF position.
- Disconnect: Antenna C2086
- Operate the audio system in radio tuner (AM/FM) mode.



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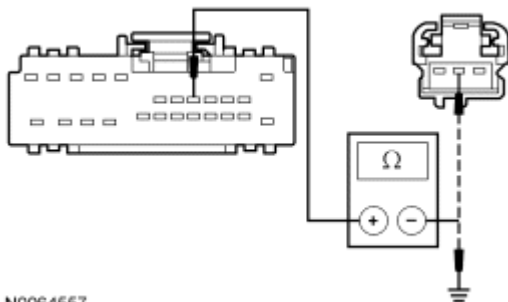
Fig. 3: Measuring Voltage Between Antenna C2086-2, Circuit CME44 (YE/GN), Harness Side & Ground

Courtesy of FORD MOTOR CO.

- Measure the voltage between the antenna C2086-2, circuit CME44 (YE/GN), harness side and ground.
- **Is the voltage greater than 10 volts?**
YES : Go to A8.
NO : Go to A7.

A7 CHECK CIRCUIT CME44 (YE/GN) FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Disconnect: ACM C240a



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Fig. 4: Measuring Resistance Between Audio Unit C240A-5, Circuit CME44 (YE/GN) & Antenna C2086-2

Courtesy of FORD MOTOR CO.

- Measure the resistance between the ACM C240a-5, circuit CME44 (YE/GN), harness side and the antenna C2086-2, circuit CME44 (YE/GN), harness side; and between the ACM C240a-5, circuit CME44 (YE/GN), harness side and ground.
- **Is the resistance less than 5 ohms between the ACM and the antenna, and greater than 10,000 ohms between the ACM and ground?**
YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.
NO : REPAIR the circuit. TEST the system for normal operation.

A8 CHECK THE ANTENNA COAXIAL CABLE

- Key in OFF position.
- Disconnect: Antenna Coaxial Cable (At ACM)
- Measure the resistance of the antenna coaxial cable between the antenna and the ACM.
- **Is the resistance less than 2 ohms?**

YES : Go to A9.

NO : INSTALL a new antenna cable. REFER to **Antenna Cable - AM/FM**. CLEAR the DTCs. REPEAT the self-test.

A9 ISOLATE THE ANTENNA

- Connect: Antenna Coaxial Cable (At ACM)
- Substitute the antenna for a known good antenna.
- Operate the audio system in radio tuner (AM/FM) mode.
- **Is the reception OK?**

YES : INSTALL a new antenna. REFER to **Antenna - AM/FM**. CLEAR the DTCs. REPEAT the self-test.

NO : INSTALL the original antenna. Go to A10.

A10 SUBSTITUTE THE ANTENNA CABLE

- Key in OFF position.
- Substitute a known good AM/FM antenna cable.
- Operate the audio system in radio tuner (AM/FM) mode.
- **Is the reception OK?**

YES : INSTALL a new antenna cable. REFER to **Antenna Cable - AM/FM**. CLEAR the DTCs. REPEAT the self-test.

NO : Go to A11.

A11 CHECK THE ACM FOR CORRECT OPERATION

- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the ACM connectors and make sure they seat correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Except THX® Audio System

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

The audio control module (ACM) directs audio signals to the front door speakers and to the rear entertainment module (RETM) (if equipped) in the form of a fluctuating AC voltage. The tweeter speakers (if equipped) are wired in series with the front door speakers. The rear door speakers receive audio signals through the RETM (if equipped) or directly from the ACM.

If the vehicle is equipped with the DVD entertainment system, the RETM mutes the speaker output to the rear speakers when the audio system is in dual-play mode. Make sure the audio system is in single-play mode before diagnosing a symptom of no sound to the rear speakers.

For the navigation system, if the voice guidance setting is set to zero rather than to off, it may appear that the front speakers intermittently produce no sound. Make sure the voice guidance setting is not set to zero before addressing a concern with the front speakers.

- DTC B2965 (Audio System Speaker Circuit Fault) - sets when an open, a short to ground, or a short to voltage is detected on the front or rear door speaker circuits. The ACM checks for faults once every 500ms during normal operation, or for any duration during the ACM self-test. The following should also be noted:
 - The Audiophile ACM does not check for opens in the front door speakers.
 - The navigation ACM does not check for opens.

For some ACMs, a short to ground or short to voltage in the circuitry to one of the speakers will cause all speakers to lose sound due to the built-in overload protection feature of the ACM. In this case, DTC B2965 will set in the ACM, and this pinpoint test should be followed to isolate the damaged circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Speaker
- RETM (if equipped)
- ACM

PINPOINT TEST B: POOR QUALITY/DISTORTED/NO SOUND FROM ONE OR MORE SPEAKERS (NOT ALL SPEAKERS) - EXCEPT THX® AUDIO SYSTEM

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

B1 ISOLATE THE SUSPECT SPEAKER

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NOTE: Make sure the audio system is in single-play mode before beginning this pinpoint test, or the results will not be accurate.

- Key in ON position.
- Carry out the speaker walk-around test.
- **Is the concern with a high-mounted tweeter speaker only?**

YES : Go to B13.

NO : Go to B2.

B2 CHECK THE SPEAKER CIRCUITS FOR VOLTAGE

- Key in OFF position.
- Disconnect: Suspect Speaker
- Operate the audio system in radio tuner (AM/FM) mode.
- Measure the AC voltage between the suspect speaker pin 1 and pin 2, harness side as follows:

Suspect Speaker	Connector-Pin/Circuit	Connector-Pin/Circuit
LH front	C536-1 or C570a-1 VME07 (WH)	C536-2 or C570a-2 RME07 (WH/BN)
RH front	C628-1 or C612a-1 VME10 (WH/VT)	C628-2 or C612a-2 RME10 (WH/OG)
LH rear	C702-1 VME09 (WH/GN)	C702-2 RME09 (BN/YE)
RH rear	C802-1 VME12 (BN/WH)	C802-2 RME12 (BN/BU)

- **Is a fluctuating AC voltage present?**

YES : INSTALL a new speaker for the suspect speaker. REFER to **Door Speaker**. CLEAR any DTCs present. TEST the system for normal operation.

NO : For a front speaker concern, go to B11.

For a rear speaker concern, go to B3.

B3 VERIFY THE PRESENCE OF A DVD ENTERTAINMENT SYSTEM

- Key in OFF position.
- Check for the presence of a DVD entertainment system.
- **Is the vehicle equipped with a DVD entertainment system?**

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YES : Go to B6.

NO : Go to B4.

B4 CHECK THE REAR SPEAKER CIRCUITS FOR A SHORT TO VOLTAGE

- Disconnect: ACM C240a
- Key in ON position.
- Measure the voltage between the suspect speaker pins, harness side and ground as follows:

Suspect Speaker	Connector-Pin	Circuit
LH rear	C702-1	VME09 (WH/GN)
	C702-2	RME09 (BN/YE)
RH rear	C802-1	VME12 (BN/WH)
	C802-2	RME12 (BN/BU)

- **Is any voltage present?**

YES : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

NO : Go to B5.

B5 CHECK THE REAR SPEAKER CIRCUITS FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the suspect speaker, harness side and the ACM, harness side; and between the suspect speaker, harness side and ground as follows:

Suspect Speaker	Speaker Connector-Pin	ACM Connector-Pin	Circuit
LH rear	C702-1	C240a-9	VME09 (WH/GN)
	C702-2	C240a-22	RME09 (BN/YE)
RH rear	C802-1	C240a-10	VME12 (BN/WH)
	C802-2	C240a-23	RME12 (BN/BU)

- **Is the resistance less than 5 ohms between the suspect speaker and the ACM, and greater than 10,000 ohms between the suspect speaker and ground?**

YES : Go to B17.

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NO : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

B6 CHECK THE ACM OUTPUT SIGNALS

- Disconnect: RETM C949
- Operate the audio system in radio tuner (AM/FM) mode.
- Measure the AC voltage between the RETM pins, harness side as follows:

Suspect Speaker	Connector-Pin/Circuit	Connector-Pin/Circuit
LH rear	C949-27 VME09 (WH/GN)	C949-28 RME09 (BN/YE)
RH rear	C949-11 VME12 (BN/WH)	C949-12 RME12 (BN/BU)

- Is a fluctuating AC voltage present?

YES : Go to B9.

NO : Go to B7.

B7 CHECK THE ACM OUTPUT CIRCUITS FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: ACM C240a
- Key in ON position.
- Measure the voltage between the RETM, harness side and ground as follows:

Suspect Speaker	Connector-Pin	Circuit
LH rear	C949-27	VME09 (WH/GN)
	C949-28	RME09 (BN/YE)
RH rear	C949-11	VME12 (BN/WH)
	C949-12	RME12 (BN/BU)

- Is any voltage present?

YES : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

NO : Go to B8.

B8 CHECK THE ACM OUTPUT CIRCUITS FOR AN OPEN OR SHORT TO GROUND

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- Key in OFF position.
- Measure the resistance between the ACM and the RETM, harness side; and between the ACM, harness side and ground as follows:

Suspect Speaker	ACM Connector-Pin	RETM Connector-Pin	Circuit
LH rear	C240a-9	C949-27	VME09 (WH/GN)
	C240a-22	C949-28	RME09 (BN/YE)
RH rear	C240a-10	C949-11	VME12 (BN/WH)
	C240a-23	C949-12	RME12 (BN/BU)

- **Is the resistance less than 5 ohms between the ACM and the RETM, and greater than 10,000 ohms between the ACM and ground?**

YES : Go to B17.

NO : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

B9 CHECK THE RETM OUTPUT CIRCUITS FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: Suspect Speaker
- Key in ON position.
- Measure the voltage between the suspect speaker, harness side and ground as follows:

Suspect Speaker	Connector-Pin	Circuit
LH rear	C702-1	VME09 (WH/GN)
	C702-2	RME09 (BN/YE)
RH rear	C802-1	VME12 (BN/WH)
	C802-2	RME12 (BN/BU)

- **Is any voltage present?**

YES : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

NO : Go to B10.

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B10 CHECK THE RETM OUTPUT CIRCUITS FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the suspect speaker, harness side and the RETM, harness side; and between the suspect speaker, harness side and ground as follows:

Suspect Speaker	Speaker Connector-Pin	RETM Connector-Pin	Circuit
LH rear	C949-30	C702-1	VME09 (WH/GN)
	C949-29	C702-2	RME09 (BN/YE)
RH rear	C949-14	C802-1	VME12 (BN/WH)
	C949-13	C802-2	RME12 (BN/BU)

- Is the resistance less than 5 ohms between the suspect speaker and the RETM, and greater than 10,000 ohms between the suspect speaker and ground?

YES : Go to B18.

NO : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

B11 CHECK THE FRONT SPEAKER CIRCUITS FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: ACM C240a
- Key in ON position.
- Measure the voltage between the suspect speaker, harness side and ground as follows:

Suspect Speaker	Connector-Pin/Circuit	Connector-Pin/Circuit
LH front	C536-1 or C570a-1	VME07 (WH)
	C536-2 or C570a-2	RME07 (WH/BN)
RH front	C628-1 or C612a-1	VME10 (WH/VT)
	C628-2 or C612a-2	RME10 (WH/OG)

- Is any voltage present?

YES : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

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NO : Go to B12.

B12 CHECK THE FRONT SPEAKER CIRCUITS FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the ACM, harness side and the suspect speaker, harness side; and between the ACM, harness side and ground as follows:

Suspect Speaker	ACM Connector-Pin	Speaker Connector-Pin	Circuit
LH front	C240a-8	C536-1 or C570a-1	VME07 (WH)
	C240a-21	C536-2 or C570a-2	RME07 (WH/BN)
RH front	C240a-11	C628-1 or C612a-1	VME10 (WH/VT)
	C240a-12	C628-2 or C612a-2	RME10 (WH/OG)

- **Is the resistance less than 5 ohms between the ACM and the suspect speaker, and greater than 10,000 ohms between the ACM and ground?**

YES : Go to B17.

NO : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

B13 CHECK FOR VOLTAGE TO THE SUSPECT TWEETER SPEAKER

- Key in OFF position.
- Disconnect: Suspect Tweeter Speaker
- Operate the audio system in radio tuner (AM/FM) mode.
- Measure the AC voltage between the suspect tweeter speaker pin 1 and pin 2, harness side as follows:

Suspect Speaker	Connector-Pin/Circuit	Connector-Pin/Circuit
LH tweeter	C569-1 VME08 (GN/OG)	C569-2 RME08 (GY/OG)
RH tweeter	C613-1 VME11 (VT/OG)	C613-2 RME11 (YE/OG)

- **Is a fluctuating AC voltage present?**

YES : INSTALL a new speaker for the suspect tweeter speaker. REFER to **Door Speaker**. TEST the system for normal operation.

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NO : Go to B14.

B14 CHECK THE TWEETER SPEAKER CIRCUITS FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: LF Speaker C570b or RF Speaker C612b
- Key in ON position.
- Measure the voltage between the suspect tweeter speaker, harness side and ground as follows:

Suspect Tweeter Speaker	Tweeter Speaker Connector-Pin	Circuit
LH tweeter	C569-1	VME08 (GN/OG)
	C569-2	RME08 (GY/OG)
RH tweeter	C613-1	VME11 (VT/OG)
	C613-2	RME11 (YE/OG)

- **Is any voltage present?**

YES : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

NO : Go to B15.

B15 CHECK THE TWEETER SPEAKER CIRCUITS FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the suspect tweeter speaker, harness side and the respective door speaker, harness side; and between the suspect tweeter speaker, harness side and ground as follows:

Suspect Tweeter Speaker	Tweeter Speaker Connector-Pin	Door Speaker Connector-Pin	Circuit
LH tweeter	C569-1	C570b-1	VME08 (GN/OG)
	C569-2	C570b-2	RME08 (GY/OG)
RH tweeter	C613-1	C612b-1	VME11 (VT/OG)
	C613-2	C612b-2	RME11 (YE/OG)

- **Is the resistance less than 5 ohms between the suspect tweeter speaker and the door speaker.**

and greater than 10,000 ohms between the suspect tweeter speaker and ground?

YES : Go to B16.

NO : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

B16 ISOLATE THE DOOR SPEAKER

- Connect: Suspect Tweeter Speaker
- Install a known good door speaker for the door with the suspect tweeter speaker.
- Operate the audio system in radio tuner (AM/FM) mode.
- **Does the tweeter speaker operate correctly?**

YES : INSTALL a new door speaker. REFER to **Door Speaker**. TEST the system for normal operation.

NO : INSTALL a new tweeter speaker. REFER to **Door Speaker**. TEST the system for normal operation.

B17 CHECK FOR CORRECT ACM OPERATION

- Key in OFF position.
- Connect: All Disconnected Connectors
- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the ACM connectors and make sure they seat correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

B18 CHECK FOR CORRECT RETM OPERATION

- Key in OFF position.
- Connect: All Disconnected Connectors
- Disconnect the RETM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the RETM connector and make sure it seats correctly.
- Operate the system and determine if the concern is still present.

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- **Is the concern still present?**

YES : INSTALL a new RETM. REFER to **Rear Entertainment Module (RETM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test C: Poor Quality/Distorted/No Sound From One Or More Speakers (Not All Speakers) - THX® Audio System

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

The audio control module (ACM) directs the audio signals to the audio digital signal processing (DSP) module, which processes the audio signals and sends them to the speakers. Each speaker has a signal and return circuit.

DTC Description	Fault Set Conditions
B1158 - Subwoofer #2 Not Connected	May set when an open is detected in the subwoofer circuit.
B2913 - Audio Subwoofer Not Connected	May set when an open is detected in the subwoofer circuit.
B2925 - Subwoofer Speaker Short Circuit	Sets when a short to ground is detected in the subwoofer circuit.
B2965 - Audio System Speaker Circuit Fault	May set when an open or a short to ground is detected in the front door or rear door speaker circuits, or in the subwoofer.

The following should also be noted:

- The ACM does not set DTC B2965 when the vehicle is equipped with the THX® audio system. Only the DSP module can set this DTC.
- This DTC may not set due to a fault in the circuitry to the rear tweeters.
- An open or short to ground in the rear tweeter circuitry may cause both rear tweeters to lose sound.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Speaker
- Subwoofer
- DSP module
- ACM

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PINPOINT TEST C: POOR QUALITY/DISTORTED/NO SOUND FROM ONE OR MORE SPEAKERS (NOT ALL SPEAKERS) - THX® AUDIO SYSTEM

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

C1 CHECK THE AUDIO SIGNALS TO THE SUSPECT SPEAKER

- Key in OFF position.
- Disconnect: Suspect Speaker
- Operate the audio system in radio tuner (AM/FM) mode.
- Measure the AC voltage between the suspect speaker pins, harness side as follows:

Suspect Speaker	Speaker Connector-Pin/Circuit	Speaker Connector-Pin/Circuit
LR door	C702-1 VME09 (WH/GN)	C702-2 RME09 (BN/YE)
LF door	C536-1 VME28 (BN/VT)	C536-2 RME28 (BN/GN)
LH tweeter	C569-1 VME07 (WH)	C569-2 RME07 (WH/BN)
LH front imaging	C2359-1 VME07 (WH)	C2359-4 RME07 (WH/BN)
Center front imaging	C2359-2 VME06 (GN)	C2359-5 RME06 (GY/YE)
RH front imaging	C2359-3 VME10 (WH/VT)	C2359-6 RME10 (WH/OG)
RH tweeter	C613-1 VME11 (VT/OG)	C613-2 RME11 (WH/OG)
RF door	C628-1 VME29 (GN/WH)	C628-2 RME29 (BU/WH)
RR door	C802-1 VME12 (BN/WH)	C802-2 RME12 (BN/BU)
Subwoofer 1	C4212-1 VME01	C4212-2 RME01

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	(GN/VT)	(GY)
Subwoofer 2	C4212-3 VME02 (VT)	C4212-4 RME02 (YE)
LH rear center imaging	C4337-1 VME30 (GN/BN)	C4337-2 RME30 (VT/BN)
RH rear center imaging	C4438-1 VME30 (GN/BN)	C4438-2 RME30 (VT/BN)

- **Is a fluctuating AC voltage present?**

YES : INSTALL a new speaker for the suspect speaker. REFER to **Door Speaker**, **Instrument Panel Speaker**, or **Subwoofer Speaker** . CLEAR any DTCs present. TEST the system for normal operation.

NO : Go to C2.

C2 CHECK THE CIRCUITS TO THE SUSPECT SPEAKER FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: DSP Module C4348b or C4348c
- Key in ON position.
- Measure the voltage between the suspect speaker, harness side and ground as follows:

Suspect Speaker	Speaker Connector-Pin	Circuit
LR door	C702-1	VME09 (WH/GN)
	C702-2	RME09 (BN/YE)
LF door	C536-1	VME28 (BN/VT)
	C536-2	RME28 (BN/GN)
LH tweeter	C569-1	VME07 (WH)
	C569-2	RME07 (WH/BN)
LH front imaging	C2359-1	VME07 (WH)
	C2359-4	RME07 (WH/BN)
Center front	C2359-2	VME06 (GN)

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

imaging	C2359-5	RME06 (GY/YE)
RH front imaging	C2359-3	VME10 (WH/VT)
	C2359-6	RME10 (WH/OG)
RH tweeter	C613-1	VME11 (VT/OG)
	C613-2	RME11 (WH/OG)
RF door	C628-1	VME29 (GN/WH)
	C628-2	RME29 (BU/WH)
RR door	C802-1	VME12 (BN/WH)
	C802-2	RME12 (BN/BU)
Subwoofer 1	C4212-1	VME01 (GN/VT)
	C4212-2	RME01 (GY)
Subwoofer 2	C4212-3	VME02 (VT)
	C4212-4	RME02 (YE)
LH rear center imaging	C4337-1	VME30 (GN/BN)
	C4337-2	RME30 (VT/BN)
RH rear center imaging	C4338-1	VME30 (GN/BN)
	C4338-2	RME30 (VT/BN)

- **Is any voltage present?**

YES : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

NO : Go to C3.

C3 CHECK THE CIRCUITS TO THE SUSPECT SPEAKER FOR AN OPEN OR SHORT TO GROUND

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- Measure the resistance between the suspect speaker, harness side and the DSP module, harness side; and between the suspect speaker, harness side and ground as follows:

Suspect Speaker	Speaker Connector- Pin	DSP Module Connector- Pin	Circuit
LR door	C702-1	C4348c-7	VME09 (WH/GN)
	C702-2	C4348c-6	RME09 (BN/YE)
LF door	C536-1	C4348c-3	VME28 (BN/VT)
	C536-2	C4348c-2	RME28 (BN/GN)
LH tweeter	C569-1	C4348b-9	VME08 (GN/OG)
	C569-2	C4348b-10	RME08 (GY/OG)
LH front imaging	C2359-1	C4348b-2	VME07 (WH)
	C2359-4	C4348b-1	RME07 (WH/BN)
Center front imaging	C2359-2	C4348b-4	VME06 (GN)
	C2359-5	C4348b-3	RME06 (GY/YE)
RH front imaging	C2359-3	C4348b-6	VME10 (WH/VT)
	C2359-6	C4348b-5	RME10 (WH/OG)
RH tweeter	C613-1	C4348b-11	VME11 (VT/OG)
	C613-2	C4348b-12	RME11 (YE/OG)
RF door	C628-1	C4348c-5	VME29 (GN/WH)
	C628-2	C4348c-4	RME29 (BU/WH)
RR door	C802-1	C4348c-18	VME12 (BN/WH)
	C802-2	C4348c-19	RME12 (BN/BU)
Subwoofer 1	C4212-1	C4348c-14	VME01 (GN/VT)
	C4212-2	C4348c-15	RME01 (GY)
Subwoofer 2	C4212-3	C4348c-16	VME02 (VT)
	C4212-4	C4348c-17	RME02 (YE)
LH rear center imaging	C4337-1	C4348b-13	VME30 (GN/BN)
	C4337-2	C4348b-14	RME30 (VT/BN)
RH rear center imaging	C4338-1	C4348b-13	VME30 (GN/BN)
	C4338-2	C4348b-14	RME30 (VT/BN)

- Is the resistance less than 5 ohms between the suspect speaker and the DSP module, and greater than 10,000 ohms between the suspect speaker and ground?

YES : Go to C4.

NO : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

C4 ISOLATE THE SUSPECT SPEAKER

- Key in OFF position.
- Connect: DSP Module C4348b and C4348c

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- Substitute a known good speaker for the suspect speaker.
- Operate the audio system in radio tuner (AM/FM) mode.
- **Is the concern still present?**

YES : Go to C5.

NO : INSTALL a new speaker for the suspect speaker. REFER to **Door Speaker**, **Instrument Panel Speaker**, or **Subwoofer Speaker** . CLEAR any DTCs present. TEST the system for normal operation.

C5 CHECK FOR CORRECT DSP MODULE OPERATION

- Key in OFF position.
- Connect: All Disconnected Connectors
- Disconnect all the DSP module connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the DSP module connectors and make sure they seat correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new DSP module. REFER to **Audio Digital Signal Processing (DSP) Module**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test D: Poor Quality/Distorted/No Sound From All Speakers - THX® Audio System

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

The THX® amplifier receives voltage through circuits SBP38 (BN/RD) and SBP18 (YE/RD), and ground through circuit GD148 (BK/YE).

Left and right channel audio signals are sent from the audio control module (ACM) to the audio digital signal processing (DSP) module through circuits VME07 (WH), RME07 (WH/BN), VME10 (WH/VT), and RME10 (WH/OG). If there is a fault on one of these circuits, the audio quality will be reduced by approximately 25% for all speakers.

To enable the DSP module, the ACM sends voltage through the enable/clip circuit SME23 (VT/RD). The circuit acts as both an output (to enable the amplifier) and an input (to detect an amplifier overload condition). A variable resistor in the DSP module modifies the voltage signal from the ACM. In the event of an overload, the ACM clips the audio output signal to the amplifier (heard as distortion), in order to prevent damage to the

amplifier and speakers.

The clip/enable status is based on the following voltages, as detected by the ACM:

- Less than 0.4 volts: amplifier disabled
- Between 3.8 and 6.7 volts: amplifier enabled
- Greater than 8.5 volts: amplifier clipped

NOTE: **Despite their names, DTCs B2925 and B292A apply to the DSP module clip/enable circuit, not to the subwoofer.**

- DTC B2925 (Subwoofer Speaker Short Circuit) - sets with DTCs B2965 and B292A when a short to ground is detected in circuit SME23 (VT/RD).
- DTC B292A (Subwoofer #2 Speaker Short Circuit) - sets when an open or short to ground is detected in circuit SME23 (VT/RD). If an open is detected, DTC B2965 will also be set. If a short to ground is detected, DTCs B2965 and B2925 will also be set.
- DTC B2965 (Audio System Speaker Circuit Fault) - sets when an open or short to ground is detected in circuit SME23 (VT/RD).

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- DSP module
- ACM

PINPOINT TEST D: POOR QUALITY/DISTORTED/NO SOUND FROM ALL SPEAKERS - THX® AUDIO SYSTEM

NOTE: **Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.**

D1 CHECK THE DSP MODULE DTCs

- Carry out the DSP module continuous and on-demand self-tests.
- **Is DTC B292A present?**
 YES : Go to D5.
 NO : Go to D2.

D2 CHECK CIRCUITS SBP18 (YE/RD) AND SBP38 (BN/RD) FOR VOLTAGE

- Key in OFF position.
- Disconnect: DSP Module C4348c
- Measure the voltage between the DSP module, harness side and ground as follows:

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Connector-Pin	Circuit
C4348c-9	SBP38 (BN/RD)
C4348c-10	SBP38 (BN/RD)
C4348c-11	SBP18 (YE/RD)
C4348c-12	SBP18 (YE/RD)

- **Are the voltages greater than 10 volts?**

YES : Go to D3.

NO : VERIFY the smart junction box (SJB) fuse 38 (20A) or fuse 18 (20A) is OK. If OK, REPAIR the circuit in question. TEST the system for normal operation. If not OK, REFER to appropriate SYSTEM WIRING DIAGRAMS article to identify the possible sources of the circuit short.

D3 CHECK CIRCUIT GD148 (BK/YE) FOR AN OPEN

- Measure the resistance between the DSP module, harness side and ground as follows:

Connector-Pin	Circuit
C4348c-22	GD148 (BK/YE)
C4348c-23	GD148 (BK/YE)
C4348c-24	GD148 (BK/YE)

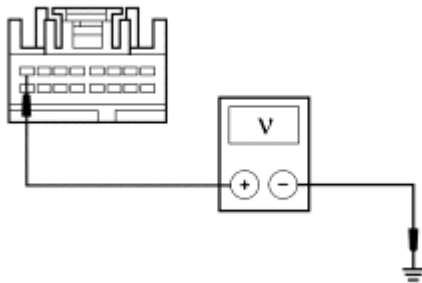
- **Are the resistances less than 5 ohms?**

YES : Go to D4.

NO : REPAIR the circuit in question. TEST the system for normal operation.

D4 CHECK CIRCUIT SME23 (VT/RD) FOR VOLTAGE

- Connect: DSP Module C4348c
- Disconnect: DSP Module C4348a
- Operate the audio system in radio tuner (AM/FM) mode.



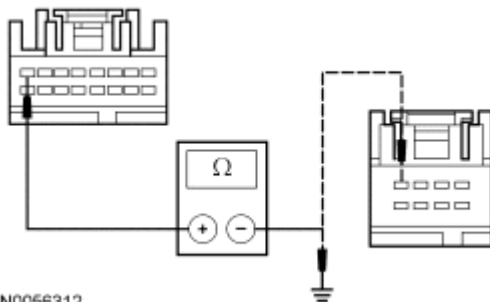
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Fig. 5: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between the DSP module C4348a-8, circuit SME23 (VT/RD), harness side and ground.
- **Is the voltage between 3.8 and 6.7 volts?**
YES : Go to D6.
NO : Go to D5.

D5 CHECK CIRCUIT SME23 (VT/RD) FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Disconnect: ACM C240c



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Fig. 6: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between the DSP module C4384a-8, circuit SME23 (VT/RD), harness side and the ACM C240c-4, circuit SME23 (VT/RD), harness side; and between the DSP module C4384a-8, circuit SME23 (VT/RD), harness side and ground.
- **Is the resistance less than 5 ohms between the DSP module and the ACM, and greater than 10,000 ohms between the DSP module and ground?**
YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.
NO : REPAIR the circuit. CLEAR and DTCs present. TEST the system for normal operation.

D6 CHECK THE AUDIO SIGNALS TO THE DSP MODULE

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- Operate the audio system in radio tuner (AM/FM) mode.
- Measure the AC voltage between the audio circuits at the DSP module as follows:

Connector-Pin	Circuit
C4348a-1	VME07 (WH)
C4348a-2	RME07 (WH/BN)
C4348a-3	VME10 (WH/VT)
C4348a-4	RME10 (WH/OG)

- **Is a fluctuating AC voltage present?**

YES : Go to D10.

NO : Go to D7.

D7 CHECK THE AUDIO CIRCUITS TO THE DSP MODULE FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: ACM C240a
- Key in ON position.
- Measure the voltage between the DSP module, harness side and ground as follows:

Connector-Pin	Circuit
C4348a-1	VME07 (WH)
C4348a-2	RME07 (WH/BN)
C4348a-3	VME10 (WH/VT)
C4348a-4	RME10 (WH/OG)

- **Is any voltage present?**

YES : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

NO : Go to D8.

D8 CHECK THE AUDIO CIRCUITS TO THE DSP MODULE FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the DSP module, harness side and the ACM, harness side; and

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between the DSP module, harness side and ground as follows:

DSP Module Connector- Pin	ACM Connector- Pin	Circuit
C4348a-1	C240a-8	VME07 (WH)
C4348a-2	C240a-21	RME07 (WH/BN)
C4348a-3	C240a-11	VME10 (WH/VT)
C4348a-4	C240a-12	RME10 (WH/OG)

- **Is the resistance less than 5 ohms between the DSP module and the ACM, and greater than 10,000 ohms between the DSP module and ground?**

YES : Go to D9.

NO : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

D9 CHECK FOR CORRECT ACM OPERATION

- Key in OFF position.
- Connect: All Disconnected Connectors
- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the ACM connectors and make sure they seat correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. CLEAR any DSP module DTCs present. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR any DSP module DTCs present.

D10 CHECK FOR CORRECT DSP MODULE OPERATION

- Key in OFF position.
- Connect: All Disconnected Connectors
- Disconnect all the DSP module connectors.
- Check for:
 - corrosion

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- damaged pins
- pushed-out pins
- Connect the DSP module connectors and make sure they seat correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new DSP module. REFER to **Audio Digital Signal Processing (DSP) Module**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test E: The Subwoofer Is Inoperative/Does Not Operate Correctly - Audiophile

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

The subwoofer amplifier receives voltage through circuit SBP38 (BN/RD) and ground through circuit GD148 (BK/YE). The subwoofer speakers and subwoofer amplifier are integral to the enclosure and cannot be repaired individually.

To enable the subwoofer amplifier, the audio control module (ACM) sends voltage through the enable/clip circuit SME23 (VT/RD). The circuit acts as both an output (to enable the amplifier) and an input (to detect an amplifier overload condition). A variable resistor in the subwoofer amplifier modifies the voltage signal from the ACM. In the event of an overload, the ACM clips the audio output signal to the amplifier (heard as distortion), in order to prevent damage to the amplifier and speakers.

The clip/enable status is based on the following voltages, as detected by the ACM:

- Less than 0.4 volts: amplifier disabled
- Between 3.8 and 6.7 volts: amplifier enabled
- Greater than 8.5 volts: amplifier clipped

An open in circuit SME23 (VT/RD) will cause the subwoofer to produce no sound, because the enable signal will not reach the subwoofer amplifier. A short to ground or short to voltage can cause severe distortion to be heard in the subwoofer.

- DTC B1154 (Subwoofer #1 Short to Ground) - sets when a short to ground is detected on circuit RME22 (GN/WH) or circuit VME22 (VT/GN). This DTC only sets during the on-demand self-test.
- DTC B1155 (Subwoofer #1 Short to Battery) - sets when a short to voltage is detected on circuit RME22 (GN/WH) or circuit VME22 (VT/GN). This DTC only sets during the on-demand self-test.
- DTC B2913 (Audio Subwoofer Not Connected) - sets when the ACM does not detect the subwoofer through circuit SME23 (VT/RD) during the subwoofer connection test (part of the ACM self-test).

This pinpoint test is intended to diagnose the following:

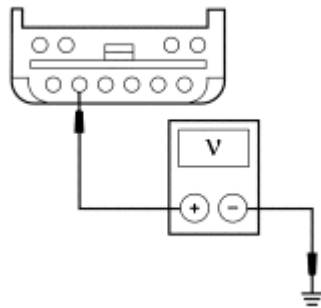
- Fuse
- Wiring, terminals or connectors
- Subwoofer
- ACM

PINPOINT TEST E: THE SUBWOOFER IS INOPERATIVE/DOES NOT OPERATE CORRECTLY - AUDIOPHILE

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

E1 CHECK CIRCUIT SBP38 (BN/RD) FOR VOLTAGE

- Key in OFF position.
- Disconnect: Subwoofer C4208



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Fig. 7: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between the subwoofer C4208-5, circuit SBP38 (BN/RD), harness side and ground.
- **Is the voltage greater than 10 volts?**
YES : Go to E2.
NO : VERIFY the smart junction box (SJB) fuse 38 (20A) is OK. If OK, REPAIR the circuit. CLEAR any DTCs present. TEST the system for normal operation. If not OK, REFER to appropriate SYSTEM WIRING DIAGRAMS article to identify the possible sources of the circuit short.

E2 CHECK CIRCUIT GD148 (BK/YE) FOR AN OPEN

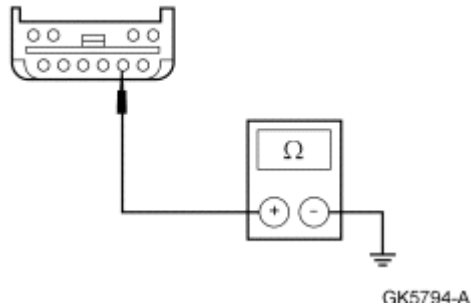


Fig. 8: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between the subwoofer C4208-2, circuit GD148 (BK/YE), harness side and ground.
- **Is the resistance less than 5 ohms?**
YES : Go to E3.
NO : REPAIR the circuit. CLEAR any DTCs present. TEST the system for normal operation.

E3 CHECK CIRCUIT SME23 (VT/RD) FOR VOLTAGE

- Operate the audio system in radio tuner (AM/FM) mode.

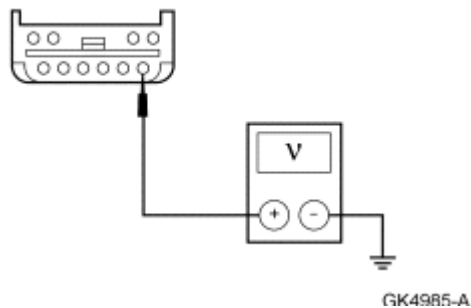


Fig. 9: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between the subwoofer C4208-1, circuit SME23 (VT/RD), harness side and ground.
- **Is the voltage between 3.8 and 6.7 volts?**
YES : Go to E6.
NO : Go to E4.

E4 CHECK CIRCUIT SME23 (VT/RD) FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: ACM C240c
- Key in ON position.

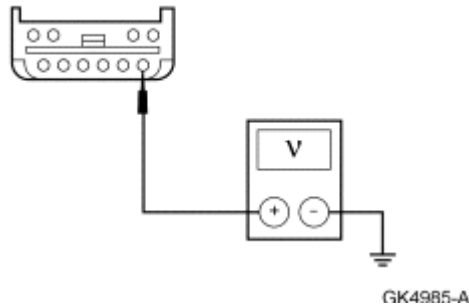


Fig. 10: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between the subwoofer C4208-1, circuit SME23 (VT/RD), harness side and ground.
- **Is any voltage present?**
YES : Go to E5.
NO : REPAIR the circuit. CLEAR any DTCs present. TEST the system for normal operation.

E5 CHECK CIRCUIT SME23 (VT/RD) FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.

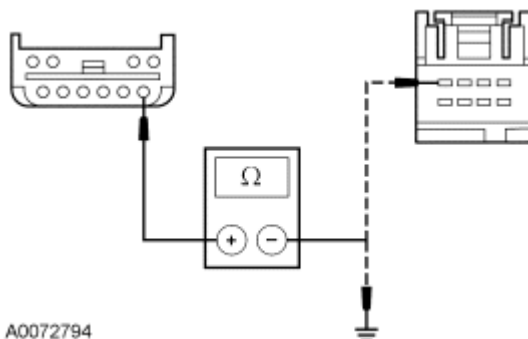
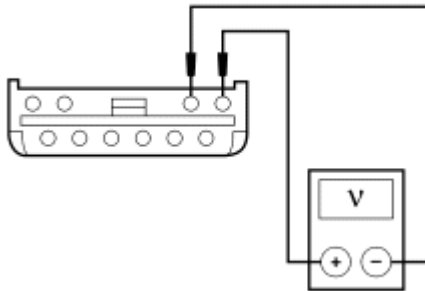


Fig. 11: Checking Circuit For Open Or Short To Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between the subwoofer C4208-1, circuit SME23 (VT/RD), harness side and the ACM C240c-4, circuit SME23 (VT/RD), harness side; and between the subwoofer C4208-1, circuit SME23 (VT/RD), harness side and ground.
- **Is the resistance less than 5 ohms between the subwoofer and the ACM, and greater than 10,000 ohms between the subwoofer and ground?**
YES : Go to E9.
NO : REPAIR the circuit. CLEAR any DTCs present. TEST the system for normal operation.

E6 CHECK THE AUDIO SIGNAL TO THE SUBWOOFER

- Key in OFF position.
- Disconnect: Subwoofer C4208
- Operate the audio system in radio tuner (AM/FM) mode.



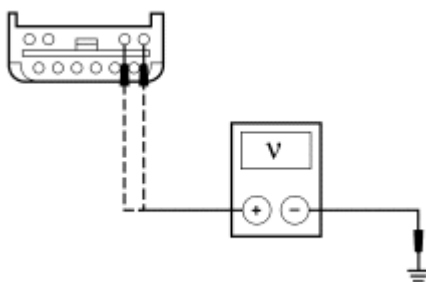
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Fig. 12: Checking Audio Signal
Courtesy of FORD MOTOR CO.

- Measure the AC voltage between the subwoofer C4208-7, circuit VME22 (VT/GN), harness side and the subwoofer C4208-8, circuit RME22 (GN/WH), harness side.
- **Is a fluctuating AC voltage present?**
YES : INSTALL a new subwoofer. REFER to **Subwoofer Speaker**. CLEAR any DTCs present. TEST the system for normal operation.
NO : Go to E7.

E7 CHECK THE AUDIO CIRCUITS FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: ACM C240c
- Key in ON position.



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Fig. 13: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between the subwoofer C4208-7, circuit VME22 (VT/GN), harness side and ground; and between the subwoofer C4208-8, circuit RME22 (GN/WH), harness side and ground.
- **Is any voltage present?**
YES : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.
NO : Go to E8.

E8 CHECK THE AUDIO CIRCUITS FOR AN OPEN OR SHORT TO GROUND

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- Key in OFF position.
- Measure the resistance between the subwoofer, harness side and the ACM, harness side; and between the subwoofer, harness side and ground as follows:

Subwoofer Connector-Pin	ACM Connector-Pin	Circuit
C4208-7	C240c-1	VME22 (VT/GN)
C4208-8	C240c-2	RME22 (GN/WH)

- **Is the resistance less than 5 ohms between the subwoofer and the ACM, and greater than 10,000 ohms between the subwoofer and ground?**

YES : Go to E9.

NO : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

E9 CHECK FOR CORRECT ACM OPERATION

- Key in OFF position.
- Connect: All Disconnected Connectors
- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the ACM connectors and make sure they seat correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test F: The Steering Wheel Controls Are Inoperative/Do Not Operate Correctly

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

There are 4 different steering wheel controls configurations:

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- Non-navigation, without the SYNC system
 - VOL-, VOL+, SEEK-, SEEK+, MEDIA
- Non-navigation, with the SYNC system
 - VOL-, VOL+, SEEK-, SEEK+, VOICE, MEDIA, PHONE, OK
- Navigation, without the SYNC system
 - VOL-, VOL+, SEEK-, SEEK+, VOICE, MEDIA
- Navigation, with the SYNC system
 - VOL-, VOL+, SEEK-, SEEK+, VOICE, MEDIA, PHONE

The VOL-, VOL+, SEEK+, SEEK-, and MEDIA buttons are always wired to the audio control module (ACM). Voltage is sent from the ACM to the steering wheel controls through circuit VME14 (GY/YE), and the ACM grounds the reference voltage through circuit RME24 (BU/WH).

For vehicles equipped with navigation, but not the SYNC system, the VOICE button is wired to the ACM.

For vehicles equipped with the SYNC system, the VOICE (an icon of someone speaking), PHONE (an icon of a phone with an up and down arrow), and OK buttons are wired directly to the accessory protocol interface module (APIM) unless the vehicle is equipped with navigation, in which case there is no OK button, and the VOICE and PHONE buttons are wired to the ACM. Voltage is sent from the APIM (or ACM if equipped with navigation) to the steering wheel controls through circuit VME54 (BU/OG), and the APIM (or ACM if equipped with navigation) grounds the reference voltage through circuit RME54 (WH/VT).

When a switch is pressed, the voltage is routed through a specific resistor value for each function. The ACM (or APIM) then uses this reference voltage to determine which control input function has been selected.

- DTC B1117 (Audio Steering Wheel Button Stuck) - set by the ACM or the APIM (depending on vehicle configuration) when a steering wheel control button is detected as active for greater than 120 seconds during normal operation, or for any duration during the self-test.
- DTC B1136 (Audio Steering Wheel Switch #2 Circuit Failure) - set by the ACM or the APIM (depending on vehicle configuration) when the reference voltage to the steering wheel controls is out-of-range. This can be caused by a number of different failures on the steering wheel controls circuit, except for a short to ground in circuit RME54 (WH/VT). This DTC only applies to the VOICE, PHONE, and OK buttons.
- DTC B2404 (Audio Steering Wheel Switch Circuit Fault) - set by the ACM when the reference voltage to the steering wheel controls is out-of-range. This can be caused by a number of different failures on the steering wheel controls circuit, except for a short to ground in circuit RME24 (BU/WH). This DTC only applies to the VOL-, VOL+, SEEK+, SEEK-, and MEDIA buttons.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Clockspring
- Steering wheel controls
- ACM

- APIM (if equipped)

PINPOINT TEST F: THE STEERING WHEEL CONTROLS ARE INOPERATIVE/DO NOT OPERATE CORRECTLY

F1 DETERMINE THE INOPERATIVE STEERING WHEEL CONTROLS

- Determine which steering wheel controls are inoperative by pressing each steering wheel control button individually while operating the audio system in various modes (AM/FM, SYNC, etc.).

- **Is the concern with the VOL-, VOL+, SEEK-, SEEK+, or MEDIA button?**

YES : Go to F2.

NO : Go to F3.

F2 MONITOR THE STEERING WHEEL CONTROL PIDs (EXCEPT NAVIGATION/SYNC SWITCHES)

- Enter the following diagnostic mode on the diagnostic tool: ACM DataLogger
- Monitor the steering wheel controls PIDs while pressing each steering wheel controls button as follows:

Steering Wheel Control Button	PID
VOL-	VOL_DN
VOL+	VOL_UP
SEEK-	SEEK_DN
SEEK+	SEEK_UP
MEDIA	MODE_SW

- **Do the PID values agree with the switch positions?**

YES : Go to F8.

NO : If only one PID value is incorrect or the PID value always reads one particular switch position, INSTALL new steering wheel controls. REFER to **Steering Wheel Controls**. CLEAR any DTCs present. TEST the system for normal operation.

Otherwise, go to F3.

F3 MONITOR THE STEERING WHEEL CONTROL PIDs (NAVIGATION/SYNC SWITCHES)

- Enter the following diagnostic mode on the diagnostic tool: ACM DataLogger (With Navigation) or APIM DataLogger (Without Navigation)
- Monitor the steering wheel controls PIDs while pressing each steering wheel controls button as follows:

Steering	
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Wheel Controls Button	PID
VOICE	VOICE_SW
PHONE	PHONE_SW
OK (without navigation only)	OK_SW

- **Do the PID values agree with the switch positions?**

YES : For vehicles with navigation, go to F8.

For vehicles without navigation, go to F9.

NO : If only one PID value is incorrect or the PID value always reads one particular switch position, **INSTALL** new steering wheel controls. REFER to **Steering Wheel Controls**. **CLEAR** any DTCs present. **TEST** the system for normal operation.

Otherwise, go to F4.

F4 CHECK THE REFERENCE VOLTAGE CIRCUIT TO THE CLOCKSPRING FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: Clockspring C218a
- Disconnect: ACM C240a or APIM C3342
- Key in ON position.

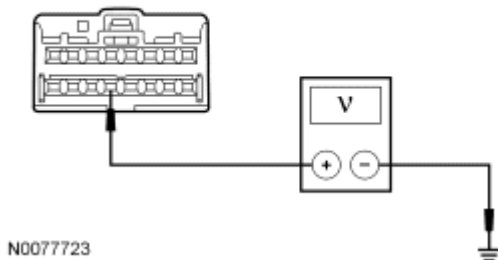


Fig. 14: Measuring Voltage Between Clockspring C218A-13, Circuit VME14 (GY/YE)
Courtesy of FORD MOTOR CO.

- For a concern with the VOL-, VOL+, SEEK-, SEEK+, or MEDIA buttons, measure the voltage between the clockspring C218a-13, circuit VME14 (GY/YE), harness side and ground.

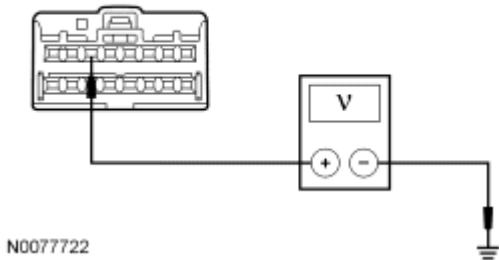


Fig. 15: Measuring Voltage Between Clockspring C218A-6, Circuit VME54 (BU/OG)
Courtesy of FORD MOTOR CO.

- For a concern with the VOICE, PHONE (or SEND), or OK (or END) buttons, measure the voltage between the clockspring C218a-6, circuit VME54 (BU/OG), harness side and ground.
- **Is any voltage present?**
YES : REPAIR the circuit. CLEAR any DTCs present. TEST the system for normal operation.
NO : Go to F5.

F5 CHECK THE CIRCUITS TO THE CLOCKSPRING FOR AN OPEN OR SHORT TO GROUND

NOTE: **Circuits RME14 (BU/WH) and RME54 (WH/VT) do not need to be checked for a short to ground.**

- Key in OFF position.
- For a concern with the VOL-, VOL+, SEEK-, SEEK+, or MEDIA buttons, measure the resistance between the ACM, harness side and the clockspring, harness side; and between the ACM, harness side and ground as follows:

ACM Connector-Pin	Clockspring Connector-Pin	Circuit
C240a-18	C218a-13	VME14 (GY/YE)
C240a-19	C218a-12	RME24 (BU/WH)

- For a concern with the VOICE, PHONE, or OK buttons, measure the resistance between the ACM (with navigation) or APIM (with the SYNC system, without navigation), harness side and the clockspring, harness side; and between the ACM (with navigation) or APIM (with the SYNC system, without navigation), harness side and ground as follows:

ACM or APIM Connector-	Clockspring Connector-	Circuit
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Pin	Pin	
C240a-17 or C3342-14	C218a-6	VME54 (BU/OG)
C240a-16 or C3342-15	C218a-4	RME54 (WH/VT)

- Is the resistance less than 5 ohms between the ACM (or APIM) and the clockspring, and greater than 10,000 ohms between the ACM (or APIM) and ground?

YES : Go to F6.

NO : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

F6 CHECK THE REFERENCE VOLTAGE CIRCUIT TO THE STEERING WHEEL CONTROLS FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: Clockspring C218b
- Disconnect: Steering Wheel Controls C2999
- Key in ON position.
- For a concern with the VOL-, VOL+, SEEK-, SEEK+, or MEDIA buttons, measure the voltage between the clockspring C218b-10, circuit VME14 (GY/YE), harness side and ground.
- For a concern with the VOICE, PHONE, or OK buttons, measure the voltage between the clockspring C218b-3, circuit VME54 (BU/OG), harness side and ground.
- **Is any voltage present?**

YES : REPAIR the circuit. CLEAR any DTCs present. TEST the system for normal operation.

NO : Go to F7.

F7 CHECK THE CIRCUITS TO THE STEERING WHEEL CONTROLS FOR AN OPEN OR SHORT TO GROUND

NOTE: Circuits RME14 (BU/WH) and RME54 (WH/VT) do not need to be checked for a short to ground.

- Key in OFF position.
- For a concern with the VOL-, VOL+, SEEK-, SEEK+, or MEDIA buttons, measure the resistance between the clockspring, harness side and the steering wheel controls, harness side; and between the clockspring, harness side and ground as follows:

Clockspring Connector- Pin	Steering Wheel Controls Connector- Pin	Circuit

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

C218b-10	C2999-2	VME14 (GY/YE)
C218b-11	C2999-4	RME24 (BU/WH)

- For a concern with the VOICE, PHONE, or OK buttons, measure the resistance between the clockspring, harness side and the steering wheel controls, harness side; and between the clockspring, harness side and ground as follows:

Clockspring Connector- Pin	Steering Wheel Controls Connector- Pin	Circuit
C218b-3	C2999-6	VME54 (BU/OG)
C218b-5	C2999-7	RME54 (WH/VT)

- **Is the resistance less than 5 ohms between the clockspring and the steering wheel controls, and greater than 10,000 ohms between the clockspring and ground?**

YES : INSTALL a new clockspring. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. CLEAR the DTCs. REPEAT the self-test.

NO : REPAIR the circuit in question. CLEAR any DTCs present. TEST the system for normal operation.

F8 CHECK FOR CORRECT ACM OPERATION

- Key in OFF position.
- Connect: All Disconnected Connectors
- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

F9 CHECK FOR CORRECT APIM OPERATION

- Key in OFF position.

2008 Ford Edge SE

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- Disconnect the APIM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pints
- Connect the APIM connector and make sure it seats correctly.
- Wait 2 minutes for the APIM to re-initialize.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new APIM. REFER to **Accessory Protocol Interface Module (APIM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test G: The Audio Input Jack Is Inoperative/Does Not Operate Correctly

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation - Vehicles Without the SYNC System

Audio signals are sent from the audio input jack to the audio control module (ACM). There are no external power or ground circuits to the audio input jack.

Vehicles With The SYNC System

Audio signals are sent from the audio input jack to the accessory protocol interface module (APIM). The signals are then sent to the ACM. There are no external power or ground circuits to the audio input jack.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Audio input jack
- ACM
- APIM (if equipped)

PINPOINT TEST G: THE AUDIO INPUT JACK IS INOPERATIVE/DOES NOT OPERATE CORRECTLY

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

2008 Ford Edge SE

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NOTE: Before carrying out this pinpoint test, make sure the MP3 device is operating correctly.

G1 CHECK THE AUDIO INPUT JACK AUDIO

- Connect: Known Good MP3 Device (to Audio Input Jack)
- Using the known good MP3 device, attempt to play a file using the audio input jack.
- **Does the file play correctly?**
YES : The system is operating correctly at this time. The concern may be with the customer device.
NO : Go to G2.

G2 DETERMINE THE VEHICLE CONTENT

- Determine if the vehicle is equipped with the SYNC system.
- **Is the vehicle equipped with the SYNC system?**
YES : Go to G3.
NO : Go to G4.

G3 CHECK THE APIM AUDIO OUTPUT

- Attempt to play an audio file using either the universal serial bus (USB) port or a Bluetooth® device.
- **Is the audio output OK for devices connected through the USB port or Bluetooth®?**
YES : Go to G4.
NO : Go to **Pinpoint Test S**.

G4 CHECK THE AUDIO INPUT JACK CIRCUITS FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: Audio Input Jack C3312
- Disconnect: ACM C240b or APIM C3342 (if equipped)
- Key in ON position.
- Measure the voltage between the audio input jack, harness side and ground as follows:

Connector-Pin	Circuit
C3312-1	VME46 (BU/GN)
C3312-2	RME46 (WH/GN)
C3312-3	RME45 (YE/GN)
C3312-4	VME45 (BU)

- **Is any voltage present?**
YES : REPAIR the circuit in question. TEST the system for normal operation.

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NO : Go to G5.

G5 CHECK THE AUDIO INPUT JACK CIRCUITS FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the audio input jack, harness side and the ACM (or APIM, if equipped), harness side; and between the audio input jack, harness side and ground as follows:

Audio Input Jack Connector-Pin	ACM or APIM Connector-Pin	Circuit
C3312-1	C240b-6 or C3342-47	VME46 (BU/GN)
C3312-2	C240b-14 or C3342-48	RME46 (WH/GN)
C3312-3	C240b-8 or C3342-46	RME45 (YE/GN)
C3312-4	C240b-7 or C3342-45	VME45 (BU)

- Is the resistance less than 5 ohms between the audio input jack and the ACM (or APIM), and greater than 10,000 ohms between the audio input jack and ground?

YES : Go to G6.

NO : REPAIR the circuit in question. TEST the system for normal operation.

G6 SUBSTITUTE THE AUDIO INPUT JACK

- Install a known good audio input jack.
- Operate the audio system using a device plugged into the audio input jack.
- **Does the system operate correctly?**

YES : INSTALL a new audio input jack. REFER to Audio Input Jack. TEST the system for normal operation.

NO : INSTALL the original audio input jack. For vehicles without the SYNC system, go to G7.

For vehicles with the SYNC system, go to G8.

G7 CHECK FOR CORRECT ACM OPERATION

- Key in OFF position.
- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins

2008 Ford Edge SE

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- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

G8 CHECK FOR CORRECT APIM OPERATION

- Key in OFF position.
- Disconnect the APIM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the APIM connector and make sure it seats correctly.
- Wait 2 minutes for the APIM to re-initialize.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new APIM. REFER to **Accessory Protocol Interface Module (APIM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test H: The Satellite Audio Is Inoperative/Does Not Operate Correctly - Vehicles Without DVD Entertainment

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

The satellite digital audio receiver system (SDARS) module receives voltage through circuit CBP13 (GY/BN) and ground through circuit GD148 (BK/YE). Digital signals are received by the satellite radio antenna and sent to the SDARS module, which converts the signals to analog and send them to either the audio control module (ACM) or the accessory protocol interface module (APIM) (if equipped).

If there is an open in the satellite antenna circuit, there will be no satellite audio. The ACM display will indicate a satellite antenna concern (the method varies according to the type of ACM) when the audio system is operated in satellite radio mode. To diagnose an open in the satellite antenna circuit, go to **Pinpoint Test J** .

The SDARS module transmits left and right channel audio signals. When there is a fault on any of these circuits, the audio quality is reduced. If the circuits for a specific channel are shorted together, loss of audio on

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

that entire channel will occur. This can be detected by using the SDARS module tones test.

- DTC U0184 (Lost Communication With Radio [ACM]) - set by the SDARS module when certain medium speed controller area network (MS-CAN) messages are missing from the ACM.
- DTC U0193 (Lost Communication With Digital Audio Control Module [SDARS]) - set by the ACM when certain MS-CAN messages are missing from the SDARS module.
- DTC U0197 (Lost Communication With Telephone Control Module) - set by the SDARS module when certain MS-CAN messages are missing from the APIM.

This pinpoint test is intended to diagnose the following:

- Subscription status
- Wiring, terminals or connectors
- SDARS module
- APIM (if equipped)
- ACM

PINPOINT TEST H: THE SATELLITE AUDIO IS INOPERATIVE/DOES NOT OPERATE CORRECTLY - VEHICLES WITHOUT DVD ENTERTAINMENT

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

H1 VERIFY AN ACTIVE SUBSCRIPTION

- Operate the audio system in satellite radio mode and observe the display.
- **Does the display read CALL SIRIUS?**

YES : The subscription has expired. INFORM the customer to contact Sirius to re-activate the subscription.

NO : If the display indicates a satellite antenna fault, go to **Pinpoint Test J** .

Otherwise, go to H2.

H2 CHECK FOR NON-NETWORK DTCs

- Carry out the SDARS module self-test using the scan tool.
- **Are any non-network DTCs present?**

YES : For DTC B1031 or DTC B1032, go to **Pinpoint Test J** .

For all other DTCs, REFER to DTC Charts in this service information.

NO : Go to H3.

H3 CHECK FOR COMMUNICATION WITH THE ACM AND APIM

- Carry out the network test using the scan tool.

- **Does the scan tool communicate with the ACM and the APIM (if equipped)?**

YES : Go to H4.

NO : REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose no communication with the ACM or APIM.

H4 CHECK FOR LOST COMMUNICATION DTCs

- Key in ON position.
- Using the scan tool, clear any ACM and SDARS module continuous DTCs.
- Key in OFF position.
- Key in ON position.
- Wait at least 10 seconds.
- Retrieve the ACM and SDARS module continuous DTCs.
- **Is DTC U0184 (SDARS module), U0193 (ACM), or U0197 (SDARS module, if equipped with the SYNC system) present?**

YES : For DTC U0184, go to H13.

For DTC U0193, go to H11.

For DTC U0197, go to H12.

NO : Go to H5.

H5 CHECK THE SATELLITE AUDIO OUTPUT

- Enter the following diagnostic mode on the diagnostic tool: SDARS Module Tones Test
- Carry out the SDARS module tones test.
- **Are alternating LH/RH tones audible?**

YES : Go to H11.

NO : If the vehicle is equipped with the SYNC system, go to H6.

Otherwise, go to H9.

H6 CHECK THE OPERATION OF THE SYNC SYSTEM AUDIO

- Operate the audio system in radio tuner (AM/FM) mode.
- Connect: Known Good MP3 Device (To Audio Input Jack)
- Using the ACM buttons, press AUX until "SYNC LINE IN" appears on the display (except navigation), or press MEDIA, select "USER DEVICE", and press "SOURCE" repeatedly until "LINE IN" appears (navigation).
- Using the known good device, attempt to play a file through the audio input jack.
- **Does the SYNC system audio operate correctly?**

YES : Go to H7.

NO : Go to **Pinpoint Test S** .

H7 CHECK THE AUDIO CIRCUITS TO THE APIM FOR A SHORT TO VOLTAGE

2008 Ford Edge SE

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- Key in OFF position.
- Disconnect: APIM C3342
- Disconnect: SDARS Module C4344
- Key in ON position.
- Measure the voltage between the SDARS module, harness side and ground as follows:

Connector-Pin	Circuit
C4344-5	VME41 (GN/OG)
C4344-6	VME42 (VT/BN)
C4344-11	RME41 (BU/BN)
C4344-12	RME42 (YE/BU)

- **Is any voltage present?**

YES : REPAIR the circuit in question. TEST the system for normal operation.

NO : Go to H8.

H8 CHECK THE AUDIO CIRCUITS TO THE APIM FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the APIM, harness side and the SDARS module, harness side; and between the APIM, harness side and ground as follows:

APIM Connector- Pin	SDARS Module Connector- Pin	Circuit
C3342-10	C4344-5	VME41 (GN/OG)
C3342-7	C4344-6	VME42 (VT/BN)
C3342-11	C4344-11	RME41 (BU/BN)
C3342-8	C4344-12	RME42 (YE/BU)

- **Is the resistance less than 5 ohms between the APIM and the SDARS module, and greater than 10,000 ohms between the APIM and ground?**

YES : Go to H11.

NO : REPAIR the circuit in question. TEST the system for normal operation.

H9 CHECK THE AUDIO CIRCUITS TO THE ACM FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: ACM C240b
- Disconnect: SDARS Module C4344
- Key in ON position.
- Measure the voltage between the ACM, harness side and ground as follows:

Connector-Pin	Circuit
C240b-1	VME52 (BU)
C240b-2	RME52 (GY/OG)
C240b-9	VME53 (VT/GN)
C240b-10	RME53 (BN/WH)

- **Is any voltage present?**

YES : REPAIR the circuit in question. TEST the system for normal operation.

NO : Go to H10.

H10 CHECK THE AUDIO CIRCUITS FROM THE SDARS MODULE FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the SDARS module, harness side and the ACM, harness side; and between the SDARS module, harness side and ground as follows:

SDARS Module Connector-Pin/Circuit	ACM Connector-Pin/Circuit
C4344-5 VME41 (GN/OG)	C240b-1 VME52 (BU)
C4344-6 VME42 (VT/BN)	C240b-9 RME52 (GY/OG)
C4344-11 RME41 (BU/BN)	C240b-2 VME53 (VT/GN)
C4344-12 RME42	C240b-10 RME53

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(YE/BU) (BN/WH)

- **Is the resistance less than 5 ohms between the SDARS module and the ACM, and greater than 10,000 ohms between the SDARS module and ground?**

YES : Go to H11.

NO : REPAIR the circuit in question. TEST the system for normal operation.

H11 ISOLATE THE SDARS MODULE

- Key in OFF position.
- Connect: ACM C240b
- Connect: APIM C3342 (If Equipped)
- Install a new SDARS module. Refer to Satellite Digital Audio Receiver System (SDARS) Module.
- Operate the audio system in satellite radio mode.
- **Does the system operate correctly?**

YES : The concern was caused by an inoperative SDARS module. The system is operating correctly at this time.

NO : If equipped with the SYNC system, go to H12.

Otherwise, go to H13.

H12 CHECK FOR CORRECT APIM OPERATION

- Disconnect the APIM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the APIM connector and make sure it seats correctly.
- Wait 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new APIM. REFER to Accessory Protocol Interface Module (APIM). TEST the system for normal operation. If the concern is still present, go to H13.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

H13 CHECK FOR CORRECT ACM OPERATION

- Key in OFF position.
- Disconnect all the ACM connectors.
- Check for:
 - corrosion

2008 Ford Edge SE

2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

- damaged pins
- pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test I: The Satellite Audio Is Inoperative/Does Not Operate Correctly - Vehicles With DVD Entertainment

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

The satellite digital audio receiver system (SDARS) module receives voltage through circuit CBP13 (GY/BN) and ground through circuit GD148 (BK/YE). Digital signals are received by the satellite radio antenna and sent to the SDARS module, which converts the signals to analog and sends them to either the rear entertainment module (RETM), or the accessory protocol interface module (APIM) (if equipped).

The satellite audio signals utilize separate circuitry from the remaining audio signals. This allows the audio control module (ACM) to distinguish between satellite audio and SYNC or DVD audio, in order to provide for more dual-play options.

If there is an open in the satellite antenna circuit, there will be no satellite audio. The ACM display will indicate a satellite antenna concern (the method varies according to the type of ACM) when the audio system is operated in satellite radio mode. To diagnose an open in the satellite antenna circuit, go to **Pinpoint Test J**.

Circuits VME41 (GN/OG), RME41 (BU/BN), VME42 (VT/BN), and RME42 (YE/BU) transmit left and right channel audio signals. When there is a fault on any of these circuits, the audio quality is reduced. If the circuits for a specific channel are shorted together, loss of audio on that entire channel will occur. This can be detected by using the SDARS module tones test.

DTC Description	Fault Set Conditions
U0184 - Lost Communication With Radio (ACM)	Set by the SDARS module when certain medium speed controller area network (MS-CAN) messages are missing from the ACM.
U0193 - Lost Communication With Digital Audio Control Module (SDARS)	Set by the ACM or the RETM when certain MS-CAN messages are missing from the SDARS module.
U0196 - Lost Communication With Entertainment Control	Set by the SDARS module when certain

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Module - Rear (AUX)	MS-CAN messages are missing from the RETM.
U0197 - Lost Communication With Telephone Control Module	Set by the SDARS module when certain MS-CAN messages are missing from the APIM.

This pinpoint test is intended to diagnose the following:

- Subscription status
- Wiring, terminals or connectors
- SDARS module
- RETM
- APIM (if equipped)
- ACM

PINPOINT TEST I: THE SATELLITE AUDIO IS INOPERATIVE/DOES NOT OPERATE CORRECTLY - VEHICLES WITH DVD ENTERTAINMENT

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

I1 VERIFY AN ACTIVE SUBSCRIPTION

- Operate the audio system in satellite radio mode and observe the display.
- **Does the display read CALL SIRIUS?**

YES : The subscription has expired. INFORM the customer to contact Sirius to re-activate the subscription.

NO : If the display indicates a satellite antenna fault, go to **Pinpoint Test J** .

Otherwise, go to I2.

I2 CHECK FOR NON-NETWORK DTCs

- Carry out the SDARS module self-test using the scan tool.
- **Are any non-network DTCs present?**

YES : For DTC B1031 or DTC B1032, go to **Pinpoint Test J** .

For all other DTCs, REFER to DTC Charts in this service information.

NO : Go to I3.

I3 CHECK FOR COMMUNICATION WITH THE ACM AND APIM

- Carry out the network test using the scan tool.
- **Does the scan tool communicate with the ACM, the RETM, and the APIM (if equipped)?**

YES : Go to I4.

NO : REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose no communication with the ACM or APIM.

I4 CHECK FOR LOST COMMUNICATION DTCs

- Key in ON position.
- Using the scan tool, clear any ACM and SDARS module continuous DTCs.
- Key in OFF position.
- Key in ON position.
- Wait at least 10 seconds.
- Retrieve the ACM and SDARS module continuous DTCs.
- **Is DTC U0184 (SDARS module), U0193 (ACM), U0196 (SDARS module), or U0197 (SDARS module, if equipped with the SYNC system) present?**

YES : For DTC U0184, go to I16.

For DTC U0193, go to I13.

For DTC U0197, go to I14.

For DTC U0196, go to I15.

NO : Go to I5.

I5 CHECK THE SATELLITE AUDIO OUTPUT

- Enter the following diagnostic mode on the diagnostic tool: SDARS Module Tones Test
- Carry out the SDARS module tones test.
- **Are alternating LH/RH tones audible?**

YES : Go to I13.

NO : Go to I6.

I6 CHECK THE OPERATION OF THE DVD AUDIO

- Insert a known good DVD, and operate the audio system in DVD mode.
- Observe the audio output from the DVD.
- **Does the DVD audio operate correctly?**

YES : If the vehicle is equipped with the SYNC system, go to I7.

Otherwise, go to I11.

NO : Go to **Symptom Chart - Audio System** to diagnose the observed symptom.

I7 CHECK THE AUDIO CIRCUITS TO THE APIM FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: SDARS Module C4344
- Disconnect: APIM C3342
- Key in ON position.

2008 Ford Edge SE

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- Measure the voltage between the SDARS module, harness side and ground as follows:

Connector-Pin	Circuit
C4344-5	VME41 (GN/OG)
C4344-6	VME42 (VT/BN)
C4344-11	RME41 (BU/BN)
C4344-12	RME42 (YE/BU)

- **Is any voltage present?**

YES : REPAIR the circuit in question. TEST the system for normal operation.

NO : Go to I8.

I8 CHECK THE AUDIO CIRCUITS TO THE APIM FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the SDARS module, harness side and the APIM, harness side; and between the SDARS module, harness side and ground as follows:

SDARS Module Connector-Pin	APIM Connector-Pin	Circuit
C4344-5	C3342-10	VME41 (GN/OG)
C4344-6	C3342-7	VME42 (VT/BN)
C4344-11	C3342-11	RME41 (BU/BN)
C4344-12	C3342-8	RME42 (YE/BU)

- **Is the resistance less than 5 ohms between the SDARS module and the APIM, and greater than 10,000 ohms between the SDARS module and ground?**

YES : Go to I9.

NO : REPAIR the circuit in question. TEST the system for normal operation.

I9 CHECK THE AUDIO CIRCUITS FROM THE APIM TO THE RETM FOR A SHORT TO VOLTAGE

- Disconnect: RETM C949
- Key in ON position.

2008 Ford Edge SE

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- Measure the voltage between the RETM, harness side and ground as follows:

Connector-Pin	Circuit
C949-5	VME41 (GN/OG)
C949-6	RME41 (BU/BN)
C949-21	VME42 (VT/BN)
C949-22	RME42 (YE/BU)

- **Is any voltage present?**

YES : REPAIR the circuit in question. TEST the system for normal operation.

NO : Go to I10.

I10 CHECK THE AUDIO CIRCUITS FROM THE APIM TO THE RETM FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the APIM, harness side and the RETM, harness side; and between the APIM, harness side and ground as follows:

APIM Connector-Pin	RETM Connector-Pin	Circuit
C3342-40	C949-5	VME41 (GN/OG)
C3342-41	C949-6	RME41 (BU/BN)
C3342-42	C949-21	VME42 (VT/BN)
C3342-43	C949-22	RME42 (YE/BU)

- **Is the resistance less than 5 ohms between the APIM and the RETM, and greater than 10,000 ohms between the APIM and ground?**

YES : Go to I13.

NO : REPAIR the circuit in question. TEST the system for normal operation.

I11 CHECK THE AUDIO CIRCUITS FROM THE SDARS MODULE FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: SDARS Module C4344

2008 Ford Edge SE

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- Disconnect: RETM C949
- Key in ON position.
- Measure the voltage between the SDARS module, harness side and ground as follows:

Connector-Pin	Circuit
C4344-5	VME41 (GN/OG)
C4344-6	VME42 (VT/BN)
C4344-11	RME41 (BU/BN)
C4344-12	RME42 (YE/BU)

- **Is any voltage present?**

YES : REPAIR the circuit in question. TEST the system for normal operation.

NO : Go to I12.

I12 CHECK THE AUDIO CIRCUITS FROM THE SDARS MODULE FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the SDARS module, harness side and the RETM, harness side; and between the SDARS module, harness side and ground as follows:

SDARS Module Connector-Pin	RETM Connector-Pin	Circuit
C4344-5	C949-5	VME41 (GN/OG)
C4344-6	C949-21	VME42 (VT/BN)
C4344-11	C949-6	RME41 (BU/BN)
C4344-12	C949-22	RME42 (YE/BU)

- **Is the resistance less than 5 ohms between the SDARS module and the RETM, and greater than 10,000 ohms between the SDARS module and ground?**

YES : Go to I13.

NO : REPAIR the circuit in question. TEST the system for normal operation.

I13 ISOLATE THE SDARS MODULE

2008 Ford Edge SE

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- Key in OFF position.
- Connect: RETM C949
- Connect: APIM C3342 (If Equipped)
- Install a new SDARS module. Refer to **Satellite Digital Audio Receiver System (SDARS) Module**.
- Operate the audio system in satellite radio mode.
- **Does the system operate correctly?**

YES : The concern was caused by an inoperative SDARS module. The system is operating correctly at this time.

NO : If equipped with the SYNC system, go to I14.

Otherwise, go to I15.

I14 CHECK FOR CORRECT APIM OPERATION

- Disconnect the APIM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the APIM connector and make sure it seats correctly.
- Wait 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new APIM. REFER to **Accessory Protocol Interface Module (APIM)**. TEST the system for normal operation. If the concern is still present, go to I15.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

I15 CHECK FOR CORRECT RETM OPERATION

- Disconnect the RETM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the RETM connector and make sure it seats correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new RETM. REFER to **Rear Entertainment Module (RETM)**. TEST the system for normal operation. If the concern is still present, go to I16.

NO : The system is operating correctly at this time. The concern may have been caused by a loose

or corroded connector.

I16 CHECK FOR CORRECT ACM OPERATION

- Key in OFF position.
- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test J: Poor Reception/No Sound - Satellite Audio**Normal Operation**

Digital signals are received by the satellite radio antenna and sent to the satellite digital audio receiver system (SDARS) module, which then converts the signals and provides analog audio signals to the audio control module (ACM), the accessory protocol interface module (APIM), or to the rear entertainment module (RETM), depending on vehicle configuration.

A short to ground typically has no effect on the antenna signal. However, an open in the antenna circuit will result in no sound from the satellite radio system. A short to voltage in the antenna circuit can have varying effects on the system.

- DTC B1031 (SDARS Satellite Antenna Open) - sets when an open or high resistance is detected in the satellite antenna circuit. This DTC can be either continuous or on-demand.
- DTC B1032 (SDARS Satellite Antenna Short) - sets when a short to ground or short to voltage is detected in the satellite antenna circuit. This DTC can be either continuous or on-demand.

This pinpoint test is intended to diagnose the following:

- Obstructions to the line of sight
- Satellite radio antenna cable
- Satellite radio antenna
- SDARS module

PINPOINT TEST J: POOR RECEPTION/NO SOUND - SATELLITE AUDIO

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

J1 CHECK THE OPERATION OF THE SATELLITE AUDIO SYSTEM

- Drive the vehicle to an open location, free of obstacles.
- Operate the audio system in satellite audio mode.
- **Is the reception OK?**

YES : The system is OK at this time. ADVISE the customer that the satellite signal can be affected by obstructions to the satellite radio antenna line of sight.

NO : Go to J2.

J2 CHECK FOR A SATELLITE ANTENNA FAULT MESSAGE

- Operate the audio system in satellite audio mode and observe the ACM display.
- **Does the ACM display indicate a satellite radio antenna fault?**

YES : Go to J3.

NO : For vehicles without a DVD entertainment system, go to **Pinpoint Test H**.

For vehicles with a DVD entertainment system, go to **Pinpoint Test I**.

J3 CHECK THE SATELLITE RADIO ANTENNA CABLE

- Key in OFF position.
- Disconnect: Satellite Radio Antenna Connection (At SDARS Module)
- Disconnect: Satellite Radio Antenna Connection (At Satellite Radio Antenna)
- Measure the resistance of the satellite radio antenna cable between the SDARS module and the satellite radio antenna connection.
- **Is the resistance less than 1 ohm?**

YES : Go to J4.

NO : INSTALL a new satellite radio antenna cable. REFER to **Antenna Cable - Satellite Radio**. CLEAR any DTCs present. TEST the system for normal operation.

J4 SUBSTITUTE THE SATELLITE RADIO ANTENNA

- Install a known good satellite radio antenna.
- Operate the audio system in satellite audio mode.
- **Is the reception OK?**

YES : INSTALL a new satellite radio antenna. REFER to **Antenna - Satellite Radio**. CLEAR any DTCs present. TEST the system for normal operation.

NO : Go to J5.

J5 CHECK FOR CORRECT SDARS MODULE OPERATION

- Key in OFF position.
- Disconnect the SDARS module connector.
- Check for:
 - corrosion
 - damaged pins

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- pushed-out pins
- Connect the SDARS module connector and make sure it seats correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SDARS module. REFER to Satellite Digital Audio Receiver System (SDARS) Module. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test K: The Video System Is Inoperative/Does Not Operate Correctly

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

The rear entertainment module (RETM) routes audio signals to the audio control module (ACM). In single-play mode, the ACM plays the DVD audio on all speakers. In dual-play mode, the audio to the rear speakers is muted and the RETM continues to control the rear audio source to the headphones.

If the vehicle is equipped with satellite audio and/or the SYNC system, the audio signals from these sources are routed through the RETM to the ACM. This allows for single-play and dual-play to include these audio sources.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- RETM
- ACM

PINPOINT TEST K: THE VIDEO SYSTEM IS INOPERATIVE/DOES NOT OPERATE CORRECTLY

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Before diagnosing the RETM, make sure a known good DVD is installed.

K1 IDENTIFY THE SYMPTOM

- Operate the audio system in DVD single-play mode and observe the DVD audio.
- **Does the DVD audio operate correctly?**

YES : Go to K2.

NO : If the vehicle is equipped with satellite audio or the SYNC system, go to K3.

Otherwise, go to K4.

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K2 CHECK THE OPERATION OF THE DVD VIDEO

- Operate the audio system in DVD single-play mode.
- **Does the DVD video operate correctly?**

YES : The system is operating correctly at this time. The concern was caused by an incorrect system setting. **INFORM** the customer that pressing the power button or the VIDEO button while a DVD is playing can cause the display to go blank.

NO : Go to K6.

K3 CHECK THE RETM AUDIO PASS-THROUGH

- Operate the audio system in satellite audio or SYNC mode.
- **Does the satellite audio (or SYNC) audio operate correctly?**

YES : Go to K6.

NO : Go to K4.

K4 CHECK THE AUDIO CIRCUITS TO THE ACM FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: ACM C240b
- Disconnect: RETM C949
- Key in ON position.
- Measure the voltage between the ACM, harness side and ground as follows:

Connector-Pin	Circuit
C240b-1	VME17 (GN)
C240b-2	RME17 (GY)
C240b-9	VME18 (VT)
C240b-10	RME18 (YE)

- **Is any voltage present?**

YES : **REPAIR** the circuit in question. **TEST** the system for normal operation.

NO : Go to K5.

K5 CHECK THE AUDIO CIRCUITS TO THE ACM FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the ACM, harness side and the RETM, harness side; and between the ACM, harness side and ground as follows:

ACM Connector-	RETM Connector-	Circuit
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Pin	Pin	
C240b-1	C949-10	VME17 (GN)
C240b-2	C949-9	RME17 (GY)
C240b-9	C949-26	VME18 (VT)
C240b-10	C949-25	RME18 (YE)

- **Is the resistance less than 5 ohms between the ACM and the RETM, and greater than 10,000 ohms between the ACM and ground?**

YES : Go to K6.

NO : REPAIR the circuit in question. TEST the system for normal operation.

K6 CHECK FOR CORRECT RETM OPERATION

- Disconnect all the RETM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the RETM connectors and make sure they seat correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new RETM. REFER to **Rear Entertainment Module (RETM)**. TEST the system for normal operation. If the concern is still present, Go to K7

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

K7 CHECK FOR CORRECT ACM OPERATION

- Key in OFF position.
- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose

or corroded connector.

Pinpoint Test L: No Auxiliary Headphone Audio - THX®

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

The audio control module (ACM) receives audio input from the rear entertainment module (RETM). The audio signals are then routed to the audio digital signal processing (DSP) module. In dual-play mode, the rear speaker audio is muted and the RETM controls the rear audio source to the headphones.

It is important to note that if the vehicle is equipped with the SYNC system, the headphones must be set to channel A in order to listen to SYNC media. Channel B only allows audio from the RETM, which can be either DVD audio or audio from a peripheral device plugged into the RETM.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Headphones
- RETM

PINPOINT TEST L: NO AUXILIARY HEADPHONE AUDIO - THX®

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Before diagnosing this condition, make sure the headphone batteries are OK.

L1 CHECK THE OPERATION OF THE BASE AUDIO SYSTEM

- Operate the audio system in radio tuner (AM/FM) mode.
- **Does the audio system operate correctly?**

YES : Go to L2.

NO : Go to Symptom Chart - Audio System.

L2 CHECK THE ACM OUTPUT TO THE RETM

- Key in OFF position.
- Disconnect: RETM C949

NOTE: Make sure the headphones are set to "Radio" as their audio source.

- Operate the audio system in radio tuner (AM/FM) dual-play mode.
- Measure the AC voltage between the RETM pins, harness side as follows:

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Connector-Pin/Circuit	Connector-Pin/Circuit
C949-11 VME12 (BN/WH)	C949-12 RME12 (BN/BU)
C949-27 VME09 (WH/GN)	C949-28 RME09 (BN/YE)

- **Is a fluctuating AC voltage present?**

YES : Go to L5.

NO : Go to L3.

L3 CHECK THE AUDIO CIRCUITS FROM THE ACM FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: ACM C240a
- Key in ON position.
- Measure the voltage between the ACM, harness side and ground as follows:

Connector-Pin	Circuit
C240a-9	VME09 (WH/GN)
C240a-10	VME12 (BN/WH)
C240a-22	RME09 (BN/YE)
C240a-23	RME12 (BN/BU)

- **Is any voltage present?**

YES : REPAIR the circuit in question. TEST the system for normal operation.

NO : Go to L4.

L4 CHECK THE AUDIO CIRCUITS FROM THE ACM FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the ACM, harness side and the RETM, harness side; and between the ACM, harness side and ground as follows:

ACM Connector-Pin	RETM Connector-Pin	Circuit
		VME09

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C240a-9	C949-27	(WH/GN)
C240a-10	C949-11	VME12 (BN/WH)
C240a-22	C949-28	RME09 (BN/YE)
C240a-23	C949-12	RME12 (BN/BU)

- **Is the resistance less than 5 ohms between the ACM and the RETM, and greater than 10,000 ohms between the ACM and ground?**

YES : Go to L5.

NO : REPAIR the circuit in question. TEST the system for normal operation.

L5 ISOLATE THE HEADPHONES

- Connect: ACM C240a
- Connect: RETM C949
- Install a set of known good headphones.

NOTE: **Make sure the headphones are set to "Radio" as their audio source.**

- Operate the audio system in radio tuner (AM/FM) dual-play mode and listen through the headphones.
- **Do the headphones operate correctly?**

YES : The concern was caused by inoperative headphones. The system is OK at this time.

NO : Go to L6.

L6 CHECK FOR CORRECT RETM OPERATION

- Key in OFF position.
- Disconnect the RETM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the RETM connector and make sure it seats correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new RETM. REFER to **Rear Entertainment Module (RETM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Normal Operation

The global positioning system (GPS) antenna provides information from the GPS satellite system to the GPS receiver in the navigation module (part of the navigation audio control module [ACM]). This information is used to calculate vehicle position and direction of travel.

- DTC B2204 (GPS Antenna Connection Open or Short) - sets when an open or short to ground is detected in the GPS antenna circuit.

This pinpoint test is intended to diagnose the following:

- GPS antenna
- ACM

PINPOINT TEST M: NO GLOBAL POSITIONING SYSTEM (GPS) ANTENNA SIGNAL

M1 CHECK THE OPERATION OF THE NAVIGATION SYSTEM

- Move the vehicle outside of any enclosed structure to an area that is unobstructed by trees, tall buildings, and bridges.



N0026638

Fig. 16: Checking Operation Of Navigation System
Courtesy of FORD MOTOR CO.

- Operate the audio system in navigation mode.
- **Does the GPS icon appear, then disappear?**
YES : The system is operating correctly at this time. NOTIFY the customer that obstructions to the line of sight may affect the GPS antenna.
NO : Go to M2.

M2 CHECK THE ACM DTCs

- Retrieve the ACM continuous and on-demand DTCs.
- **Is DTC B2204 recorded?**
YES : Go to M4.
NO : Go to M3.

M3 CHECK THE GPS ANTENNA MOUNTING

- Verify the GPS antenna is installed correctly and no aftermarket equipment is obstructing the antenna.

- **Is the GPS antenna mounted correctly?**

YES : Go to M4.

NO : Correctly INSTALL the GPS antenna. REFER to **Global Positioning System (GPS) Antenna**. TEST the system for normal operation.

M4 SUBSTITUTE THE GPS ANTENNA

- Install a known good GPS antenna.
- Move the vehicle outside of any enclosed structure to an area that is unobstructed by trees, tall buildings, and bridges.



N0026638

Fig. 17: Checking Operation Of Navigation System
Courtesy of FORD MOTOR CO.

- Operate the audio system in navigation mode.
- **Does the GPS icon appear, then disappear?**
YES : INSTALL a new GPS antenna. REFER to **Global Positioning System (GPS) Antenna**. CLEAR any DTCs present. TEST the system for normal operation.

NO : INSTALL the original GPS antenna. Go to M5.

M5 CHECK FOR CORRECT ACM OPERATION

- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test N: The Audible Switch Feedback Is Inoperative

Normal Operation

The audible switch feedback settings are controlled by the audio control module (ACM). The audible switch feedback can be set to ALL, TOUCH SCREEN, or NONE, depending on user preference.

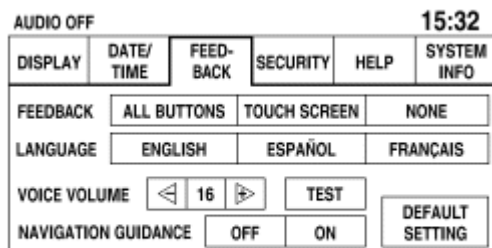
This pinpoint test is intended to diagnose the following:

- Audible switch feedback setting
- ACM

PINPOINT TEST N: THE AUDIBLE SWITCH FEEDBACK IS INOPERATIVE

N1 CHECK THE SWITCH DISPLAY

- Turn the ACM on.
- Press the MENU button.



N0026637

Fig. 18: Checking Switch Display
Courtesy of FORD MOTOR CO.

- Select the FEEDBACK tab.
- **Is NONE highlighted on the audible feedback settings?**
YES : SELECT "ALL BUTTONS". TEST the system for normal operation.
NO : Go to N2.

N2 CHECK FOR CORRECT ACM OPERATION

- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test O: The Voice Guidance Is Inoperative/Does Not Operate Correctly

Normal Operation

The voice guidance settings are controlled by the audio control module (ACM). The voice guidance volume can be adjusted or turned off depending on user preference.

For the THX® audio system, the ACM sends the voice guidance signal to the audio digital signal processing (DSP) module through circuits VME22 (VT/GN) and RME22 (GN/WH).

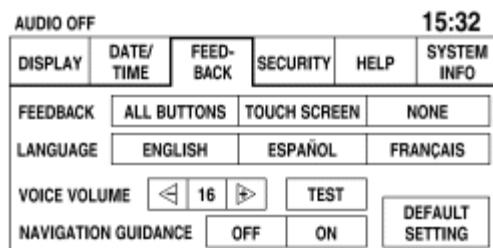
This pinpoint test is intended to diagnose the following:

- Incorrect system setting
- Wiring, terminals or connectors (THX® only)
- DSP module (if equipped)
- ACM

PINPOINT TEST O: THE VOICE GUIDANCE IS INOPERATIVE/DOES NOT OPERATE CORRECTLY

O1 CHECK THE VOICE GUIDANCE SETTINGS

- Operate the audio system in navigation mode.
- Press the MENU button.



N0026637

Fig. 19: Checking Switch Display
Courtesy of FORD MOTOR CO.

- Select the FEEDBACK tab.
- Select DEFAULT SETTING.

- Verify the operation of the voice guidance.
- **Does the voice guidance operate correctly?**
YES : The system is operating correctly at this time. The concern was caused by a customer-selected setting.
NO : If the vehicle is equipped with THX® audio, go to O2.

Otherwise, go to O4.

O2 CHECK THE VOICE SIGNAL CIRCUITS TO THE DSP MODULE FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: ACM C240c
- Disconnect: DSP Module C4348a
- Key in ON position.

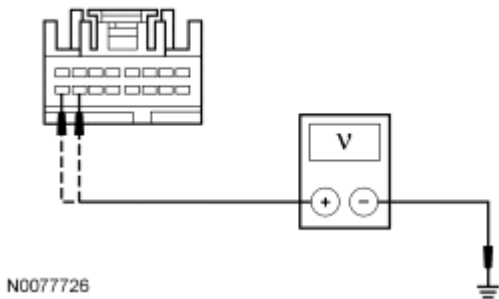


Fig. 20: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between the DSP module C4348a-15, circuit VME22 (VT/GN), harness side and ground; and between the DSP module C4348a-16, circuit RME22 (GN/WH), harness side and ground.
- **Is any voltage present?**
YES : REPAIR the circuit in question. TEST the system for normal operation.
NO : Go to O3.

O3 CHECK THE VOICE SIGNAL CIRCUITS TO THE DSP MODULE FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the DSP module and the ACM, harness side; and between the DSP module, harness side and ground as follows:

DSP Module Connector-	ACM Connector- Pin	Circuit
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Pin		
C4348a-15	C240c-1	VME22 (VT/GN)
C4348a-16	C240c-2	RME22 (GN/WH)

- Is the resistance less than 5 ohms between the DSP module and the ACM, and greater than 10,000 ohms between the DSP module and ground?

YES : Go to O5.

NO : REPAIR the circuit in question. TEST the system for normal operation.

O4 CHECK FOR CORRECT ACM OPERATION

- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to Audio Control Module (ACM). TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

O5 CHECK FOR CORRECT DSP MODULE OPERATION

- Key in OFF position.
- Disconnect all the DSP module connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the DSP module connectors and make sure they seat correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new DSP module. REFER to Audio Digital Signal Processing (DSP) Module. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test P: The Position Cursor Is Inaccurate

Normal Operation

The primary source of vehicle location for the navigation system is the audio control module (ACM) receiving the position signal through the global positioning system (GPS) antenna.

In addition, the ACM receives the wheel pulse signal from the instrument cluster (IC), which gateways the signal from the ABS module. This secondary signal is used to calculate vehicle position when the GPS signal is lost. It also supports the adaptive learning function of the ACM, whereby the ACM can compensate for long-term differences between the GPS signal location, and the actual distance traveled by the vehicle.

- DTC B2204 (GPS Antenna Connection Open or Short) - sets when an open, short to ground, or short to voltage is detected in the GPS antenna circuit. If DTC B2204 is present, go to **Pinpoint Test M**.
- DTC U0155 (Lost Communication With Instrument Panel Cluster [IC] Control Module) - sets when the ACM is missing messages from the IC for greater than 5 seconds. For this symptom, the message in question is the wheel pulse data message.
- DTC U2473 (Unexpected Vehicle Speed [VSS]) - sets when the calculated vehicle distance traveled based on the wheel pulse signal does not agree with the GPS antenna location. To do this, the ACM compares the wheel pulse signal during a 4-second range and compares it with the change in GPS antenna location. If the ACM finds that the variation is greater than 0.5% after carrying out this check 4 times, it sets DTC U2473. When DTC U2473 is set, the adaptive learn function is disabled. This DTC is also set when the navigation rolling wheel count signal is missing for greater than 2 seconds with the key in RUN.

This pinpoint test is intended to diagnose the following:

- VSS signal concern
- ACM

PINPOINT TEST P: THE POSITION CURSOR IS INACCURATE

P1 CHECK THE OPERATION OF THE NAVIGATION SYSTEM

- Move the vehicle outside of any enclosed structure to an area that is unobstructed by trees, tall buildings, and bridges.



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Fig. 21: Checking Operation Of Navigation System
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- Operate the audio system in navigation mode.
- **Does the GPS icon appear, then disappear?**

YES : Go to P2.

NO : Go to **Pinpoint Test M** .

P2 CHECK THE ACM DTCs

- Retrieve the continuous DTCs from the ACM self-test.
- **Is DTC U2473 or U0155 present?**

YES : Go to P3.

NO : Go to **Pinpoint Test M** .

P3 CHECK FOR COMMUNICATION WITH THE IC

- Carry out the network test using the scan tool.
- **Does the scan tool communicate with the IC?**

YES : Go to P4.

NO : REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose no communication with the IC.

P4 CHECK FOR CORRECT ACM OPERATION

- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test Q: The Audio Does Not Reduce When The Parking Aid Tone Sounds

Normal Operation

The parking aid module (PAM) sends the volume cutback message to the audio control module (ACM) via the medium speed controller area network (MS-CAN). When the ACM receives this message, it reduces the speaker output so that the parking aid tone can be heard more clearly.

- DTC U0159 (Lost Communication With Parking Assist Control Module [PAM]) - sets when the volume cutback message is missing for greater than 5 seconds. When the message is missing, the ACM defaults to no volume cutback when the parking aid tone is sounding.

This pinpoint test is intended to diagnose the following:

- Parking aid system concern
- ACM

PINPOINT TEST Q: THE AUDIO DOES NOT REDUCE WHEN THE PARKING AID TONE SOUNDS

Q1 CHECK THE ACM CONFIGURATION

- Apply the parking brake.
- Place an object behind the vehicle within the range of the parking aid sensors. Refer to **PARKING AID** article.
- Key in ON position.
- Place the transmission in REVERSE (R).
- Enter the following diagnostic mode on the diagnostic tool: ACM DataLogger
- Monitor the RPA_STAT (Parking Aid Input Status) PID while the parking aid tone is sounding.
- **Does the RPA_STAT PID read "Enabled" while the parking aid tone is sounding?**

YES : Go to Q2.

NO : REFER to **PARKING AID** article to continue diagnosis of the parking aid system.

Q2 CHECK FOR COMMUNICATION WITH THE PAM

- Carry out the network test using the scan tool.
- **Does the scan tool communicate with the PAM?**

YES : Go to Q3.

NO : REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose no communication with the PAM.

Q3 CHECK FOR CORRECT ACM OPERATION

- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

SYNC System

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

When the VOICE switch is pressed, it changes the reference voltage from the audio control module (ACM) on circuit VMN54 (BU/OG), and the ACM enters voice recognition mode. A microphone located in the interior rear view mirror receives the voice command and sends a signal to the ACM through circuits VMM13 (YE/GN) and RMM13 (BU). Power to the interior rear view mirror is provided through circuit CBP41 (BU). Ground is provided through circuit GD133 (BK).

The microphone test is available through the integrated diagnostic system (IDS). Running this test causes the ACM to produce a test tone. If the system is operating correctly, the microphone detects the tone and produces a signal to the ACM indicating the tone was heard.

- DTC B2633 (Driver-Front Microphone Circuit Failure) - set by the ACM during the microphone test when the ACM does not receive an adequate signal from the microphone when the test tone is sounded.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Steering wheel controls
- Microphone (part of the interior rear view mirror)
- ACM

PINPOINT TEST R: VOICE RECOGNITION IS INOPERATIVE/DOES NOT OPERATE CORRECTLY - VEHICLES WITHOUT THE SYNC SYSTEM

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

R1 CHECK THE OPERATION OF THE STEERING WHEEL CONTROLS

- Operate the audio system in radio tuner (AM/FM) mode.
- Press the VOICE button on the steering wheel controls.
- **Does the audio system enter voice recognition mode?**

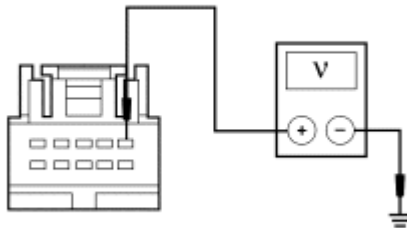
YES : Go to R2.

NO : Go to **Pinpoint Test F ** to diagnose the steering wheel controls system.

R2 CHECK CIRCUIT CBP41 (BU) FOR VOLTAGE

- Key in OFF position.
- Disconnect: Interior Rear View Mirror C911

- Key in ON position.



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Fig. 22: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

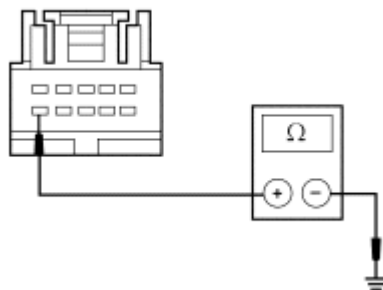
- Measure the voltage between the interior rear view mirror C911-1, circuit CBP41 (BU), harness side and ground.
- **Is the voltage greater than 10 volts?**

YES : Go to R3.

NO : VERIFY the smart junction box (SJB) fuse 41 (15A) is OK. If OK, REPAIR the circuit. TEST the system for normal operation. If not OK, REFER to appropriate SYSTEM WIRING DIAGRAMS article to identify the possible sources of the circuit short.

R3 CHECK CIRCUIT GD133 (BK) FOR AN OPEN

- Key in OFF position.



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Fig. 23: Checking Circuit For Open
Courtesy of FORD MOTOR CO.

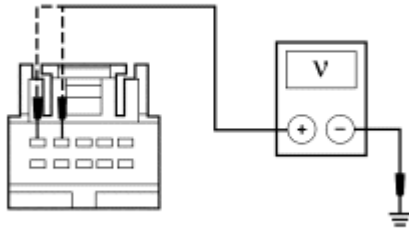
- Measure the resistance between the interior rear view mirror C911-10, circuit GD133 (BK), harness side and ground.
- **Is the resistance less than 5 ohms?**

YES : Go to R4.

NO : REPAIR the circuit. TEST the system for normal operation.

R4 CHECK THE MICROPHONE CIRCUITS FOR A SHORT TO VOLTAGE

- Disconnect: ACM C240b
- Key in ON position.



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Fig. 24: Checking Circuits For Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between the interior rear view mirror C911-4, circuit VMM13 (YE/GN), harness side and ground; and between the interior rear view mirror C911-5, circuit RMM13 (BU), harness side and ground.
- **Is any voltage present?**
YES : REPAIR the circuit in question. TEST the system for normal operation.
NO : Go to R5.

R5 CHECK THE MICROPHONE CIRCUITS FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the inside rear view mirror, harness side and the ACM, harness side; and between the interior rear view mirror, harness side and ground as follows:

Interior Rear View Mirror Connector-Pin/Circuit	ACM Connector-Pin/Circuit
C911-4 VMM13 (YE/GN)	C240b-11 VMN01 (BU/BN)
C911-5 RMM13 (BU)	C240b-12 RMN01 (GN)

- **Is the resistance less than 5 ohms between the interior rear view mirror and the ACM, and greater than 10,000 ohms between the interior rear view mirror and ground?**
YES : Go to R6.
NO : REPAIR the circuit in question. TEST the system for normal operation.

R6 ISOLATE THE INTERIOR REAR VIEW MIRROR

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- Connect: ACM C240b
- Install a new interior rear view mirror. Refer to **REAR VIEW MIRRORS** article.
- Operate the audio system in radio tuner (AM/FM) mode.
- Press the VOICE button on the steering wheel controls, and attempt several voice commands.
- **Does the voice recognition operate correctly?**

YES : The fault was caused by an inoperative inside rear view mirror. The system is now operating correctly.

NO : Go to R7.

R7 CHECK FOR CORRECT ACM OPERATION

- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test S: The SYNC System Is Completely Inoperative

Normal Operation

The accessory protocol interface module (APIM) receives voltage at all times from circuit SBP03 (BU/RD), and ground through circuit GD148 (BK/YE). The wake-up signal is sent to the APIM by the audio control module (ACM) via the medium speed controller area network (MS-CAN).

- DTC U0184 (Lost Communication With Radio [ACM]) - set by the APIM when it is missing messages from the ACM over the MS-CAN for greater than 5 seconds.
- DTC U0196 (Lost Communication With Entertainment Control Module - Rear [AUX]) - set by the APIM when it is missing messages from the rear entertainment module (RETM) over the MS-CAN for greater than 5 seconds.
- DTC U0197 (Lost Communication With Telephone Control Module) - set by the ACM when it is missing messages from the APIM over the MS-CAN for greater than 5 seconds.

This pinpoint test is intended to diagnose the following:

- Audio system concern

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- Communication network concern
- Customer error
- Customer device
- APIM
- ACM

PINPOINT TEST S: THE SYNC SYSTEM IS COMPLETELY INOPERATIVE

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

S1 CHECK THE AUDIO SYSTEM OPERATION

- Press the AUX button repeatedly until "SYNC LINE IN" appears on the display (except navigation), or look for the presence of the "USER DEVICE" tab (navigation).
- **Is "SYNC LINE IN" available when pressing the AUX button (except navigation), or is the "USER DEVICE" tab present (navigation)?**

YES : Go to S2.

NO : Go to S3.

S2 CHECK THE SYNC SYSTEM OPERATION

- Connect: Known Good MP3 Device (To Audio Input Jack)

NOTE: Make sure no universal serial bus (USB) device is plugged-in, and no Bluetooth® device is active for this step.

- Using the ACM buttons, press AUX until "SYNC LINE IN" appears on the display (except navigation), or press MEDIA, select "USER DEVICE", and press "SOURCE" repeatedly until "LINE IN" appears (navigation).
- Using the known good MP3 device, attempt to play a file using the audio input jack.
- Disconnect: Known Good MP3 Device (From Audio Input Jack)
- Connect: Known Good USB Device (To USB Port)
- Using the ACM buttons, press AUX until "SYNC USB" appears on the display (except navigation), or press MEDIA, select "USER DEVICE", and press "SOURCE" repeatedly until "USB" appears (navigation).

NOTE: If a USB mass storage device is used to play the audio files, the SYNC system will only play audio files that do not have digital rights management (DRM) protection.

- Using a known good MP3 device or a mass storage device, attempt to play a file using the USB port.
- Disconnect: Known Good USB Device (From USB Port)

- Using a known good Bluetooth® device capable of streaming audio, attempt to pair to the APIM using Bluetooth®, and attempt to play an audio file using Bluetooth® connection. Refer to the Owner's Literature.
- **Do all of the SYNC inputs function correctly?**
YES : The SYNC system is operating correctly. REVIEW the system operation with the customer. If the customer device still does not operate correctly, the fault is with the customer device.
NO : If only some of the inputs are inoperative, go to **Symptom Chart - SYNC System** to diagnose the observed symptom.

If all inputs have poor quality/distorted sound, go to **Pinpoint Test U** .

S3 CHECK FOR SCAN TOOL COMMUNICATION

- Carry out the network test using the scan tool.
- **Does the APIM pass the network test?**
YES : Go to S4.
NO : REFER to **MODULE COMMUNICATIONS NETWORK** article.

S4 CHECK FOR CORRECT APIM OPERATION

- Key in OFF position.
- Disconnect the APIM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pints
- Connect the APIM connector and make sure it seats correctly.
- Wait 2 minutes for the APIM to re-initialize.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**
YES : INSTALL a new APIM. REFER to **Accessory Protocol Interface Module (APIM)**. TEST the system for normal operation.
NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test T: Unable To Pair Bluetooth® Device

Normal Operation

When a new Bluetooth® device is added, the accessory protocol interface module (APIM) and the Bluetooth® device must be paired together. Most Bluetooth® devices will pair with the SYNC system, although functionality will vary. To determine if a Bluetooth® device is supported, retrieve the consumer interface processor (CIP) software level using the **Accessory Protocol Interface Module (APIM) Software Level Check** , and verify the customer device is on the compatibility list for the current CIP software level.

Pairing a Bluetooth® device is accomplished through the "Add Device" selection of the phone menu. When pairing a device, the SYNC system generates a unique personal identification number (PIN) that must be entered on the Bluetooth® device in order for the pairing process to be successful. There are also some device-specific actions that must take place. For additional information on the pairing process, refer to the Owner's Literature.

This pinpoint test is intended to diagnose the following:

- Incompatible Bluetooth® device
- Customer error
- Customer Bluetooth® device
- APIM

PINPOINT TEST T: UNABLE TO PAIR BLUETOOTH® DEVICE

T1 CHECK THE BLUETOOTH® CONNECTION

NOTE: Carrying out a Master Reset will return all preference settings to the factory defaults, erase all the phone book and call histories and delete any devices paired with the SYNC system.

NOTE: Refer to www.syncmyride.com for a list of compatible devices.

- Using a known good Bluetooth® device, attempt to pair to the SYNC system using Bluetooth®. Refer to the Owner's Literature.
- Enter the following diagnostic mode on the diagnostic tool: APIM DataLogger
- Monitor the BT_PAIR (Bluetooth® device paired) and BT_CONN (Bluetooth® device connected) PIDs.
- **Do the BT_PAIR and BT_CONN PIDs both read "Yes"?**

YES : The SYNC system is operating correctly at this time. CARRY OUT a SYNC system Master Reset.

For vehicles without navigation, PRESS the "PHONE" button on the steering wheel controls, SCROLL until "SYSTEM SETTINGS" displays on the screen, PRESS "OK", SCROLL until "ADVANCED" displays on the screen, PRESS "OK", SCROLL until "MASTER RESET" displays on the screen, and PRESS "OK" twice.

For vehicles with navigation, PRESS "PHONE" on the ACM display, SELECT the "SETTINGS" tab, PRESS "ADVANCED", SELECT "Master Reset", and PRESS "YES".

REVIEW the pairing process with the customer. If the customer device still does not pair, the fault is with the customer device.

NO : Go to T2.

T2 CHECK FOR CORRECT APIM OPERATION

- Key in OFF position.
- Disconnect the APIM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pints
- Connect the APIM connector and make sure it seats correctly.
- Wait 2 minutes for the APIM to re-initialize.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new APIM. REFER to **Accessory Protocol Interface Module (APIM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test U: The Universal Serial Bus (USB) Port Is Inoperative/Does Not Operate Correctly**Normal Operation**

The universal serial bus (USB) port is connected to the accessory protocol interface module (APIM) through the USB cable. The USB port can be used for mass storage devices, or for connecting a media device.

If supported by the user device, the USB can provide charging. Because of this feature, when a USB device is plugged into the USB port, the SYNC system does not automatically switch to the device.

The USB cable and port are not serviceable separately.

- DTC U261C (USB #1 Device Error) - sets when the APIM detects an over-current or over-temperature condition in the USB circuit. This can be caused by a fault in the USB cable or port, or by the customer USB device.

This pinpoint test is intended to diagnose the following:

- Customer USB device
- USB cable and port
- APIM

PINPOINT TEST U: THE UNIVERSAL SERIAL BUS (USB) PORT IS INOPERATIVE/DOES NOT OPERATE CORRECTLY**U1 CHECK THE USB CONNECTION**

- Connect: Known Good USB Device (To USB Port)

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- Using the ACM buttons, press AUX until "SYNC USB" appears on the display (except navigation), or press AUX until "SYNC USB" appears on the display (except navigation), or press MEDIA, select "USER DEVICE", and press "SOURCE" repeatedly until "USB" appears (navigation)

NOTE: **If a USB mass storage device is used to play the audio files, SYNC will only play audio files that do not have digital rights management (DRM) protection.**

- Using a known good MP3 device or a mass storage device, attempt to play a file using the USB port.
- **Does the known good device successfully play an audio file using the USB connection?**
YES : The SYNC system is operating correctly. REVIEW the operation of the USB port with the customer. If the customer device still does not operate correctly, the fault is with the customer device.
NO : Go to U2.

U2 INSPECT THE USB CABLE

- Key in OFF position.
- Disconnect: USB Cable (At APIM)
- Inspect the USB cable for damage.
- Connect: USB Cable (At APIM)
- Key in ON position.
- Using the ACM buttons, press AUX until "SYNC USB" appears on the display (except navigation), or press AUX until "SYNC USB" appears on the display (except navigation), or press MEDIA, select "USER DEVICE", and press "SOURCE" repeatedly until "USB" appears (navigation).
- Using a known good MP3 device or a mass storage device, attempt to play a file using the USB port.
- **Does the known good device successfully play an audio file using the USB connection?**
YES : The concern was caused by the USB connection not being seated correctly. The system is now operating correctly.
NO : Go to U3.

U3 ISOLATE THE USB CABLE AND PORT

- Disconnect: Known Good USB Device (From USB Port)
- Install a new USB cable and port. Refer to **Universal Serial Bus (USB) Cable and Port.**
- Connect: Known Good USB Device (To USB Port)
- Using the ACM buttons, press AUX until "SYNC USB" appears on the display (except navigation), or press AUX until "SYNC USB" appears on the display (except navigation), or press MEDIA, select "USER DEVICE", and press "SOURCE" repeatedly until "USB" appears (navigation).
- Using a known good MP3 device or a mass storage device, attempt to play a file using the USB port.
- **Does the known good device successfully play an audio file using the USB connection?**
YES : The concern was caused by an inoperative USB cable. The system is now operating

correctly.

NO : Go to U4.

U4 CHECK FOR CORRECT APIM OPERATION

- Key in OFF position.
- Disconnect the APIM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the APIM connector and make sure it seats correctly.
- Wait 2 minutes for the APIM to re-initialize.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new APIM. REFER to **Accessory Protocol Interface Module (APIM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test V: The SYNC System Audible Prompts Are Inoperative/Do Not Operate Correctly

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

The text-to-speech (TTS) and voice prompt features speak certain text information and interaction requests in order to minimize driver distraction by having to look at the audio control module (ACM) while driving. The ring tone alerts the driver to an incoming call.

Audible prompts can range from a simple tone to more elaborate spoken text, based on the customer setting. When interaction mode is set to standard, detailed guidance is provided. When interaction mode is set to advanced, most prompts are tones only and minimal audible guidance is provided. Refer to the Owner's Literature for further information on voice interaction.

The audio signals for the TTS and voice prompt features, the ring tones, and audio from the outside device during a phone call, are sent from a accessory protocol interface module (APIM) to the ACM through circuits VMN14 (WH/VT) and RMN14 (GY/BN).

This pinpoint test is intended to diagnose the following:

- Customer setting
- Wiring, terminals or connectors
- ACM

- APIM

PINPOINT TEST V: THE SYNC SYSTEM AUDIBLE PROMPTS ARE INOPERATIVE/DO NOT OPERATE CORRECTLY

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

V1 CHECK THE AUDIBLE PROMPT SETTING

- Operate the audio system in SYNC mode.
- Verify the audible prompts are enabled. Refer to "SYNC Voice Recognition Feature" in the Owner's Literature.
- Press the VOICE button on the steering wheel controls and observe the SYNC voice prompt.
- **Does the SYNC system produce a voice prompt correctly?**
YES : The concern was caused by a customer setting. ADVISE the customer on the operation of the TTS feature.
NO : Go to V2.

V2 CHECK THE AUDIBLE PROMPT CIRCUITS FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: ACM C240b
- Disconnect: APIM C3342
- Key in ON position.

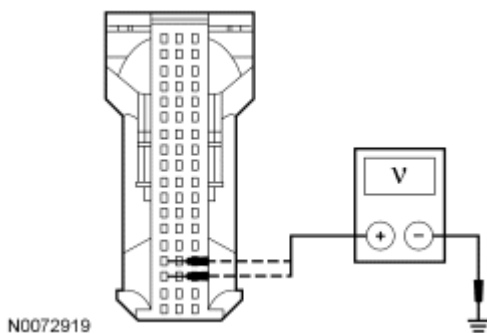


Fig. 25: Checking Audible Prompt Circuits For A Short To Voltage
 Courtesy of FORD MOTOR CO.

- Measure the voltage between the APIM C3342-3, circuit VMN14 (WH/VT), harness side and ground; and between the APIM C3342-4, circuit RMN14 (GY/BN), harness side and ground.
- **Is any voltage present?**
YES : REPAIR the circuit in question. TEST the system for normal operation.
NO : Go to V3.

V3 CHECK THE AUDIBLE PROMPT CIRCUITS FOR AN OPEN OR SHORT TO GROUND

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- Key in OFF position.
- Measure the resistance between the APIM, harness side and the ACM, harness side; and between the APIM, harness side and ground as follows:

APIM Connector- Pin	ACM Connector- Pin	Circuit
C3342-3	C240b-4	VMN14 (WH/VT)
C3342-4	C240b-5	RMN14 (GY/BN)

- **Is the resistance less than 5 ohms between the APIM and the ACM, and greater than 10,000 ohms between the APIM and ground?**

YES : Go to V4.

NO : REPAIR the circuit in question. TEST the system for normal operation.

V4 CHECK FOR A VOLTAGE SIGNAL FROM THE APIM

- Connect: ACM C240b
- Connect: APIM C3342
- Operate the audio system in SYNC mode, and verify the audible prompts are enabled. Refer to "Voice Prompts" in the Owner's Literature.
- While pressing the VOICE button repeatedly, measure the AC voltage by backprobing between the ACM C240b-4, circuit VMN14 (WH/VT), harness side and the ACM C240b-5, circuit RMN14 (GY/BN), harness side.
- **Is an AC voltage produced each time the VOICE button is pressed?**

YES : Go to V5.

NO : Go to V6.

V5 CHECK FOR CORRECT ACM OPERATION

- Key in OFF position.
- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose

or corroded connector.

V6 CHECK FOR CORRECT APIM OPERATION

- Key in OFF position.
- Disconnect the APIM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the APIM connector and make sure it seats correctly.
- Wait 2 minutes for the APIM to re-initialize.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new APIM. REFER to **Accessory Protocol Interface Module (APIM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test W: Voice Recognition Is Inoperative/Does Not Operate Correctly - Vehicles With The SYNC System

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

When the VOICE button (represented by an icon of someone speaking) is pressed, it changes the reference voltage on circuit VMN54 (BU/OG), and the audio system enters voice recognition mode. If the vehicle is equipped with navigation, the VOICE switch is wired to the audio control module (ACM). If the vehicle is not equipped with navigation, the VOICE switch is wired to the accessory protocol interface module (APIM). A microphone located in the interior rear view mirror receives the voice command and sends a signal to the APIM through circuits VMM13 (YE/GN) and RMM13 (BU). Power to the interior rear view mirror is provided through circuit CBP41 (BU). Ground is provided through circuit GD133 (BK). The microphone is also used to detect outgoing audio during a phone call.

The microphone test (with navigation only) is available through the scan tool. Running this test causes the ACM to produce a test tone. If the system is operating correctly, the microphone detects the tone and produces a signal to the ACM indicating the tone was heard.

If the vehicle is equipped with navigation, the APIM receives the microphone input, and also broadcasts it to the ACM through circuits VMN01 (BU/BN) and RMN01 (GN). This allows voice commands specific to the navigation system to be handled directly by the ACM. The first voice commands spoken determine which system handles the voice commands.

- DTC B1038 (Microphone Input Circuit Failure) - set by the APIM during the on-demand self-test if the

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APIM does not detect the microphone.

- DTC B2633 (Driver-Front Microphone Circuit Failure) - set by the ACM during the microphone test when the ACM does not receive an adequate signal from the microphone when the test tone is sounded.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Steering wheel controls
- Microphone (part of the interior rear view mirror)
- ACM (if equipped with navigation)
- APIM

PINPOINT TEST W: VOICE RECOGNITION IS INOPERATIVE/DOES NOT OPERATE CORRECTLY - VEHICLES WITH THE SYNC SYSTEM

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

W1 VERIFY THE OPERATION OF THE STEERING WHEEL CONTROLS

- Operate the audio system in radio tuner (AM/FM) mode.
- Press the VOICE button on the steering wheel controls.
- **Does the audio system enter voice recognition mode?**

YES : Go to W2.

NO : Go to **Pinpoint Test F** to diagnose the steering wheel controls system.

W2 CHECK THE OPERATION OF THE VOICE RECOGNITION

- Press the VOICE button on the steering wheel controls.
- Wait for the SYNC system to acknowledge the VOICE button being pressed.
- While still in voice recognition mode, speak the command "PHONE", and observe the audio system.
- **Does the SYNC system acknowledge the command "PHONE"?**

YES : If the vehicle is equipped with navigation, go to W3.

Otherwise, the system is operating correctly at this time. ADVISE the customer on the correct operation of the voice recognition system.

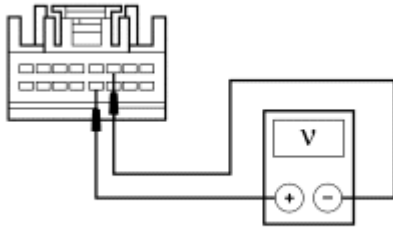
NO : Go to W6.

W3 CHECK THE MICROPHONE SIGNAL TO THE ACM

- Key in OFF position.
- Disconnect: ACM C240b

NOTE: ACM C240a must remain connected for this step to produce accurate results.

- Operate the audio system in radio tuner (AM/FM) mode.
- Press the VOICE button on the steering wheel controls.



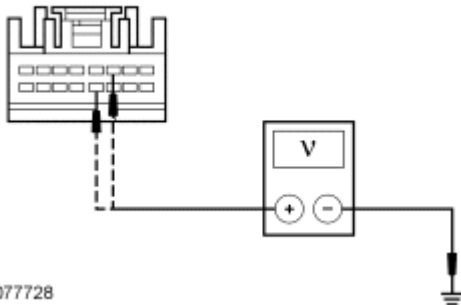
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Fig. 26: Checking Microphone Signal To ACM
Courtesy of FORD MOTOR CO.

- While speaking a command, measure the AC voltage between the ACM C240b-11, circuit VMN01 (BU/BN), harness side and the ACM C240b-12, circuit RMN01 (GN), harness side.
- **Is a fluctuating AC voltage present?**
YES : Go to W12.
NO : Go to W4.

W4 CHECK THE MICROPHONE CIRCUITS TO THE ACM FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: APIM C3342
- Key in ON position.



N0077728

Fig. 27: Checking Microphone Circuits To ACM For A Short To Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between the ACM C240b-11, circuit VMN01 (BU/BN), harness side and ground; and between the ACM C240b-12, circuit RMN01 (GN), harness side and ground.

- **Is any voltage present?**

YES : REPAIR the circuit in question. TEST the system for normal operation.

NO : Go to W5.

W5 CHECK THE MICROPHONE CIRCUITS TO THE ACM FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the ACM, harness side and the APIM, harness side; and between the ACM, harness side and ground as follows:

ACM Connector- Pin	APIM Connector- Pin	Circuit
C240b-11	C3342-12	VMN01 (BU/BN)
C240b-12	C3342-13	RMN01 (BU)

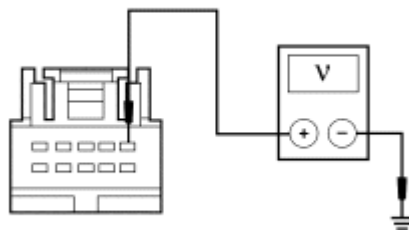
- **Is the resistance less than 5 ohms between the ACM and the APIM, and greater than 10,000 ohms between the ACM and ground?**

YES : Go to W11.

NO : REPAIR the circuit in question. TEST the system for normal operation.

W6 CHECK CIRCUIT CBP41 (BU) FOR VOLTAGE

- Key in OFF position.
- Disconnect: Interior Rear View Mirror C9012
- Key in ON position.



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Fig. 28: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

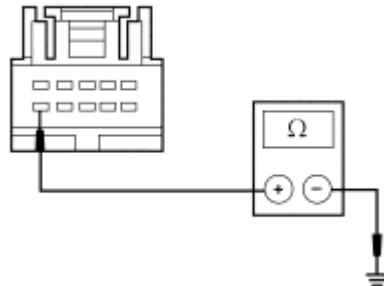
- Measure the voltage between the interior rear view mirror C9012-1, circuit CBP41 (BU), harness side and ground.
- **Is the voltage greater than 10 volts?**

YES : Go to W7.

NO : VERIFY the smart junction box (SJB) fuse 41 (15A) is OK. If OK, REPAIR the circuit. TEST the system for normal operation. If not OK, REFER to appropriate SYSTEM WIRING DIAGRAMS article to identify the possible sources of the circuit short.

W7 CHECK CIRCUIT GD133 (BK) FOR AN OPEN

- Key in OFF position.



N0057336

Fig. 29: Checking Circuit For Open
Courtesy of FORD MOTOR CO.

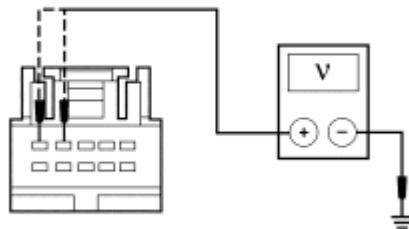
- Measure the resistance between the interior rear view mirror C9012-10, circuit GD133 (BK), harness side and ground.
- **Is the resistance less than 5 ohms?**

YES : Go to W8.

NO : REPAIR the circuit. TEST the system for normal operation.

W8 CHECK THE MICROPHONE CIRCUITS FOR A SHORT TO VOLTAGE

- Disconnect: APIM C3342
- Key in ON position.



N0057337

Fig. 30: Checking Circuits For Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between the interior rear view mirror C9012-4, circuit VMM13 (YE/GN), harness side and ground; and between the interior rear view mirror C9012-5, circuit RMM13 (BU), harness side and ground.
- **Is any voltage present?**

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YES : REPAIR the circuit in question. TEST the system for normal operation.

NO : Go to W9.

W9 CHECK THE MICROPHONE CIRCUITS FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the inside rear view mirror, harness side and the APIM, harness side; and between the interior rear view mirror, harness side and ground as follows:

Interior Rear View Mirror Connector-Pin	APIM Connector-Pin	Circuit
C9012-4	C3342-5	VMM13 (YE/GN)
C9012-5	C3342-6	RMM13 (BU)

- **Is the resistance less than 5 ohms between the interior rear view mirror and the APIM, and greater than 10,000 ohms between the interior rear view mirror and ground?**

YES : Go to W10.

NO : REPAIR the circuit in question. TEST the system for normal operation.

W10 ISOLATE THE INTERIOR REAR VIEW MIRROR

- Connect: APIM C3342
- Install a new interior rear view mirror. Refer to **REAR VIEW MIRRORS** article.
- Operate the audio system in SYNC mode.
- Press the VOICE button on the steering wheel controls, and attempt several voice commands.
- **Does the voice recognition operate correctly?**

YES : The fault was caused by an inoperative inside rear view mirror. The system is now operating correctly.

NO : Go to W11.

W11 CHECK FOR CORRECT APIM OPERATION

- Disconnect the APIM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the APIM connector and make sure it seats correctly.
- Wait 2 minutes for the APIM to re-initialize.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new APIM. REFER to **Accessory Protocol Interface Module (APIM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

W12 CHECK FOR CORRECT ACM OPERATION

- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test X: Poor Quality/Distorted/No Sound While In SYNC Mode

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

The accessory protocol interface module (APIM) sends left and right channel analog audio signals through circuits VME52 (BU), RME52 (GY/OG), VME53 (VT/GN), and RME53 (BN/WH). If the vehicle is equipped with a DVD entertainment system, these audio signals are sent to the rear entertainment module (RETM), then to the audio control module (ACM). In vehicles without a DVD entertainment system, the audio signals are sent directly to the ACM.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- RETM (if equipped)
- ACM
- APIM

PINPOINT TEST X: POOR QUALITY/DISTORTED/NO SOUND WHILE IN SYNC MODE

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

X1 VERIFY THE OPERATION OF THE SYNC AUDIO SOURCES

- Connect: Known Good MP3 Device (To Audio Input Jack)

NOTE: **Make sure no universal serial bus (USB) device is plugged-in, and no Bluetooth® device is active for this step.**

- Using the ACM buttons, press AUX until "SYNC LINE IN" appears on the display (except navigation), or press MEDIA, select "USER DEVICE", and press "SOURCE" repeatedly until "LINE IN" appears (navigation).
- Using a known good MP3 device, attempt to play a file using the audio input jack, and listen to the audio output.
- Disconnect: Known Good MP3 Device (From Audio Input Jack)
- Connect: Known Good USB Device (To USB Port)
- Using the ACM buttons, press AUX until "SYNC USB" appears on the display (except navigation), or press MEDIA, select "USER DEVICE", and press "SOURCE" repeatedly until "USB" appears (navigation).

NOTE: **If a USB mass storage device is used to play the audio files, SYNC will only play audio files that do not have digital rights management (DRM) protection.**

- Using a known good MP3 device or a mass storage device, attempt to play a file using the USB port.
- Disconnect: Known Good USB Device (From USB Port)
- Using a known good Bluetooth® device capable of streaming audio, attempt to pair to the APIM using Bluetooth®, and attempt to play an audio file using the Bluetooth® connection. Refer to the Owner's Literature.
- **Is there poor quality/distorted/no sound from each SYNC audio source?**

YES : If the vehicle is equipped with a DVD entertainment system, go to X2.

Otherwise, go to X3.

NO : If the concern is only with some of the audio sources, go to **Symptom Chart - SYNC System** to diagnose the observed symptom.

If all audio sources operate correctly, the concern is with the customer device.

X2 CHECK THE OPERATION OF THE DVD AUDIO

NOTE: **Make sure a known good DVD is inserted in the RETM before carrying out this step.**

- Operate the audio system in DVD mode.

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2008 ACCESSORIES & BODY, CAB Information and Entertainment Systems - Edge & MKX

- Observe the audio output.
- **Is there poor quality/distorted sound while in DVD mode?**

YES : Go to **Pinpoint Test K** .

NO : Go to X3.

X3 CHECK THE CIRCUITS FROM THE APIM FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: ACM C240b (Without DVD Entertainment)
- Disconnect: RETM C949 (With DVD Entertainment)
- Disconnect: APIM C3342
- Measure the voltage between the APIM pins, harness side as follows:

Connector-Pin	Circuit
C3342-23	VME53 (VT/GN)
C3342-24	RME53 (BN/WH)
C3342-25	VME52 (BU)
C3342-26	RME52 (GY/OG)

- **Is any voltage present?**

YES : REPAIR the circuit in question. TEST the system for normal operation.

NO : Go to X4.

X4 CHECK THE CIRCUITS FROM THE APIM FOR AN OPEN OR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between the APIM, harness side, and the ACM or RETM (if equipped with DVD entertainment), harness side; and between the APIM, harness side and ground as follows:

APIM Connector- Pin/Circuit	ACM or RETM Connector- Pin/Circuit	Circuit
C3342-23 VME53 (VT/GN)	C240b-9 or C949-20	VME53 (VT/GN)
C3342-23 RME53 (BN/WH)	C240b-10 or C949-19	RME53 (BN/WH)
C3342-23	C240b-1 or	VME52

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VME52 (BU)	C949-4	(BU)
C3342-23 RME52 (GN/OG)	C240b-2 or C949-3	RME52 (GN/OG)

- **Is the resistance less than 5 ohms between the APIM and the ACM (or RETM), and greater than 10,000 ohms between the APIM and ground?**

YES : REPAIR the circuit in question. TEST the system for normal operation.

NO : Go to X5.

X5 CHECK FOR CORRECT APIM OPERATION

- Connect: All Disconnected Connectors
- Disconnect the APIM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the APIM connector and make sure it seats correctly.
- Wait 2 minutes for the APIM to re-initialize.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new APIM. REFER to **Accessory Protocol Interface Module (APIM)**. TEST the system for normal operation. If the concern is still present and the vehicle is equipped with a DVD entertainment system, go to X6. If the concern is still present and the vehicle is not equipped with a DVD entertainment system, go to X7.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

X6 CHECK FOR CORRECT RETM OPERATION

- Disconnect the RETM connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the RETM connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new RETM. REFER to **Rear Entertainment Module (RETM)**. TEST the system for normal operation. If the concern is still present, go to X7.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

X7 CHECK FOR CORRECT ACM OPERATION

- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test Y: The Speed Sensitive Volume Does Not Operate Correctly

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Audio System/Navigation for schematic and connector information.

Normal Operation

The speed sensitive volume function adjusts the volume based on the vehicle speed sensor (VSS) signal from the instrument cluster (IC). The IC does not generate the VSS signal; it gateways the signal from the PCM.

For vehicles without THX® sound, the audio control module (ACM) receives the VSS signal from the IC. For vehicles with THX® sound, the audio digital signal processing (DSP) module receives the VSS signal from the IC and controls the speed sensitive volume based on the ACM setting.

- DTC U0155 (Lost Communication With Instrument Panel Cluster Control [IC] Module) - sets when the VSS signal is lost for greater than 5 seconds. When the signal is lost, the ACM turns the speed sensitive volume feature off. The smart junction box (SJB) and HVAC module also receive this signal, and should demonstrate symptoms if the VSS signal is lost. If no symptoms are present, this DTC can be ignored, as it may have been set by a low battery voltage condition.
- DTC U0184 (Lost Communication With Radio [ACM]) - set by the DSP module when expected medium speed controller area network (MS-CAN) messages are missing from the ACM lost for greater than 5 seconds. This includes the speed sensitive volume setting. If the speed sensitive volume setting message is missing, the DSP module turns the speed sensitive volume feature off.

This pinpoint test is intended to diagnose the following:

- Speed sensitive volume setting
- VSS signal concern
- ACM

PINPOINT TEST Y: THE SPEED SENSITIVE VOLUME DOES NOT OPERATE CORRECTLY

NOTE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

Y1 CHECK THE SPEEDOMETER OPERATION

- Drive the vehicle and observe the speedometer.
- **Does the speedometer operate correctly?**

YES : Go to Y2.

NO : REFER to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER, AND WARNING CHIMES** article to diagnose the inoperative speedometer.

Y2 CHECK THE SPEED SENSITIVE VOLUME SETTING

- Turn the speed sensitive volume off.
 - For vehicles without navigation, press MENU until the speed sensitive volume prompt is reached, then press TUNE DOWN until the feature is turned off.
 - For vehicles with navigation, press SOUND, select the SCV tab, and turn the feature off.
- Operate the audio system in radio tuner (AM/FM) mode.
- Drive the vehicle at various speeds and observe the speaker volume.
- Set the speed sensitive volume to maximum compensation.
 - For vehicles without navigation, press MENU until the speed sensitive volume prompt is reached, then press TUNE UP until the feature is set to setting "7".
 - For vehicles with navigation, press SOUND, select the SCV tab, then press "7".
- Operate the audio system in radio tuner (AM/FM) mode.
- Drive the vehicle at various speeds and observe the speaker volume.
- **Does the volume remain constant with the speed sensitive volume turned off, and increase and decrease with vehicle speed with the speed sensitive volume set to maximum?**

YES : The system is operating correctly at this time. ADVISE the customer on the correct operation of the speed sensitive volume feature.

NO : Go to Y3.

Y3 CHECK FOR DTC U0155 (ACM) OR U0184 (DSP)

- Key in ON position.
- Using the scan tool, clear any ACM and DSP module continuous DTCs.
- Key in OFF position.
- Key in ON position.
- Wait at least 10 seconds.
- Retrieve the ACM and DSP module continuous DTCs.
- **Is DTC U0155 (ACM) or U0184 (DSP) present?**

YES : Go to Y4.

NO : If the vehicle is equipped with THX® sound, go to Y5.

Otherwise, go to Y6.

Y4 CHECK FOR COMMUNICATION WITH THE ACM AND INSTRUMENT CLUSTER (IC)

- Carry out the network test using the scan tool.
- **Does the scan tool communicate with the ACM and the IC?**

YES : If the vehicle is equipped with THX® sound, go to Y5.

Otherwise, go to Y6.

NO : REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose no communication with the ACM or IC.

Y5 CHECK FOR CORRECT DSP MODULE OPERATION

- Disconnect all the DSP module connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the DSP module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new APIM. REFER to **Accessory Protocol Interface Module (APIM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Y6 CHECK FOR CORRECT ACM OPERATION

- Disconnect all the ACM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the ACM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ACM. REFER to **Audio Control Module (ACM)**. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

GENERAL PROCEDURES

AUDIO UNIT PART NUMBER RETRIEVAL**Audio Systems Without Navigation**

NOTE: The terms "Audio Control Module (ACM)" and "audio unit" refer to the same component.

1. Turn the ACM on.
2. Press and hold preset buttons 3 and 6 for 3 seconds. The speaker walk-around test will begin.
3. Before the speaker walk-around test is complete, press preset button 6.

NOTE: This step will cause the ACM to auto-scroll through configuration 1, configuration 2, EEPROM number, and ACM part number.

4. Press the TUNE UP button.

NOTE: Step 4 may need to be repeated until the ACM part number can be recorded.

5. Record the ACM part number.
6. Turn the ACM off.

Navigation Audio System

1. Turn the ACM on.
2. Press and hold preset buttons 3 and 6 for 3 seconds. The speaker walk-around test will begin.
3. Before the speaker walk-around test is complete, press END TEST on the display screen.
4. Select SYSTEM INFO from the menu.
5. Record the ACM part number.
6. Turn the ACM off.

SATELLITE RADIO RECEIVER ELECTRONIC SERIAL NUMBER (ESN) RETRIEVAL**Audio Systems Without Navigation**

1. Operate the audio system in satellite radio mode.
2. Press and hold the AUX button and preset button 1, simultaneously.
 - The satellite radio receiver Electronic Serial Number (ESN) will be displayed on the screen.
3. Record the satellite radio receiver ESN.
4. Turn the audio system off.

Audio Systems With Navigation

2008 Ford Edge SE

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1. Turn the audio system on.
2. Press the MENU button.
3. Select the SYSTEM INFO tab.

NOTE: The satellite radio receiver ESN is shown on the lower left corner of the screen with the text "SR ESN:".

4. Record the satellite radio receiver ESN.
5. Turn the audio system off.

ACCESSORY PROTOCOL INTERFACE MODULE (APIM) SOFTWARE LEVEL CHECK

Special Tools

Illustration	Tool Name	Tool Number
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	
 ST3051-A	Universal Serial Bus (USB) Male-A to Male-A Cable	CCMUSB2-AM-AM-10 or equivalent

1. Turn the Audio Control Module (ACM) on.
2. Connect the scan tool to the Data Link Connector (DLC).
3. Connect one end of the Universal Serial Bus (USB) male-A to male-A cable to the scan tool.
4. Connect the other end of the USB male-A to male-A cable to the vehicle USB port.
5. From the technician service publication website, run On-Line Automotive Service Information System (OASIS) using Quick Start or by manually entering the Vehicle Identification Number (VIN).
6. From the OASIS tab, select the "Sync/APIM" bullet.
7. Select the "Read APIM" button to verify the current Accessory Protocol Interface Module (APIM) software level.
 - The display shows both the Vehicle Interface Processor (VIP) and Consumer Interface Processor (CIP) software levels.
8. The scan tool displays the following information:
 - Last Recorded State - Hardware
 - VIN: vehicle identification number associated with the current APIM
 - Installed date: date the APIM was installed

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- Radio: currently identified ACM in the vehicle
 - HW Part No.: APIM hardware part number
 - Un-installed date: date (if any) the APIM was uninstalled
 - S/N: APIM serial number
 - Last Recorded State - Software
 - Date/Time: date and time of last recorded software installation
 - VIP: VIP software that was installed at that time
 - CIP: CIP software that was installed at that time
 - Description: a description of the content of the software revision
 - History - Software
 - Date/Time: date and time of any recorded software installation
 - VIP: VIP software that was installed at that time
 - CIP: CIP software that was installed at that time
 - Description: a description of the content of the software revision
 - Available Software for Programming
 - Select: allows the software package to be selected
 - Lineage: the original software release, if the software available is a revision
 - VIP: VIP software level that is available with the selection
 - CIP: CIP software level that is available with the selection
 - Description: a description of the content of the software revision
9. Click a CIP software level to view the device compatibility list associated with the CIP software level, if desired.
10. To exit the APIM software level check, disconnect the scan tool from the DLC and the USB port, or exit the OASIS screen.

REMOVAL AND INSTALLATION

AUDIO DIGITAL SIGNAL PROCESSING (DSP) MODULE

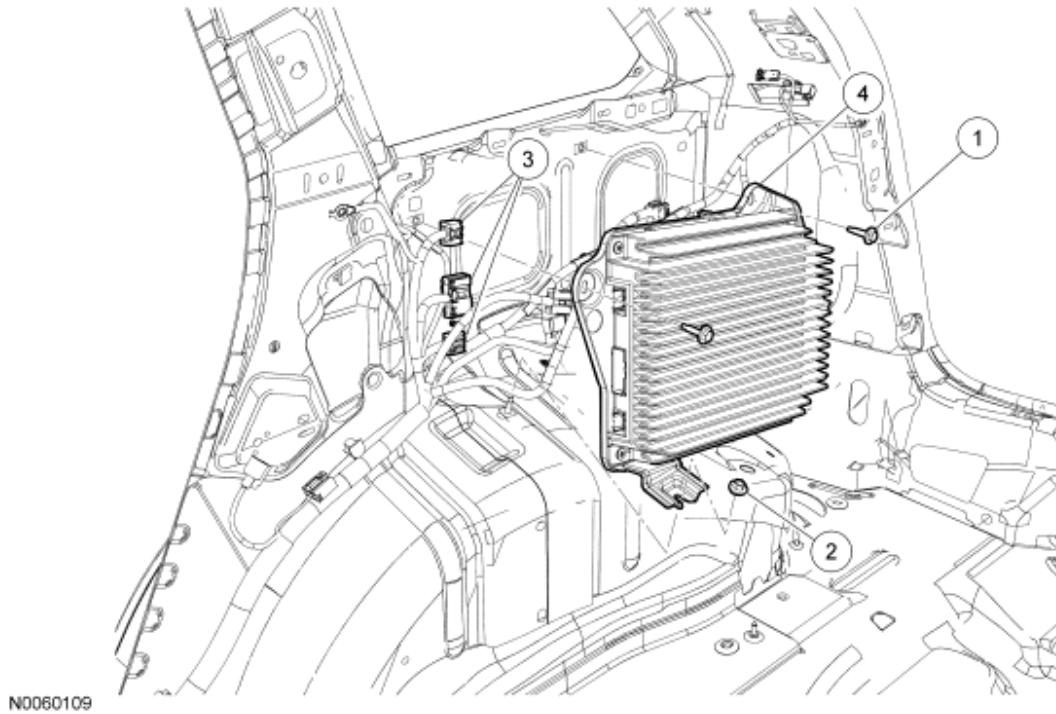


Fig. 31: Exploded View Of Audio Amplifier - THX(R)
 Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W505425-S	Audio digital signal processing (DSP) module screws (2 required)
2	N811309-S	Audio DSP module nut
3	-	Audio DSP module electrical connectors (part of 14A005)
4	18A849	Audio DSP module

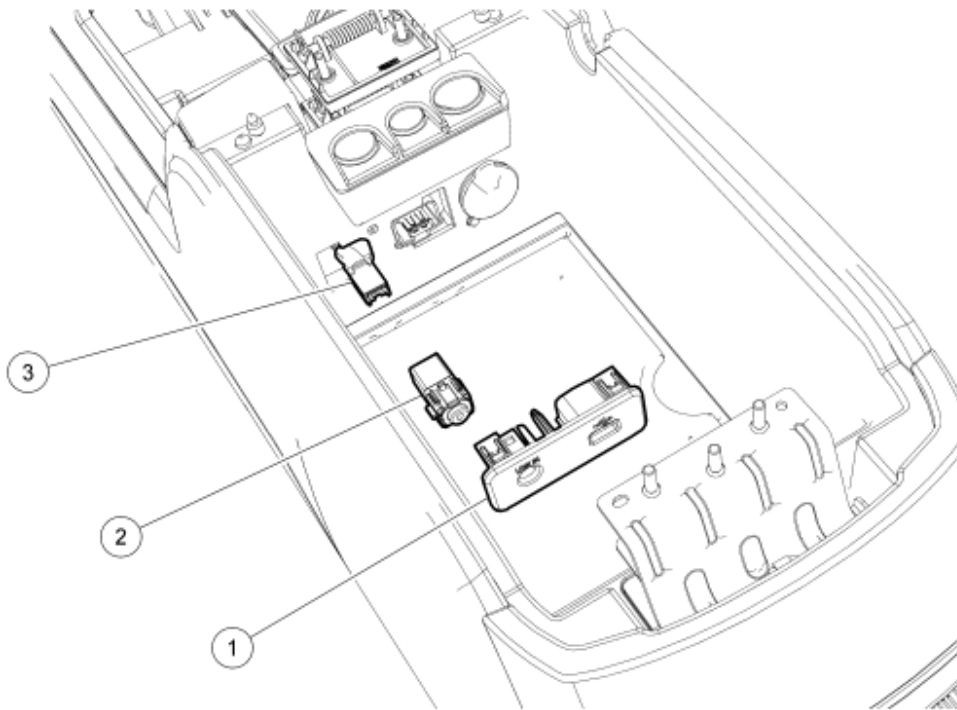
REMOVAL AND INSTALLATION

NOTE: Module configuration is only required if a new audio digital signal processing (DSP) module is being installed.

1. Upload the audio DSP module configuration information to the scan tool. For additional information, refer to Programmable Module Installation (PMI) in **MODULE CONFIGURATION** article.
2. Remove the subwoofer speaker. For additional information, refer to **Subwoofer Speaker**.
3. Disconnect the audio DSP module electrical connectors.
4. Remove the 2 screws, the nut, and the audio DSP module.
5. To install, reverse the removal procedure.
 - Download the configuration information to the audio DSP module. For additional information, refer to PMI in **MODULE CONFIGURATION** article.

AUDIO INPUT JACK

NOTE: Vehicle equipped with SYNC shown, others similar.



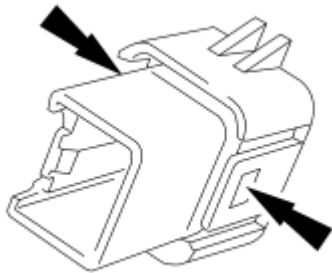
N0077732

Fig. 32: Identifying Audio Input Jack Connectors
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Audio input jack bezel (part of 78045A06)
2	19A164	Audio input jack
3	-	Audio input jack electrical connector (part of 14B079)

REMOVAL AND INSTALLATION

1. Using a suitable tool, pry the audio input jack bezel out from the floor console storage bin.
 - Disconnect the universal serial bus (USB) connector, if equipped.
2. Disconnect the audio input jack electrical connector.
3. Release the tabs and remove the audio input jack.

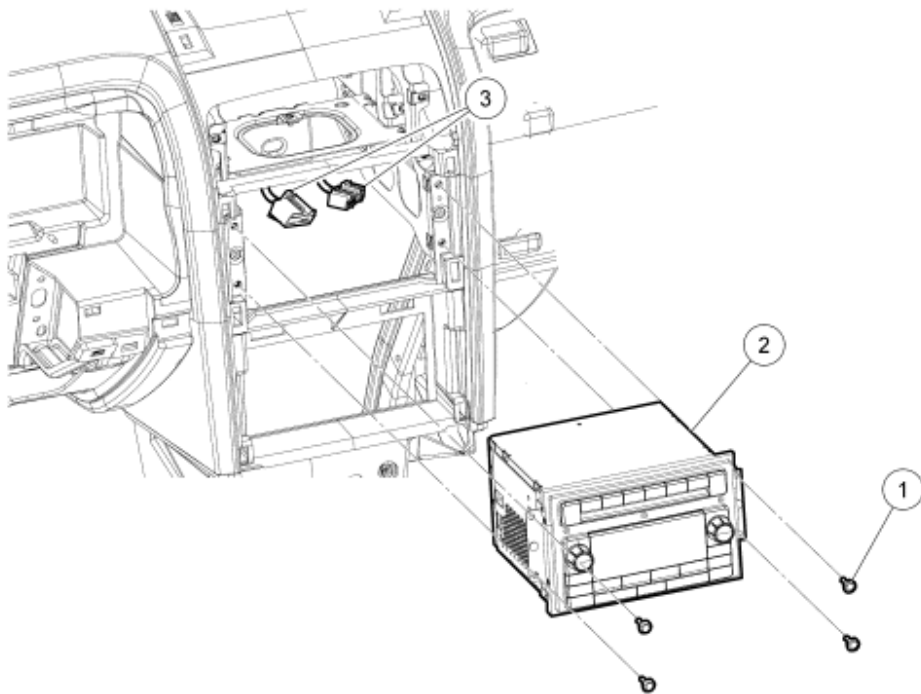


N0056277

Fig. 33: Identifying Tabs On Audio Input Jack
Courtesy of FORD MOTOR CO.

4. To install, reverse the removal procedure.

AUDIO CONTROL MODULE (ACM)



N0060099

Fig. 34: Exploded View Of Audio Unit
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W707628	Audio control module (ACM) screws (4 required)
2	18C815	ACM
3	-	ACM electrical connectors (part of 14401 or 18812)

REMOVAL AND INSTALLATION

NOTE: It is not necessary to remove the audio control module (ACM) to retrieve the part number. For additional information, refer to Audio Unit Part Number Retrieval.

NOTE: Module configuration is only required if a new ACM is being installed.

1. Upload the ACM configuration information to the scan tool. For additional information, refer to MODULE CONFIGURATION article.
2. Remove the instrument panel center finish panel. For additional information, refer to INSTRUMENT PANEL AND CONSOLE article.
3. Remove the 4 screws and the ACM.
 - Disconnect the electrical connectors and the antenna cable.
4. To install, reverse the removal procedure.
 - Download the configuration information to the ACM. For additional information, refer to MODULE CONFIGURATION article.

SATELLITE DIGITAL AUDIO RECEIVER SYSTEM (SDARS) MODULE

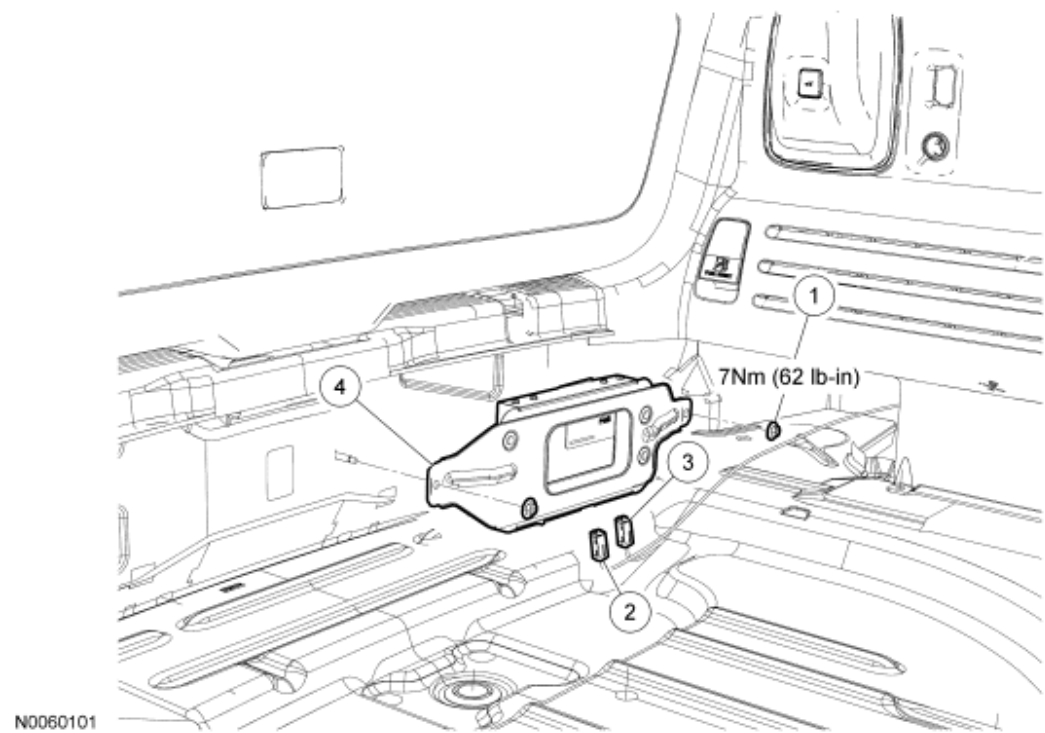


Fig. 35: Exploded View Of Satellite Radio Receiver With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
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1	W707293-S	Satellite digital audio receiver system (SDARS) module nuts (2 required)
2	19A397	Satellite radio antenna connector
3	-	SDARS module electrical connector (part of 14A005)
4	18C960	SDARS module

REMOVAL AND INSTALLATION

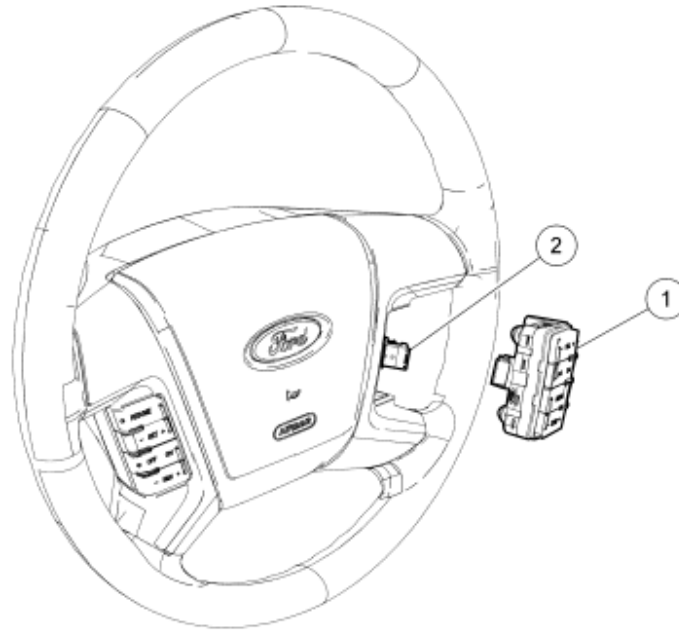
NOTE: **Module configuration is only required if a new satellite digital audio receiver (SDARS) module is being installed.**

1. Upload the SDARS module configuration information to the scan tool. For additional information, refer to Programmable Module Installation (PMI) in **MODULE CONFIGURATION** article.
2. Remove the liftgate scuff plate. For additional information, refer to Interior Trim - Exploded View in **INTERIOR TRIM AND ORNAMENTATION** article.
3. Remove the 2 nuts and the SDARS module.
 - Disconnect the satellite radio antenna connector and the electrical connector.
 - To install, tighten the nuts to 7 Nm (62 lb-in).
4. To install, reverse the removal procedure.
 - Download the configuration information to the SDARS module. For additional information, refer to PMI in **MODULE CONFIGURATION** article.

STEERING WHEEL CONTROLS

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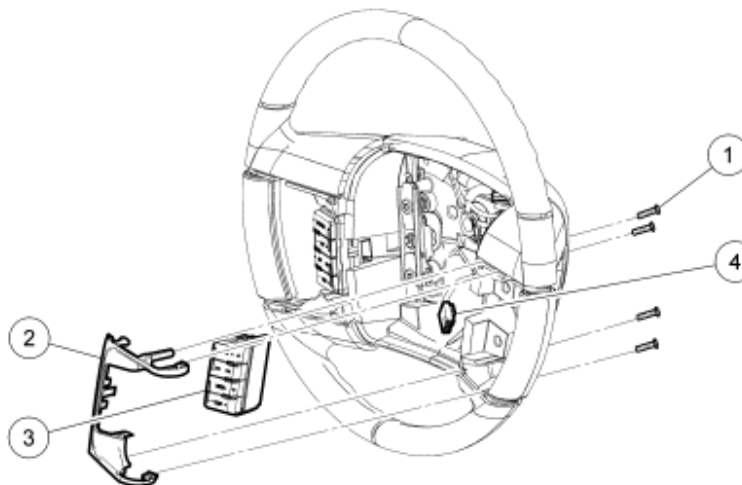
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N0035298

Fig. 36: Exploded View Of Steering Wheel Controls - Edge
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	9C888	Steering wheel controls
2	-	Steering wheel controls electrical connector (part of 14A411)



N0043419

Fig. 37: Exploded View Of Steering Wheel Controls - MKX
Courtesy of FORD MOTOR CO.

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Item	Part Number	Description
1	-	Steering wheel controls bezel screws (4 required)
2	-	Steering wheel controls bezel
3	9C888	Steering wheel controls
4	-	Steering wheel controls electrical connector (part of 14A411)

REMOVAL AND INSTALLATION

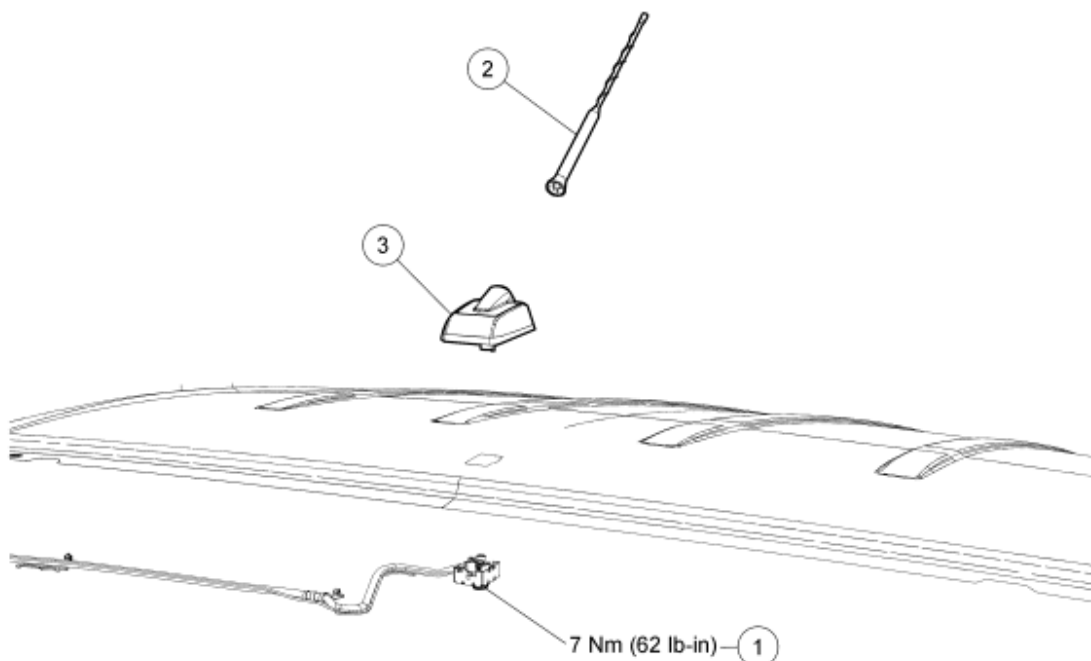
MKX

1. Remove the driver air bag module. For additional information, refer to **SUPPLEMENTAL RESTRAINT SYSTEM** article.
2. Remove the 4 screws and the steering wheel controls bezel.

All vehicles

3. Remove the steering wheel controls by pulling the switch toward the rear of the vehicle.
 - Disconnect the electrical connector.
4. To install, reverse the removal procedure.

ANTENNA - AM/FM



N0060110

Fig. 38: Exploded View Of AM/FM Antenna With Torque Specifications

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Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	AM/FM antenna base bolt (part of 18812)
2	18813	AM/FM antenna mast
3	18936	AM/FM antenna

REMOVAL AND INSTALLATION

1. Remove the AM/FM antenna mast, if desired.
2. Remove the dome lamp. For additional information, refer to **INTERIOR LIGHTING** article.
3. Loosen the AM/FM antenna base bolt.
 - To install, tighten to 7 Nm (62 lb-in).
4. Remove the AM/FM antenna.
5. To install, reverse the removal procedure.

ANTENNA - SATELLITE RADIO

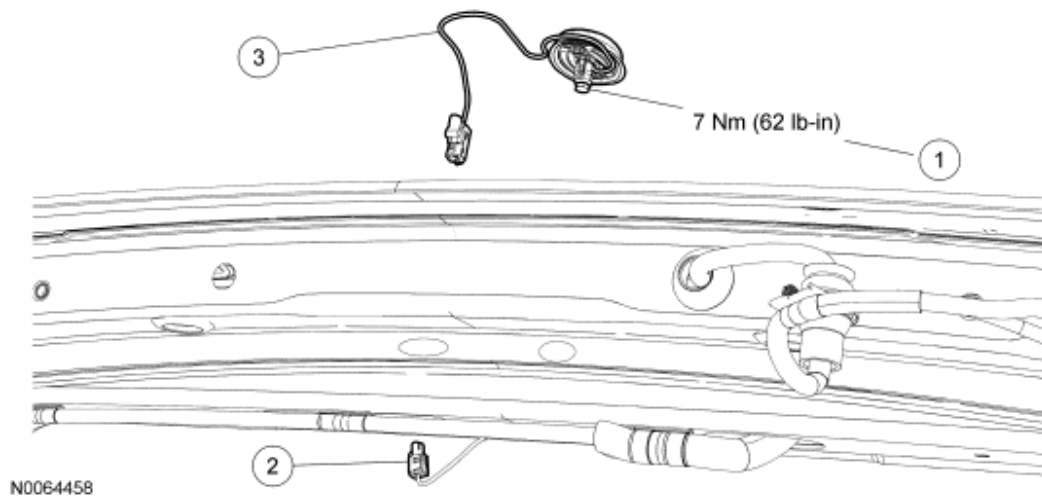
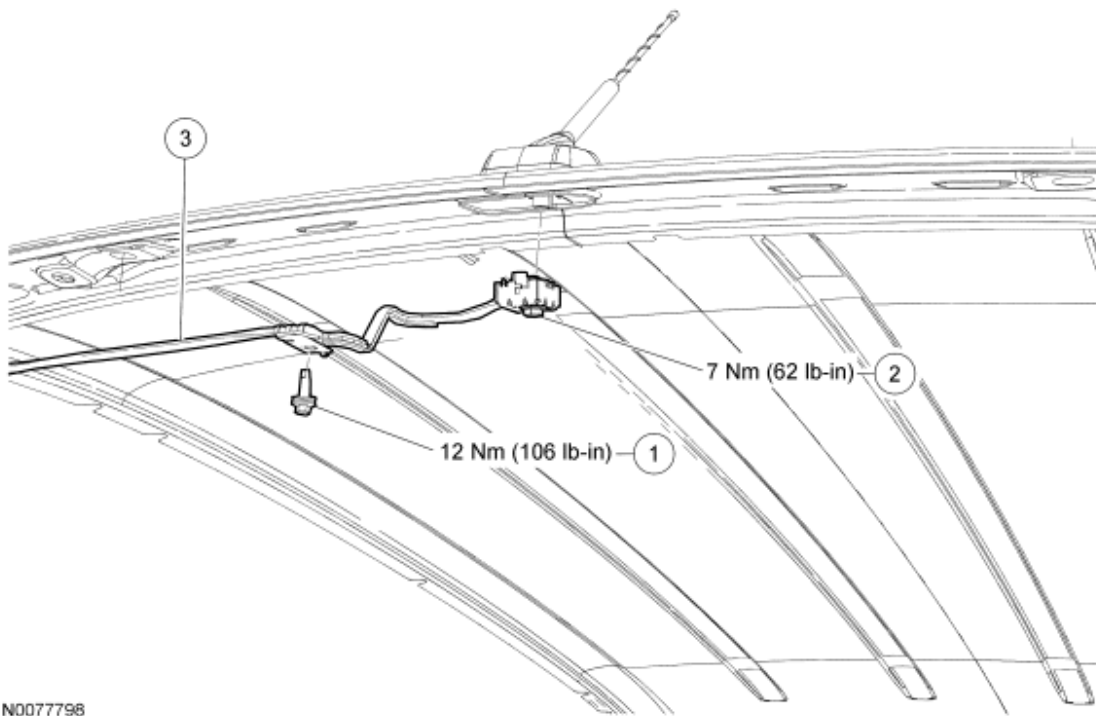


Fig. 39: Exploded View Of Satellite Radio Antenna With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Satellite radio antenna cable bolt (part of 18812)
2	-	Satellite radio antenna cable electrical connector (part of 19A397)
3	10E893	Satellite radio antenna

REMOVAL AND INSTALLATION

1. Remove the spoiler. For additional information, refer to **INTERIOR TRIM AND ORNAMENTATION** article.
 - Disconnect the satellite radio antenna electrical connector.
2. Loosen the satellite radio antenna cable bolt.
 - To install, tighten to 7 Nm (62 lb-in).
3. Remove the satellite radio antenna.
4. To install, reverse the removal procedure.

ANTENNA CABLE - AM/FM

N0077798

Fig. 40: Antenna Cable-to-Antenna With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	AM/FM antenna cable bolt
2	-	AM/FM antenna base bolt (part of 18812)
3	18812	Upper AM/FM antenna cable

2008 Ford Edge SE

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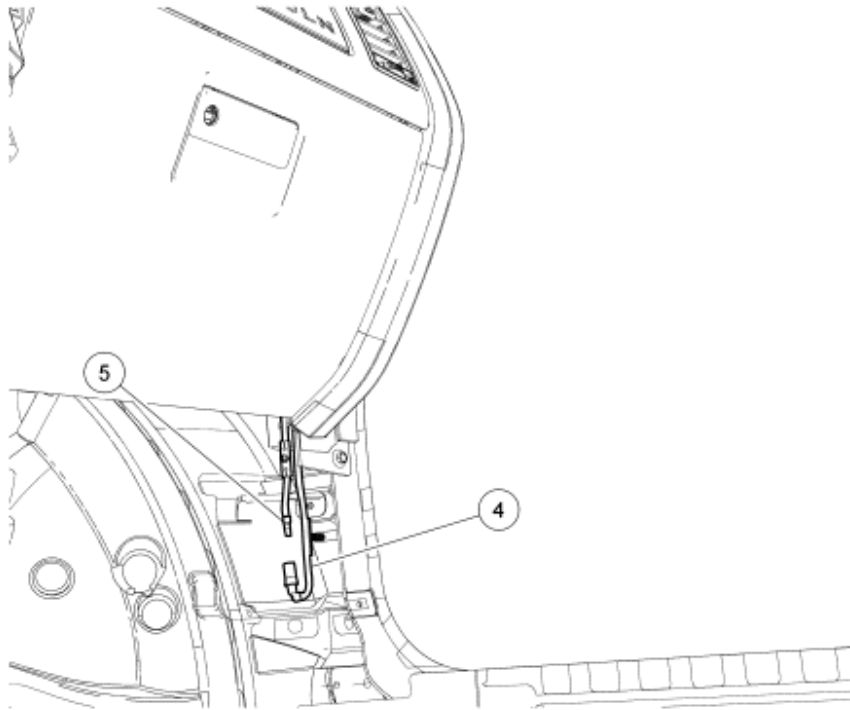


Fig. 41: Upper Antenna Cable-to-Lower Antenna Cable
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
4	18812	Upper AM/FM antenna cable
5	18812	Lower AM/FM antenna cable

REMOVAL AND INSTALLATION

Lower antenna cable

1. Remove the audio control module (ACM). For additional information, refer to **Audio Control Module (ACM)**.

Upper antenna cable

2. Remove the RH A-pillar trim panel. For additional information, refer to **INTERIOR TRIM AND ORNAMENTATION** article.
3. Remove the interior dome lamp. For additional information, refer to **INTERIOR LIGHTING** article.
4. Loosen the AM/FM antenna base bolt and disconnect the antenna cable from the antenna.
 - To install, tighten to 7 Nm (62 lb-in).
5. Remove the RH sun visor. For additional information, refer to Headliner in **INTERIOR TRIM AND ORNAMENTATION** article.

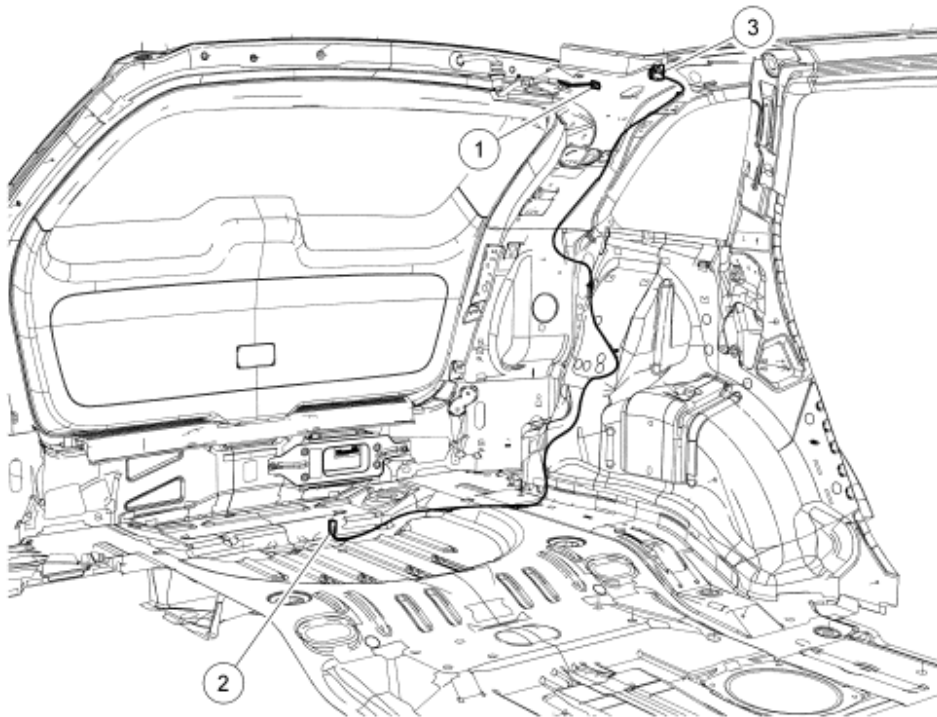
NOTE: Lower the headliner, as necessary.

6. Remove the AM/FM antenna cable bolt and the antenna cable.
 - To install, tighten the bolt to 12 Nm (106 lb-in).

Both antenna cables

7. Remove the RH front door scuff plate. For additional information, refer to Interior Trim - Exploded View in **INTERIOR TRIM AND ORNAMENTATION** article.
8. Disconnect the upper antenna cable from the lower antenna cable.
9. Release the pin-type retainers and remove the antenna cable.
10. To install, reverse the removal procedure.

ANTENNA CABLE - SATELLITE RADIO



N0061189

Fig. 42: Exploded View Of Satellite Radio Antenna Cable
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Satellite radio antenna cable-to-satellite radio antenna electrical connector (part of 19A397)
2	-	Satellite radio antenna cable-to-satellite digital audio receiver system (SDARS)

		module electrical connector (part of 19A397)
3	19A397	Satellite radio antenna cable

REMOVAL AND INSTALLATION

1. Remove the LH quarter trim panel. For additional information, refer to **INTERIOR TRIM AND ORNAMENTATION** article.
2. Remove the D-pillar trim panel. For additional information, refer to **INTERIOR TRIM AND ORNAMENTATION** article.
3. Remove the liftgate scuff plate. For additional information, refer to Interior Trim - Exploded View in **INTERIOR TRIM AND ORNAMENTATION** article.
4. Disconnect the satellite radio antenna cable from the satellite digital audio receiver system (SDARS) module.
5. Disconnect the satellite radio antenna cable from the satellite radio antenna, and remove the cable.
6. To install, reverse the removal procedure.

DOOR SPEAKER

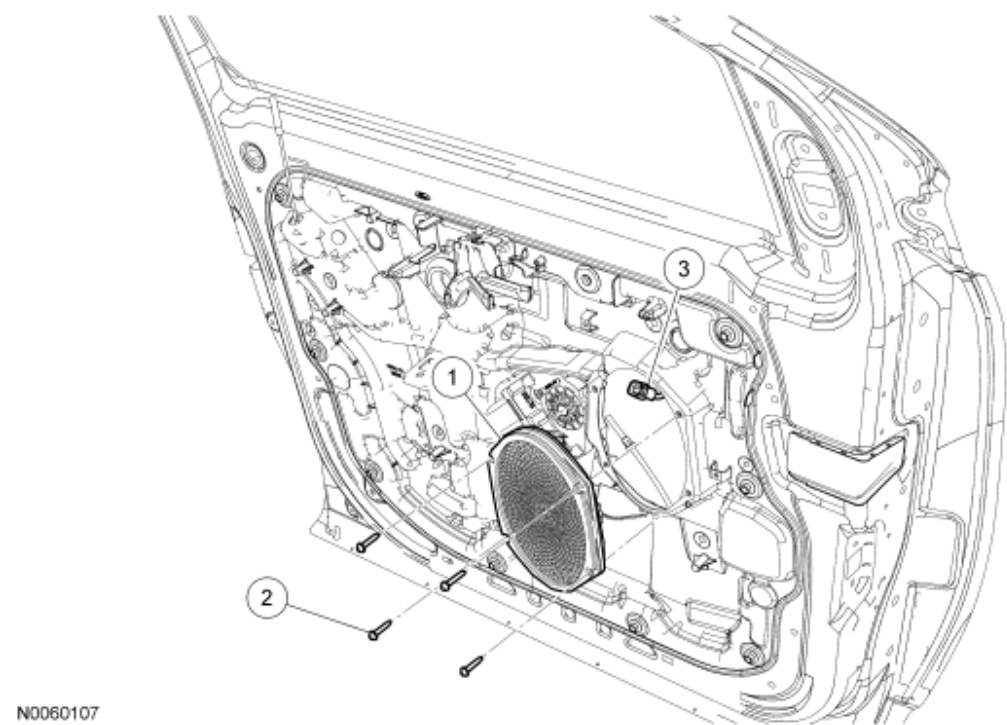
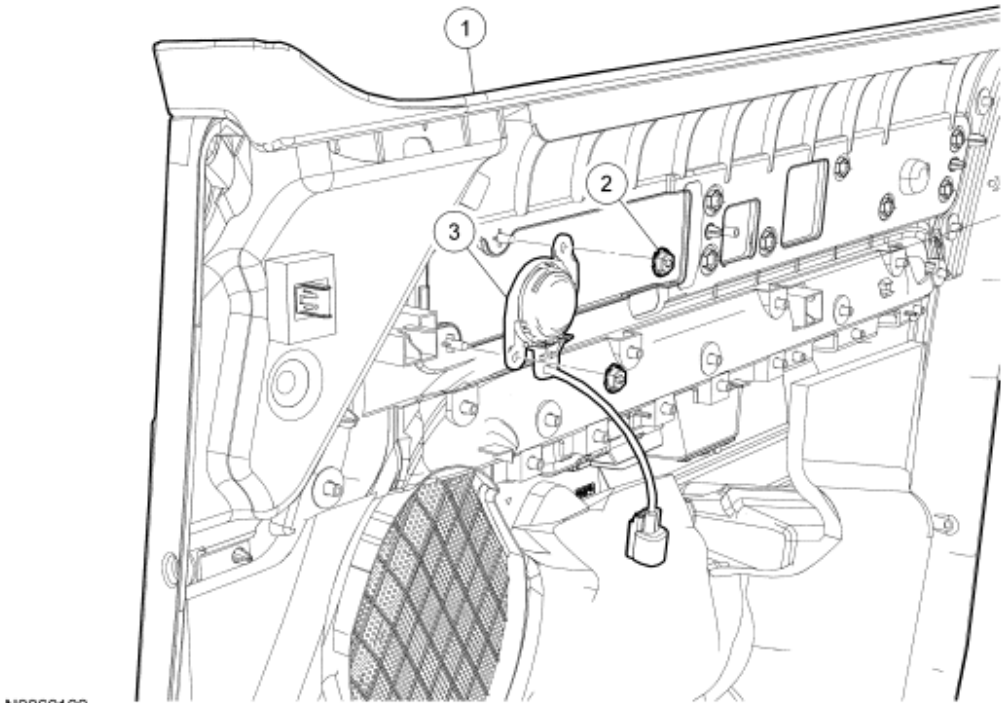


Fig. 43: Front or Rear Door
 Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	18808	Door speaker

2	973791	Door speaker screws (4 required)
3	-	Door speaker electrical connector(s) (part of door harness)



N0060108

Fig. 44: Front Door Tweeter
 Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Door trim panel
2	W706675-S	Front door tweeter speaker nuts (2 required)
3	19B170	Front door tweeter speaker

REMOVAL AND INSTALLATION

All speakers

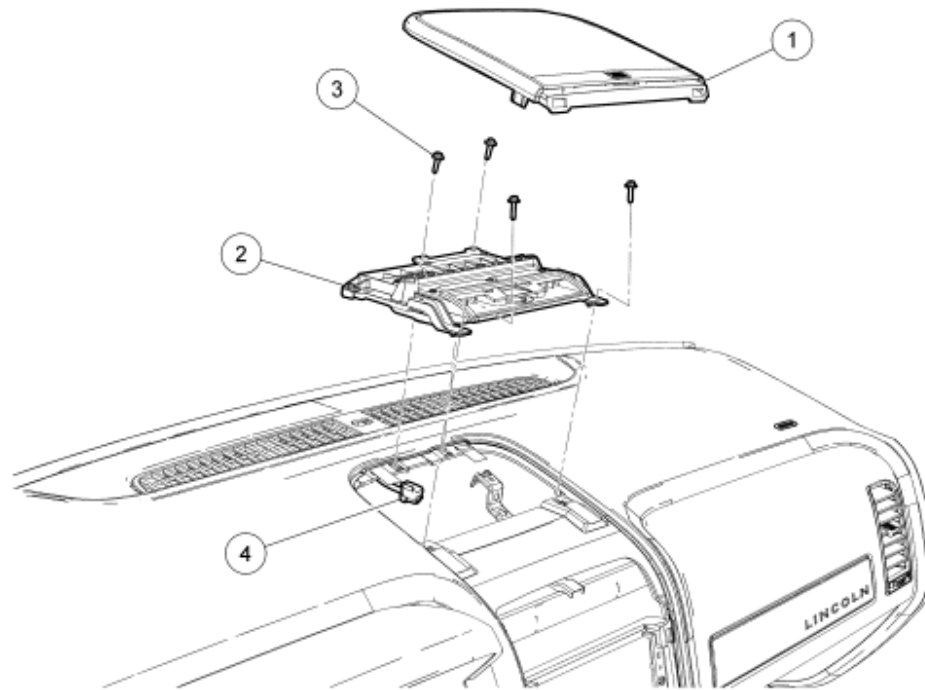
1. Remove the door trim panel. For additional information, refer to **INTERIOR TRIM AND ORNAMENTATION** article.
- Front door tweeter speaker**
2. Remove the 2 screws and the front door tweeter speaker from the door trim panel.
- Front or rear door speaker**

3. Remove the 4 screws and the door speaker.
 - Disconnect the electrical connector(s).

All speakers

4. To install, reverse the removal procedure.

INSTRUMENT PANEL SPEAKER



N0061188

Fig. 45: Exploded View Of Instrument Panel Speaker
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	78044A92	Instrument panel speaker cover
2	18808	Instrument panel speaker
3	-	Instrument panel speaker electrical connector (part of 14401)
4	W707628-S	Instrument panel speaker screws (4 required)

REMOVAL AND INSTALLATION

1. Remove the center instrument panel finish panel. For additional information, refer to **INSTRUMENT PANEL AND CONSOLE** article.
2. Remove the instrument panel speaker cover.

2008 Ford Edge SE

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3. Remove the 4 screws and the instrument panel speaker.
 - Disconnect the electrical connector.
4. To install, reverse the removal procedure.

SUBWOOFER SPEAKER

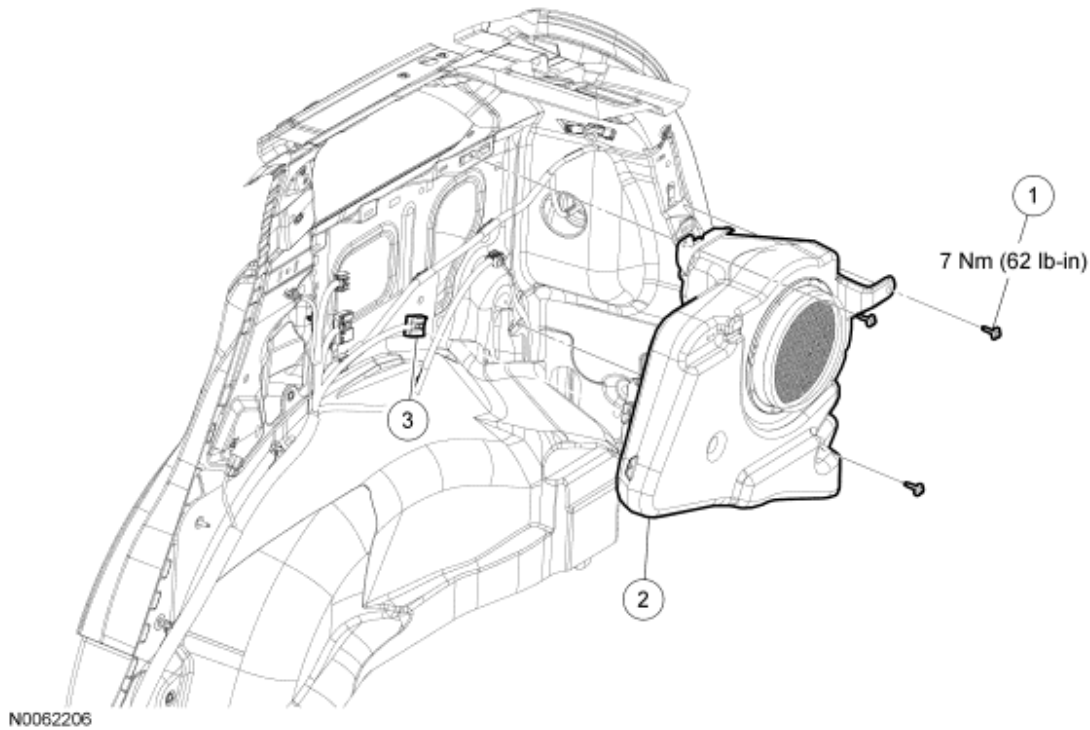


Fig. 46: Exploded View Of Subwoofer Speaker With Torque Specification - Edge
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W505425-S	Subwoofer speaker bolts (3 required)
2	18C804	Subwoofer speaker
3	-	Subwoofer speaker electrical connector (part of 14A005)

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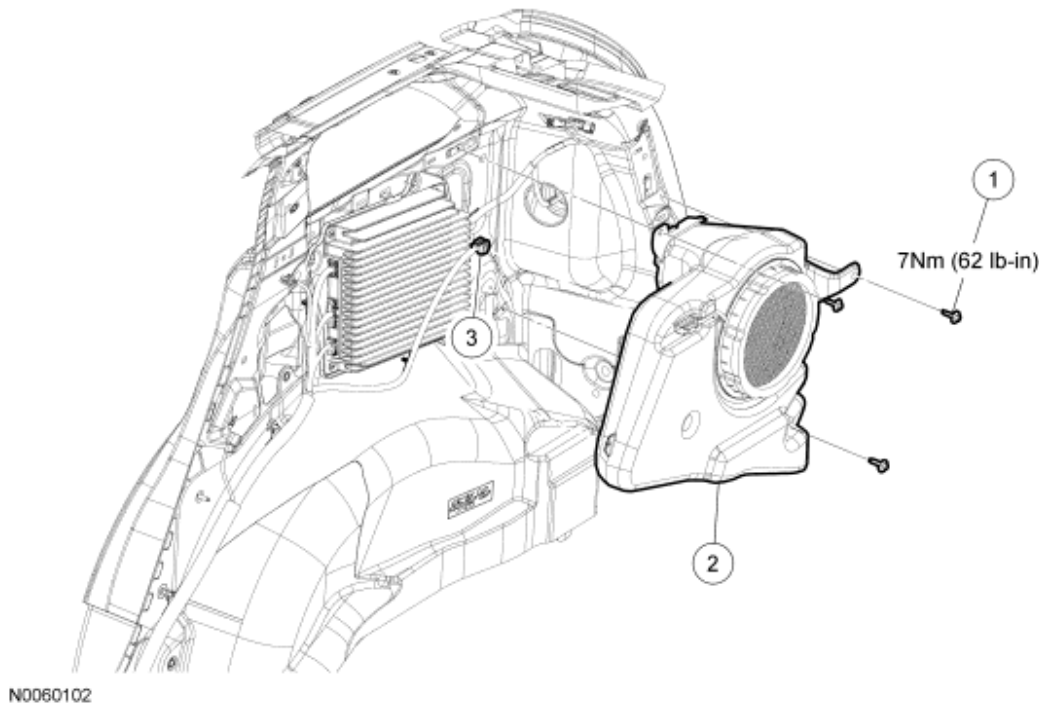


Fig. 47: Exploded View Of Subwoofer Speaker With Torque Specification - MKX
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W505425-S	Subwoofer speaker bolts (3 required)
2	18808	Subwoofer speaker
3	-	Subwoofer speaker electrical connector (part of 14A005)

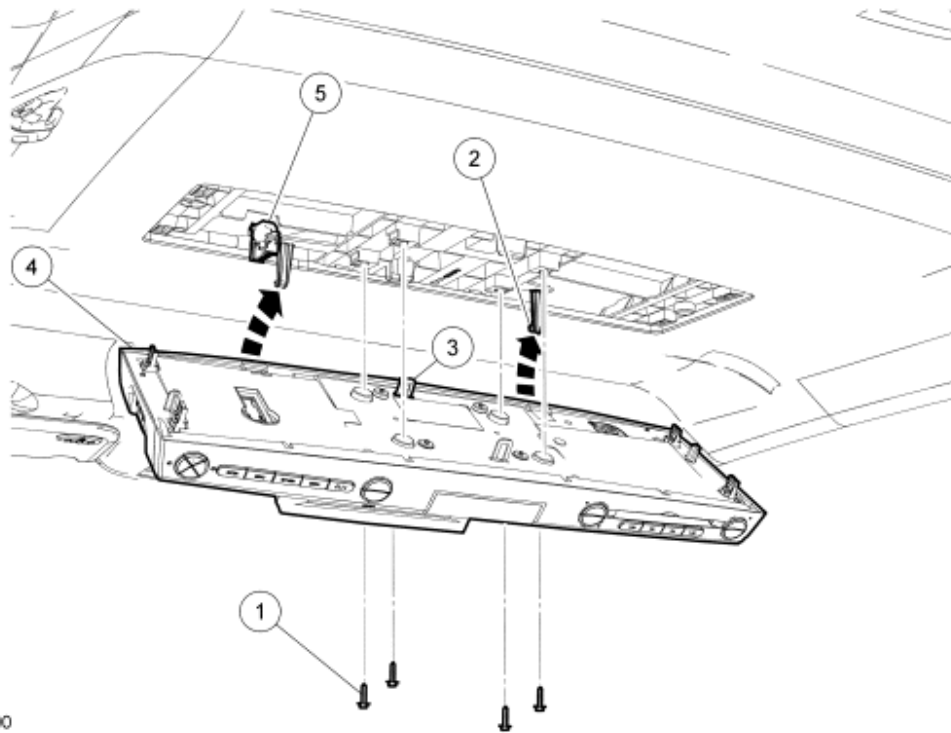
REMOVAL AND INSTALLATION

1. Remove the quarter panel trim panel. For additional information, refer to **INTERIOR TRIM AND ORNAMENTATION** article.
2. Remove the 3 bolts and the subwoofer speaker.
 - Disconnect the electrical connector.
 - To install, tighten the bolts to 7 Nm (62 lb-in).
3. To install, reverse the removal procedure.

REAR ENTERTAINMENT MODULE (RETM)

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N0060100

Fig. 48: Exploded View Of Digital Versatile Disc (DVD) Player
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W704875-S	Rear entertainment module (RETM) screws (4 required)
2	-	J-hook (part of 10E957)
3	-	U-hook (part of 10E947)
4	10E947	RETM
5	-	RETM electrical connector (part of 15A657)

REMOVAL

NOTE: Programmable module installation (PMI) only needs to be carried out if the rear entertainment module (RETM) is being replaced.

1. Upload the RETM configuration to the scan tool. For additional information, refer to **MODULE CONFIGURATION** article.

NOTE: Open the display to access the screws.

2. Remove the 4 screws.
3. Remove the RETM.

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- Pull down on the sides of the RETM to disengage the clips from the bracket.
- Using a suitable tool, push the rear most U-hook forward and release the U-hooks.
- Hang the RETM on the J-hooks and disconnect the electrical connector.
- Remove the RETM from the J-hooks.

INSTALLATION

1. Hang the RETM on the J-hooks.
2. Connect the electrical connector.
3. Remove the RETM from the J-hooks.

NOTE: **Tilt the RETM, as necessary, to align it correctly.**

4. Align the left side guide pin with the left side locating hole.
5. Lift up the right side of the RETM and engage the U-hooks.
6. Install the 4 screws.
7. Push on the left and right sides of the RETM to engage the clips.
8. Download the RETM configuration from the scan tool. For additional information, refer to **MODULE CONFIGURATION** article.

GLOBAL POSITIONING SYSTEM (GPS) ANTENNA

NOTE: **The view is shown from the back side of the instrument panel; however, the instrument panel does not have to be removed to carry out this procedure.**

2008 Ford Edge SE

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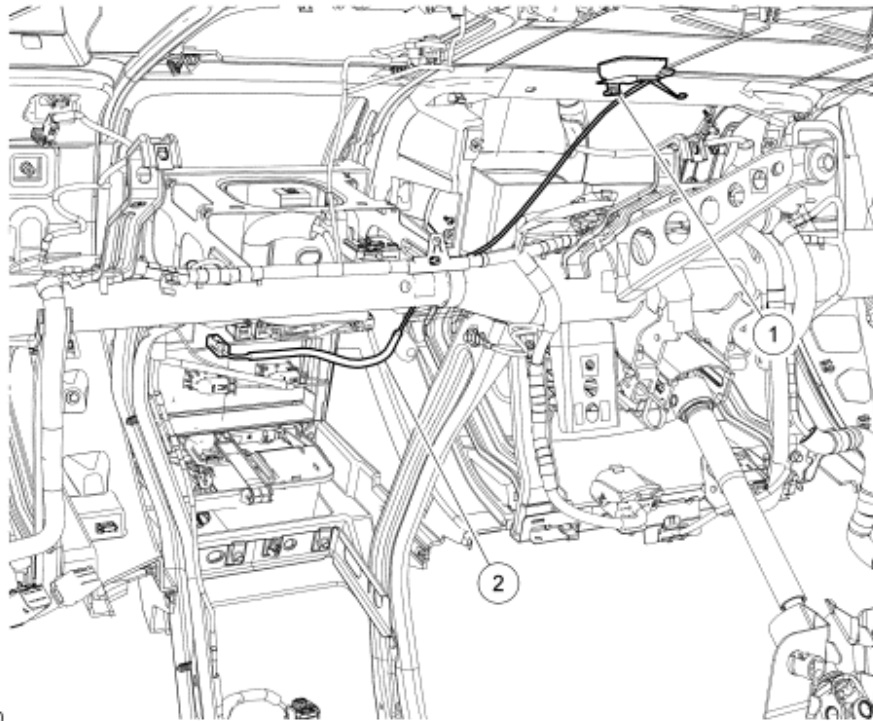


Fig. 49: Exploded View Of Global Positioning System (GPS) Antenna
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Global positioning system (GPS) antenna clip (part of 10E893)
2	10E893	GPS antenna

REMOVAL AND INSTALLATION

MKX

1. Remove the instrument cluster (IC). For additional information, refer to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER, AND WARNING CHIMES** article.

All vehicles

2. Remove the audio control module (ACM). For additional information, refer to **Audio Control Module (ACM)**.

NOTE: For vehicles without THX® audio, the vehicle will have an upper storage bin, rather than an instrument panel speaker.

3. Remove the instrument panel speaker. For additional information, refer to **Instrument Panel Speaker**.
4. Release the clip and remove the global positioning system (GPS) antenna.

5. To install, reverse the removal procedure.

ACCESSORY PROTOCOL INTERFACE MODULE (APIM)

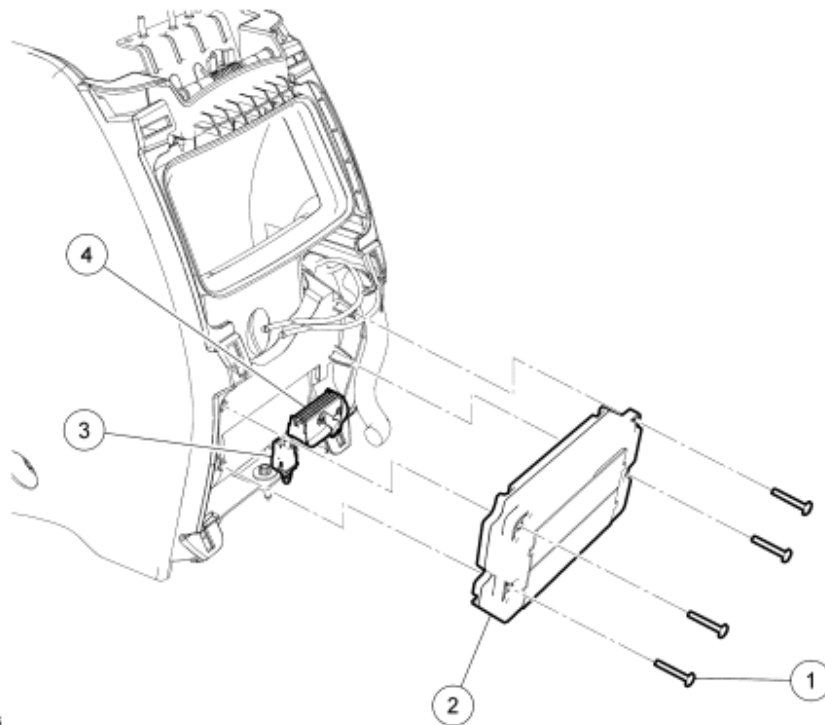


Fig. 50: Identifying Accessory Protocol Interface Module (APIM)
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Accessory protocol interface module (APIM) screws (4 required) (part of 78045A06)
2	14D212	APIM
3	14D202	Universal serial bus (USB) cable
4	-	APIM electrical connector (part of 14B079)

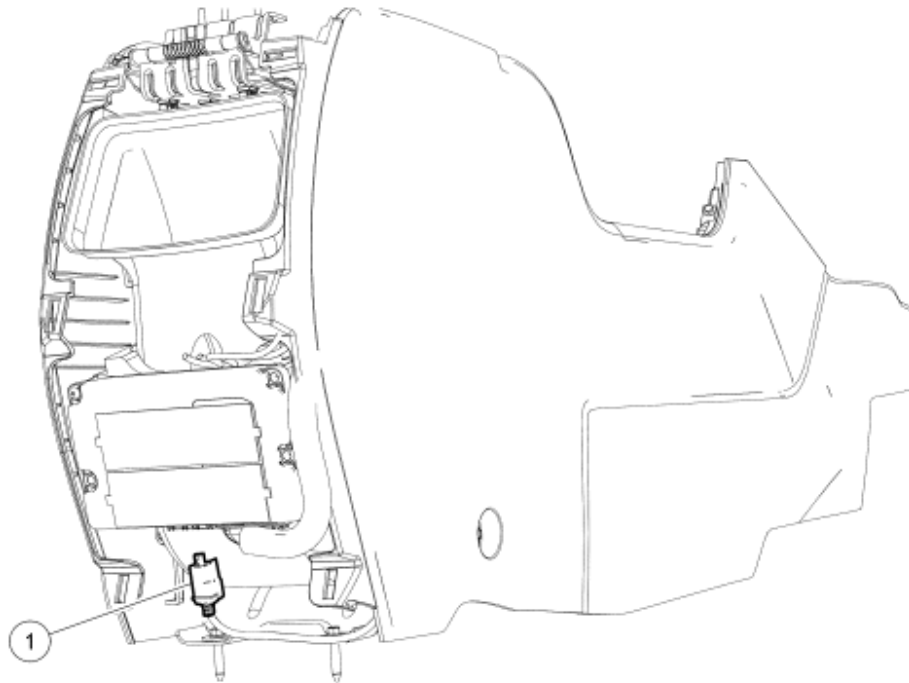
REMOVAL AND INSTALLATION

NOTE: Accessory protocol interface module (APIM) reprogramming is only necessary if a new module is being installed.

1. Verify the current APIM software and hardware level. For additional information, refer to Accessory Protocol Interface Module (APIM) Reprogramming in **MODULE CONFIGURATION** article.
2. Remove the floor console rear trim panel by pulling straight back to disengage the clips.

3. Disconnect the universal serial bus (USB) cable and the electrical connector.
4. Remove the 4 screws and the APIM.
5. To install, reverse the removal procedure.
 - Reprogram the APIM. For additional information, refer to **MODULE CONFIGURATION** article.

UNIVERSAL SERIAL BUS (USB) CABLE AND PORT



N0077734

Fig. 51: Universal Serial Bus (USB) Cable-To-Accessory Protocol Interface Module (APIM)
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Universal Serial Bus (USB) cable-to-Accessory Protocol Interface Module (APIM) connector (part of 14D202)

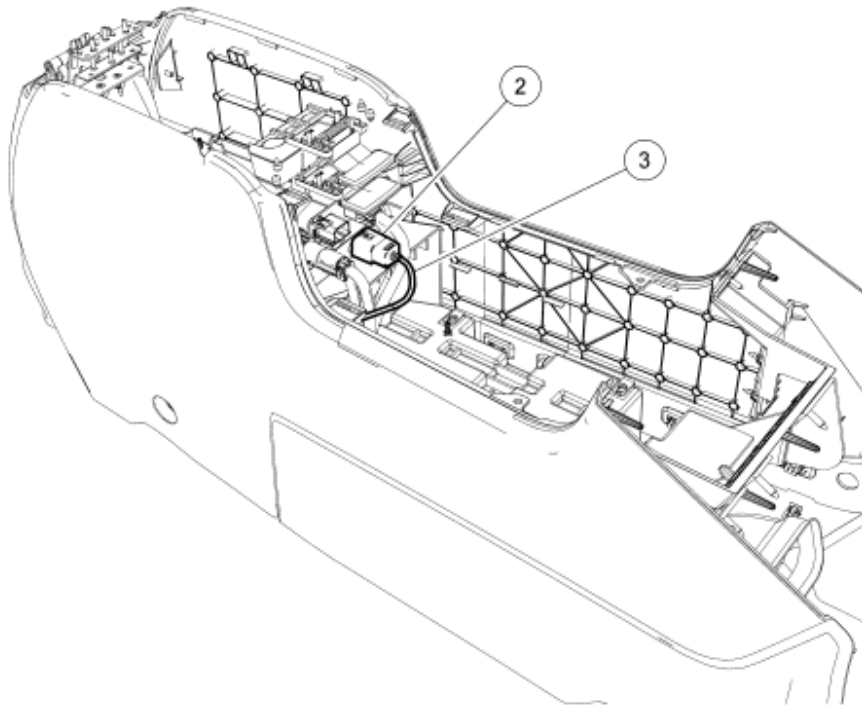
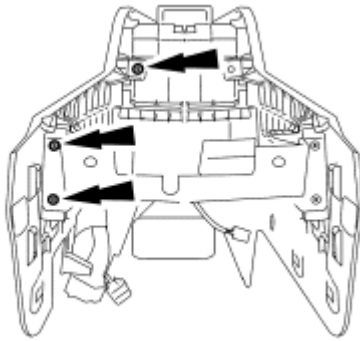


Fig. 52: Universal Serial Bus (USB)-To-Accessory Protocol Interface Module (APIM)
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
2	-	USB cable-to-bezel connector
3	14D202	USB cable and port

REMOVAL AND INSTALLATION

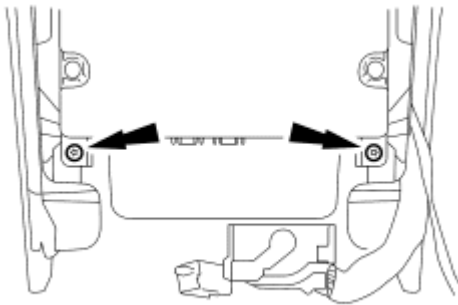
1. Remove the floor console. For additional information, refer to **INSTRUMENT PANEL AND CONSOLE** article.
2. Remove the floor console rear trim panel by pulling straight rearward to disengage the clips.
3. Disconnect the Universal Serial Bus (USB) cable-to-Accessory Protocol Interface Module (APIM) connector.
4. Disconnect the USB cable-to-bezel connector.
 - Disengage the push-pin near the connection.
5. Disconnect the USB cable-to-bezel connector.
6. Remove the 3 screws holding the front of the RH panel to the floor console.



N0084316

Fig. 53: Identifying Screws
Courtesy of FORD MOTOR CO.

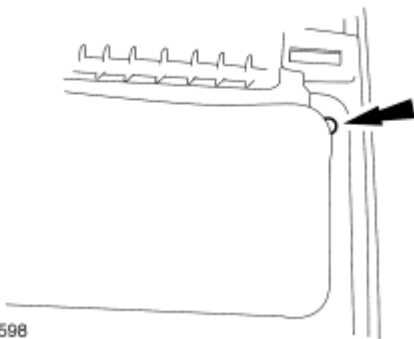
7. Remove the 2 screws and the APIM and bracket as an assembly.
 - Disconnect the electrical connector.



N0084317

Fig. 54: Identifying Screws
Courtesy of FORD MOTOR CO.

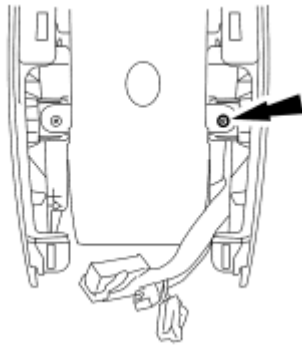
8. Remove the screw holding the upper rear of the RH panel to the floor console.



N0085598

Fig. 55: Identifying Screws
Courtesy of FORD MOTOR CO.

9. Remove the screw holding the lower rear of the RH panel to the floor console.

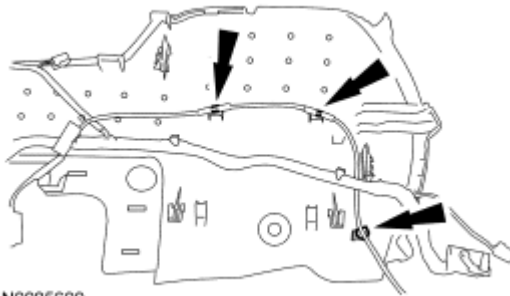


N0085599

Fig. 56: Identifying Screws

Courtesy of FORD MOTOR CO.

10. Remove the RH panel from the floor console by sliding it upward to disengage the clips.
11. Disengage the push-pins and remove the USB cable and port.



N0085600

Fig. 57: Identifying push-pins

Courtesy of FORD MOTOR CO.

12. To install, reverse the removal procedure.